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MAI(A) the Sound Be With You

MAIA-AV: Extending Multimodal AI Assessment to the Audio-Visual Domain

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Introduction

The present work was carried out during a research internship at Fondazione Bruno Kessler (FBK), in Trento. Active since 2007 and rooted in the earlier Istituto Trentino di Cultura, founded in 1962 by Bruno Kessler, FBK is a non-profit research institute working across artificial intelligence, biomedical research, digital industry, space exploration, and quantum technologies, and is widely regarded as one of the leading research institutions in Italy. The internship was hosted by the Natural Language Processing (NLP) research unit, within FBK's Augmented Intelligence Center.

Evaluating whether the remarkable success of Visual Language Models (VLMs) truly signifies a deep multimodal comprehension, or merely a reliance on surface-level correlations and linguistic biases, continues to be a significant area of inquiry. *MAIA: Multimodal AI Assessment* was introduced to address this question, as a native Italian benchmark for evaluating video-based reasoning in Vision-Language Models [22, 21]. MAIA investigates the extent to which VLMs coherently integrate visual and linguistic information. More specifically, it assesses whether model predictions are grounded in visual evidence rather than primarily driven by distributional regularities learned by the language component. This distinction is crucial because a model may answer a question correctly while failing to recognise the same information consistently when it is presented in a discriminative format. In such cases, the prediction may result from linguistic plausibility rather than genuine visual understanding. To address this issue, MAIA employs two aligned tasks: Visual Statement Verification and Open-Ended Visual Question Answering. Through these tasks, the benchmark evaluates not only model accuracy, but also the robustness and consistency of model behaviour across different evaluation formats. The benchmark further defines twelve fine-grained reasoning categories to identify which competencies are used by the models and in which areas linguistic shortcuts, unimodal collapse, or hallucinations emerge. A limitation of the original framework is that the dataset was designed and evaluated primarily on the basis of visual information. Real-world videos, however, are inherently audiovisual: relevant meaning may also be conveyed through dialogue, environmental sounds, music, and other acoustic events. Recent multimodal and omnimodal architectures can process visual, auditory, and textual inputs, but the availability of multiple modalities does not guarantee that they will be integrated effectively. Models may over-rely on a single channel or be negatively affected by redundant or weakly relevant information. Building on this limitation, *Lost in Transcription: Investigating the Role of Audio Information for Video-based Visual Statement Verification* [4] extends the original MAIA research framework to a more complex analysis of the interaction between visual, linguistic, and auditory information, while preserving its competence-oriented perspective. The study considers two complementary representations of audio: raw signal and automatic transcriptions provided as textual input. By evaluating these representations both independently and in combination with video information, the analysis investigates whether audio contributes meaningfully to task performance and whether one representation has a stronger influence on model predictions. This leads to two research questions:

RQ1: *To what extent does raw audio affect video-based Visual Statement Verification when it is jointly processed with visual and textual information?*

RQ2: *How effectively do multimodal models integrate visual, textual, and auditory information when performing video-based reasoning and understanding tasks?*

The experiments show that raw audio has a limited and strongly context-dependent effect. It is most useful when visual information is unavailable, while its addition to an already informative visual input may instead provide little benefit or introduce noise. The results also show that transcriptions can produce greater interference than raw audio, suggesting that multimodal architectures do not necessarily achieve more effective cross-modal understanding simply because additional modalities are available. These findings must nevertheless be interpreted in relation to the nature of the dataset. The one hundred original MAIA videos were selected and annotated according to their visual content, and the questions were formulated without considering the audio track. Consequently, the benchmark does not systematically require auditory information to solve its items. My thesis project begins from this limitation. The following chapters investigate whether models can integrate multiple input modalities more coherently when the dataset is specifically designed so that auditory information is at least partially relevant. The central hypothesis is that, under these conditions, models may use complementary signals across modalities to reconstruct missing information and develop a more genuinely multimodal understanding of video content, producing answers grounded in the available evidence rather than in linguistic plausibility or chance.

1. Literature review

The rapid evolution of Visual Language Models (VLMs) has progressively expanded the ability of neural systems to jointly process perceptual and linguistic information. This line of research builds on the hypothesis that linguistic meaning cannot be constructed exclusively from distributional regularities learned from textual corpora, but must also be grounded in perceptual experience [2]. Within this context, Visual Question Answering (VQA) emerged as one of the first systematic paradigms for assessing whether a model can answer questions about the content of an image [1]. Subsequent benchmarks, such as GQA, extended this setting by introducing compositional questions involving entities, attributes, and visual relations [9]. However, high task accuracy alone does not provide sufficient evidence of multimodal comprehension. Several studies have shown that models can exploit statistical regularities in answers, questions, or dataset distributions, producing correct predictions without necessarily *understanding* the associated visual content. Evaluation has therefore progressively moved beyond task performance towards the assessment of grounding, understood as the ability to associate a prediction with the relevant perceptual evidence. One widely adopted strategy for evaluating grounding relies on minimally contrastive pairs, in which a correct instance is paired with a foil obtained by modifying a semantically relevant element. *FOIL-it!*, for example, evaluates whether a model can detect localized inconsistencies between an image and its linguistic description [19]. *Winoground*, instead, presents pairs of images and descriptions containing the same lexical elements but arranged according to different compositional relations, showing that recognizing individual entities does not necessarily entail understanding the relational structure connecting them [23]. The transition from images to videos introduces additional challenges. Video understanding requires integrating information distributed over time, recognizing actions and state changes, ordering events, and establishing temporal and causal relations. *Action Genome Question Answering (AGQA)* addresses these requirements through compositional questions concerning objects, actions, and spatio-temporal relations [8]. The *Multi-modal Video Understanding Benchmark (MVBench)* similarly evaluates a broader range of competencies, including action perception, movement, temporal ordering, and dynamic reasoning [13]. Nevertheless, video benchmarks remain vulnerable to unimodal shortcuts and to strategies that rely predominantly on the most informative frame. *Video Language Model Assessment (VILMA)* addresses this limitation as a zero-shot benchmark based on video–text contrastive pairs [10]. For each complex capability, VILMA introduces a proficiency test designed to verify the preliminary perceptual ability required to solve the corresponding task. The results show that, once these prerequisite conditions are considered, part of the apparently correct performance can be attributed to accidental or spurious strategies. In particular, the examined VLMs do not display a systematic advantage over statistical image-based models on tasks requiring strict temporal grounding. Within this line of research, MAIA introduced a natively Italian benchmark designed to provide a fine-grained evaluation of video-linguistic reasoning [22]. The resource covers twelve reasoning categories related to linguistic phenomena, including causal, counterfactual, implicit, spatial, temporal, sentimental, planning, uncertainty, and out-of-scope reasoning. Its main contribution lies in aligning two tasks built on the same videos and questions: *Visual Statement Verification (VSV)*, where the model selects the correct statement from a minimally contrastive pair, and *Open-Ended Visual Question Answering (OEVQA)*, where the model freely generates the answer. This setting makes it possible to assess not only accuracy but also coherence between discriminative comprehension and linguistic generation. A model may correctly answer a multiple-choice question while failing to reproduce the same information in an open-ended setting, or may behave inconsistently across equivalent linguistic formulations. To capture this phenomenon, MAIA introduces the *Aggregate Accuracy* metric, which rewards only those instances in which the model correctly answers all true–false pairs associated with the same question and simultaneously generates the correct response in the corresponding open-ended task. The reduction in Aggregate Accuracy relative to conventional single-instance accuracy highlights the fragility of apparently strong performance and the need to jointly evaluate robustness, consistency, and grounding. Although VLMs have traditionally focused on text–vision interactions, real videos are intrinsically audiovisual. Their interpretation may depend on music, prosody, background noise, and other acoustic signals that cannot be reconstructed from visual frames alone. An increasing body of research has accordingly extended multimodal language model architectures to support joint processing of text, images, video, and audio. *ImageBind*, for instance, introduced a shared representation space spanning images, text, audio, depth, thermal information, and inertial movement [7]. Other multimodal language models (MLMs) rely on modality-specific encoders combined with projection or fusion modules that align heterogeneous representations with the linguistic space. *Video-SALMONN: Speech-Enhanced Audio-Visual Large Language Models* introduces a multi-resolution Q-former[1] to align audio signals with synchronized video [20], while *VideoLLaMA2* combines spatio-temporal modeling and acoustic processing within a generative multimodal architecture [3]. More recent omnimodal architectures process multiple modalities within a unified framework. *Qwen2.5-Omni* adopts a thinker–talker architecture that accepts text, image, video, and audio inputs and produces textual or audio responses [14]. The subsequent *Qwen3.5-Omni* architecture integrates visual and audio encoders into a common backbone and introduces explicit temporal references to improve audio–video alignment [15]. These developments demonstrate that heterogeneous signals can be processed within a single architecture. The availability of a modality, however, does not imply that it is used functionally or complementarily. The distinction between **multimodal availability** and **multimodal integration** is essential here: the former indicates that a system can accept multiple input sources, the latter requires the model to select, coordinate, and combine relevant information across modalities. A system may thus be multimodal from an architectural perspective while relying predominantly on unimodal information during prediction. This issue is evident in *Audio-Visual Question Answering* benchmarks such as MUSIC-AVQA, which evaluates spatio-temporal relations in musical performance videos and extends the analysis to objects, entities, and everyday activities [12]. Subsequent studies reveal a strong visual bias, showing that many questions can be answered using the video stream alone, without exploiting the audio channel. Under these conditions, similar audio-only and video-only performance does not demon-

[1] A multi-resolution Q-former, such as a Multi-Resolution Causal Q-Former, processes continuous audio, video, or image information at different temporal or spatial scales through distinct query banks, balancing fine-grained signals with broader contextual information.

strate genuine integration, but rather suggests a form of unimodal collapse in which one modality dominates the decision process. To address this problem, *Dave* was introduced as a benchmark in which each question jointly requires visual and acoustic information, making either modality insufficient in isolation [16]. The benchmark distinguishes among action recognition, sound classification, temporal ordering, and multimodal synchronization, a decomposition that helps determine whether an error arises from the failure to perceive an event or from the inability to establish temporal relations between sounds and visual actions. The results show that models may achieve strong performance on individual components while encountering greater difficulty in audio–visual synchronization, indicating that cross-modal integration remains a central limitation. The increasing attention to audio has also led to benchmarks specifically designed to assess acoustic intelligence. *MMAU-Pro* evaluates 49 competencies distributed across speech, environmental sounds, music, and their combinations [11]. These tasks include long-audio comprehension, multi-track reasoning, spatial localization, prosody interpretation, and the integration of dialogue, music, and background noise. The persistently limited performance of current models indicates that acoustic processing cannot be reduced to dialogue recognition alone, since audio also conveys spatial, temporal, emotional, and contextual information that must be evaluated independently. A complementary perspective is provided by *SONIC-01*, a benchmark for omnimodal models based on audio–visual interactions drawn from real conversational and everyday contexts [17]. SONIC-01 combines open-ended summarization, multiple-choice questions, and temporal event localization, while also showing that replacing raw audio with transcription removes paralinguistic information related to tone and communicative intention. Its results identify temporal localization as one of the most challenging tasks, confirming that the integration of verbal content, acoustic signals, and visual sequences remains an open problem. The comparison between native audio and transcription is also central to my recent work, *Lost in Translation: Investigating the Role of Audio Information for Video-based Visual Statement Verification* [4]. The study extends the MAIA framework by introducing audio into the VSV task and comparing two representations of the acoustic modality: raw audio directly processed by *Qwen2.5-Omni-7B* and automatic transcriptions supplied as additional textual input to *Qwen2.5-VL-7B*. To characterize the audio content, the videos are divided into four classes:

Music: Instances characterized by repetitive vocalizations, onomatopoeic patterns, strongly repetitive tokens, or explicit lexical cues associated with non-speech audio.

Dialogue: Videos whose transcriptions display sufficiently coherent linguistic content, acceptable log-probability values, low degeneracy, and limited evidence of no-speech segments.

Silence/Noise: Videos with empty or highly unreliable transcriptions, particularly when associated with low log-probability values and relatively high no-speech probabilities.

Unknown: Residual cases that appear speech-related but do not satisfy the dialogue criteria with sufficient confidence, thereby preserving potentially informative content while explicitly marking its uncertainty.

Class labels and speech transcriptions were obtained by combining the outputs of *Whisper-1* and *GPT-4o-transcribe*, together with an assessment of output plausibility. This classification distinguishes results according to the informativeness of the available audio content. The results reveal a marked dominance of the visual channel. Audio provides only a limited benefit, mainly when no visual information is available. Once rich visual input is provided, audio's contribution becomes negligible and may even introduce noise or errors, with limited robustness observed when a single black frame is added instead. Under visually rich conditions, the difference between with-audio and without-audio settings is minimal, suggesting that audio functions more as a surrogate channel than as information fully integrated with vision. Audio transcriptions exhibit a different behavior. When added to already rich visual input, they reduce accuracy relative to the visual-only condition. This shows that transcription is not a neutral transformation: once projected into the same representational space as the questions and answer options, it can compete with them, increasing the influence of textual regularities and introducing redundant or irrelevant information. *Lost in Translation* thus shows that different representations of the same acoustic content can affect the model's decision process differently. Raw audio can be partially ignored when it is not useful, whereas transcription interacts directly with the linguistic component of the model. Introducing an additional modality does not therefore guarantee an improvement, and may instead compromise predictions that were already correct on the basis of reliable visual evidence. These findings must be interpreted in light of the original MAIA design. The benchmark was created to investigate visual content, and annotators were explicitly instructed not to consider audio information, so its questions and answers were not designed to exploit acoustic evidence. The limited contribution of audio may therefore reflect both the models' modality-integration mechanisms and the limited relevance of audio within the original dataset. Recent literature further emphasizes the importance of identifying the distinct sources of multimodal errors. *MIRAGE* disentangles hallucinations caused by perceptual failures from those arising from incorrect reasoning over correctly represented information [5]. Although primarily focused on visual reasoning, its methodological principle extends naturally to audio–visual settings, where an incorrect response may derive from failed sound recognition, inaccurate temporal synchronization, selection of the wrong modality, or erroneous inference over otherwise correctly represented evidence. Complementary attribution approaches seek to identify which parts of the input contribute to the generation of an answer. *OmniTrace* proposes an attribution framework for omnimodal autoregressive generation that links generated tokens to textual segments, visual regions, and audio/video intervals, aggregating attention signals into cross-modal explanations associated with linguistically interpretable units. This type of analysis is particularly relevant for determining whether a model actually exploits audio information or whether its prediction is driven primarily by text and vision. The current state of the art thus reveals a clear tension between the rapid development of omnimodal architectures and their effective ability to integrate heterogeneous modalities. Contemporary models can receive multiple input types within the same context, yet diagnostic benchmarks provide no guarantee that these sources are used complementarily. Performance may instead depend on a dominant modality, linguistic shortcuts, dataset correlations, or additional signals that introduce noise. A central unresolved limitation is the construction of benchmarks in which audio and visual information are explicitly supervised. Assessing genuine audio–visual comprehension requires items in which audio contributes relevant information and the correct answer cannot be determined from a single modality alone. Such a design would distinguish complementarity, dominance, substitution, and interference, while evaluating whether the model selects the appropriate evidence and combines it coherently across modalities. The present study follows this direction. Building on the competence-oriented MAIA framework and the recent findings of *Lost in Translation*, it investigates model behavior on a dataset specifically designed to require a genuine contribution from audio in order to answer the questions correctly. The central hypothesis is that, when audio and video provide partial and complementary evidence, multimodal comprehension depends on the model's ability to reconstruct the missing information through cross-modal integration, avoiding both unimodal collapse and linguistic shortcuts based on plausibility.

2. Data Collection

2.1 From MAIA to MAIA-AV: Dataset Expansion and Curation

Video Selection

The dataset adopted in MAIA-AV builds upon the guidelines of Multimodal AI Assessment (MAIA), comprising 100 short videos, approximately 30 seconds each, selected from YouTube Italia to represent everyday scenarios of Italian culture. Content domains include sports, culinary arts, urban life, and cultural activities, with a strong focus on human presence and highly informative scenes. In the original formulation, video selection and annotation were oriented strictly toward visual comprehension, completely omitting audio content. Building upon this initial set, we collected an additional 100 videos, each uniquely identified from `video001` to `video100`. These identifiers map each video to its corresponding audio stream and text transcriptions. Each video was precisely trimmed to extract targeted segments, creating a controlled corpus of clips. Particular attention was paid to the alignment between dialogue content and the elements appearing in the scene. This targeted selection of relevant dialogue enables a precise investigation of multimodal interactions. By prioritizing scenarios where spoken dialogue is semantically grounded in the visual content, we ensure that the audio signal plays a genuinely informative role in the multimodal instance. This distinction represents the core departure from the original dataset: MAIA-AV-(AudioVisual) is a benchmark engineered to combine and compare distributed evidence across visual, auditory, and textual modalities.

Implementation Details The video acquisition pipeline leverages `yt_dlp` to extract specific temporal segments from chosen YouTube Italia videos. Instead of downloading entire videos for later editing, each clip is identified before acquisition using three key parameters: the source URL, the start timestamp, and the duration. These parameters define an exact temporal interval, directly controlling the retrieval process. The conversion of timestamps into seconds is managed by `parse_time`, which translates `HH:MM:SS` or `MM:SS` formatted strings into integer values.

```python
def parse_time(time_str):
    """Converts HH:MM:SS or MM:SS string to seconds."""
    parts = [int(p) for p in time_str.split(':')]

    if len(parts) == 3:
        return parts[0] * 3600 + parts[1] * 60 + parts[2]
    elif len(parts) == 2:
        return parts[0] * 60 + parts[1]
    return parts[0]
```

Adding the requested duration to the start position determines the upper boundary of the segment:

```python
start_sec = parse_time(start_time_str)
end_sec = start_sec + int(durata_str)
```

The resulting bounds, `start_sec` and `end_sec`, define the exact temporal window extracted from the source. Output paths are structured prior to downloader configuration, ensuring MP4 container compliance and consistent dataset indexing under the target directory:

```python
filename_input = input(
    "Enter output filename (e.g., clip): "
).strip()
if not filename_input.endswith(".mp4"):
    filename_input += ".mp4"
output = f"Dataset/video{filename_input}"
```

Prefixing filenames with `video` maintains a unified identifier convention across all dataset instances. Stream selection, temporal trimming, and media normalization are encapsulated within the `ydl_opts` dictionary. Format selection prioritizes high-resolution video streams alongside optimal audio feeds:

```
ydl_opts = {
    'format': 'bestvideo[height<=2160]+bestaudio/best',
}
```

This filter selects video streams up to 2160p paired with the highest quality audio stream, falling back to pre-merged streams (`/best`) if separate tracks are unavailable. Temporal constraints are applied during acquisition via `download_range_func`:

```
    'download_ranges': download_range_func(
        None,
        [(start_sec, end_sec)]
    ),
```

Embedding segment bounds directly into the retrieval mechanism eliminates separate media cropping steps and reduces bandwidth overhead. To maintain frame-accurate cutting across keyframe-based compressed video formats, the pipeline enforces keyframe generation at boundary points:

```
    'force_keyframes_at_cuts': True,
```

Forcing keyframes ensures temporal alignment at cut points, preserving action continuity for downstream multimodal evaluation in MAIA-AV. Output formats and post-processing codecs are explicitly declared to standardize the corpus:

```
    'merge_output_format': 'mp4',
    'postprocessor_args': {
        'ffmpeg': [
            '-c:v', 'libx264',
            '-preset', 'fast',
            '-c:a', 'aac'
        ]
    },
```

FFmpeg re-encodes visual streams via H.264 (`libx264`, `fast` preset) and acoustic streams via AAC, eliminating encoding heterogeneity across raw YouTube sources. The unified configuration dictionary combines acquisition parameters, formatting rules, output paths, and authentication credentials:

```
ydl_opts = {
    'format': 'bestvideo[height<=2160]+bestaudio/best',
    'download_ranges': download_range_func(
        None,
        [(start_sec, end_sec)]
    ),
    'force_keyframes_at_cuts': True,
    'merge_output_format': 'mp4',
    'postprocessor_args': {
        'ffmpeg': [
            '-c:v', 'libx264',
            '-preset', 'fast',
            '-c:a', 'aac'
        ]
    },
    'outtmpl': output,
    'cookiefile': 'src/youtube.com_cookies.txt',
}
```

Passing a session cookie file (`cookiefile`) enables retrieval of age-restricted or session-gated content without altering core extraction logic. Execution is wrapped in exception handling blocks, logging errors and returning non-zero exit codes upon failure:

```
try:
    with yt_dlp.YoutubeDL(ydl_opts) as ydl:
        ydl.download([url])
except Exception as e:
    print(f"\nError: {str(e)}")
    sys.exit(1)
```

Audio Conversion

In the MAIA-AV framework, the auditory component of each video is conceptualized as an autonomous source of information, rather than a mere ancillary element to the visual narrative. Spoken language, ambient sounds, and acoustic phenomena provide substantial evidence, enhancing visual data and facilitating the interpretation of scenes. By isolating the audio, this modality can be independently scrutinized, ensuring its contribution remains integral through subsequent processing stages. For each video, the original audio track is meticulously extracted and archived as a separate file. This process preserves the identical temporal parameters and identifier linked to the corresponding audiovisual entity. Such a one-to-one mapping maintains alignment across modalities, enabling visual, acoustic, and textual representations to be traced back to the same foundational event. The extracted audio files serve as primary inputs for further stages in the processing pipeline, including automated speech transcription and specialized audio analyses. The extraction process neither alters nor modifies the original content, rather, it separates the auditory channel, permitting independent processing while maintaining complete synchronization with the originating video content.

Implementation Details The audio extraction stage transforms each audiovisual dataset instance into a standalone acoustic representation, maintaining multimodal alignment with the source video. This process is implemented using Python's `subprocess` module, which calls FFmpeg to automate batch processing across the entire video collection. Input and output paths are clearly specified during the execution setup:

```
INPUT_FOLDER = Path("data/video")
OUTPUT_FOLDER = Path("data/audio")
VIDEO_EXTENSIONS = {".mp4"}
```

Decoupling the input and output directories serves to isolate raw video resources while also setting up an aligned audio mirror directory. By restricting `VIDEO_EXTENSIONS` to MP4, it ensures compatibility with the output format generated in the preceding acquisition stage. The main conversion process is encapsulated within `mp4_to_mp3`:

```
def mp4_to_mp3(
    input_path: Union[str, Path],
    output_path: Optional[Union[str, Path]] = None,
    bitrate: str = "192k",
) -> Path:
```

When an explicit destination path is omitted, the function retains the base filename and replaces the container extension:

```
    input_path = Path(input_path)
    output_path = (
        input_path.with_suffix(".mp3")
        if output_path is None
        else Path(output_path)
    )
```

This mapping strategy guarantees cross-modal alignment across representations (e.g., `video001.mp4` yields `video001.mp3`). Prior to invoking FFmpeg, parent directory existence is enforced:

```
    output_path.parent.mkdir(
        parents = True,
        exist_ok = True,
    )
```

This step allows nested input folder hierarchies to be automatically replicated within the output directory.

Extraction is executed using a single `subprocess.run` call:

```
    subprocess.run(
        [
            "ffmpeg",
            "-y",
            "-i",
```

```
            str(input_path),
            "-vn",
            "-c:a",
            "libmp3lame",
            "-b:a",
            bitrate,
            str(output_path),
        ],
        check=True,
    )
```

The `-vn` flag discards video streams, ensuring that FFmpeg isolates the acoustic signal rather than performing generic media transcode operations. Acoustic streams are normalized using the `libmp3lame` codec at a constant target bitrate of 192 kbit/s (`-b:a 192k`). Passing `-y` overwrites pre-existing destination files without prompt interruption, facilitating seamless pipeline re-execution. Setting `check=True` ensures runtime errors raise a `CalledProcessError` rather than failing silently. Upon completion, the function returns the confirmed target path. Batch processing begins by recursively indexing and sorting compatible media files:

```
def find_videos(folder: Path) -> list[Path]:
    """Return all supported video files found recursively."""
    return sorted(
        path
        for path in folder.rglob("*")
        if path.is_file()
        and path.suffix.lower() in VIDEO_EXTENSIONS
    )
```

Sorting file paths guarantees deterministic execution order across operating environments. Directory validity is verified before initializing conversion loops:

```
if not INPUT_FOLDER.exists():
    raise FileNotFoundError(
```

```python
        f"Input folder not found: {INPUT_FOLDER}"
    )
videos = find_videos(INPUT_FOLDER)
if not videos:
    print(f"No videos found in: {INPUT_FOLDER}")
    return
```

Validating the input directory early avoids redundant FFmpeg invocations when processing empty or invalid paths. For each video, relative path resolution preserves source subdirectory structures under `OUTPUT_FOLDER`:

```python
for index, video_path in enumerate(videos, start=1):
    relative_path = video_path.relative_to(INPUT_FOLDER)
    audio_path = (
        OUTPUT_FOLDER
        / relative_path.with_suffix(".mp3")
    )
```

Replicating source directory trees ensures structural parity between visual and acoustic datasets. Each file conversion is executed inside localized exception blocks:

```python
    try:
        mp4_to_mp3(
            video_path,
            audio_path,
        )
        print("Audio extraction completed.")
    except subprocess.CalledProcessError as error:
        print(f"FFmpeg error: {error}")
```

Capturing `CalledProcessError` exceptions locally prevents single-file execution failures from interrupting broader batch operations while logging specific processing errors.

Transcription Extraction

Another key representation in our dataset is the audio transcription, following the methodology established in our previous work [4]. Transcriptions were generated using the *Faster-Whisper* model (`large-v3`), a reimplementation of *OpenAI's Whisper* powered by the *CTranslate2* fast inference engine for Transformer architectures. The model processes raw audio content directly, with the language parameter set to Italian. Additionally, a *Voice Activity Detection* (VAD) filter is applied to isolate segments containing actual speech, thereby minimizing non-speech transcriptions. The overall transcription procedure aims to convert spoken dialogue into textual content that can be reliably recognized and processed by downstream models. This methodology represents an evolution compared to previous approaches on acoustic content analysis [4], which jointly employed *Whisper-1* and *GPT-4o-transcribe* to compare and align outputs for higher precision. In contrast, the current pipeline streamlines the transcription workflow by deploying a single, locally executed model, providing an efficient and reproducible framework.

Implementation Details The transcription stage was implemented as an independent component of the data preparation pipeline to associate each video with a textual representation of its spoken content. Powered by `Faster-Whisper` and the `large-v3` model executed locally on GPU, this setup ensures full reproducibility while eliminating dependencies on external cloud services. The execution environment specifically configures a chosen CUDA device and sets up a local cache for model files. By using a local cache, model weights are confined to the working directory, which helps prevent repetitive downloads across different system locations. The transcription model is subsequently instantiated on CUDA using `float16` precision:

```
model = WhisperModel(
    "large-v3", #MODEL_NAME
    device="cuda",
    device_index=0,
    compute_type="float16",
    download_root=str(HF_CACHE / "hub"), )
```

Prior to processing, the pipeline recursively retrieves and filters all `.mp4` files within the input directory, sorting the resulting collection to guarantee deterministic execution order:

```
def find_videos(folder: Path) -> list[Path]:
    return sorted(
        path
        for path in folder.rglob("*")
        if path.is_file()
        and path.suffix.lower() in VIDEO_EXTENSIONS
    )
```

The core transcription workflow relies on `model.transcribe()`. By directly parsing the audio embedded within the video container, Faster-Whisper eliminates intermediate extraction steps and yields chronologically ordered transcription segments:

```
segments, _ = model.transcribe(
    str(video_path),
    language="it",
    beam_size=5,
    vad_filter=True,
    condition_on_previous_text=False,
)
```

Setting `language="it"` explicitly conditions decoding on Italian. This parameter bypasses automatic language identification and directly aligns model execution with the linguistic profile of the dataset. Three key hyperparameters govern decoding precision and stability:

- `beam_size=5`: Activates beam-search decoding, evaluating multiple competing hypotheses during generation to balance accuracy and computational overhead.
- `vad_filter=True`: Enables Voice Activity Detection to strip environmental noise, music, and unvoiced pauses, preventing non-linguistic signals from reaching the ASR engine.

- `condition_on_previous_text=False`: Disables cross-segment context conditioning, isolating individual speech segments and mitigating error propagation across the audio stream.

The resulting segment iterator is aggregated into a continuous transcript by stripping whitespace, discarding empty segments, and concatenating valid text chronologically:

```
return " ".join(
    segment.text.strip()
    for segment in segments
    if segment.text.strip()
)
```

Omitting downstream linguistic post-processing or semantic filtering preserves an unadulterated representation of raw ASR performance. The corpus is processed sequentially, using each video filename as a unique primary key:

```
transcriptions = {}

for index, video_path in enumerate(videos, start=1):
try:
    transcription = transcribe_video(video_path)
    transcriptions[video_path.name] = transcription
except Exception as error:
    transcriptions[video_path.name] = (
        f"ERROR: {error}"
    )
```

Robust error handling isolates processing failures on individual files, preventing execution halts and cleanly decoupling unvoiced media from runtime exceptions.

Finally, extracted transcripts are exported to a structured CSV file indexed by `video_name` and `transcription`:

```
writer = csv.DictWriter(
```

```
        csv_file,
        fieldnames=[
        "video_name",
        "transcription"
        ],
        delimiter=";",
    )
    writer.writeheader()
    writer.writerows(
        {
            "video_name": video_name,
            "transcription": transcription,
        }
        for video_name, transcription
        in transcriptions.items()
    )
```

2.2 Expanding the Questions and Answers Dataset

The construction of linguistic data in MAIA-AV closely follows the methodological framework set by the original MAIA benchmark. The core process keeps the question as a fundamental element for collecting answers and creating an evaluation dataset. In the initial MAIA benchmark, each video had 2,400 questions. Expert annotators, working under specific category guidelines, generated these questions. They were crafted in an open-ended format to avoid simple yes or no answers. Since the original benchmark was designed for models based only on visual input, annotators were explicitly told to ignore all acoustic data during question formulation. MAIA-AV expands on the original framework by altering the informational foundation for question generation. This version includes a total of 300 questions, with each video offering three inquiries focusing on spatial, temporal, and causal aspects of the presented information. These categories align with the structure of the original MAIA benchmark. The spatial category centers on identifying locations or entities, whether visible or referenced, and detailing spatial relationships. The temporal category looks at event sequences, pinpointing when occurrences take place and establishing temporal relationships such as preceding, succeeding, or simultaneous actions. The causal category explores the reasons behind or consequences of actions, events, or behaviors observed in the video context. The main difference between the original MAIA and its follow-up, MAIA-AV, is the inclusion of audio information, which is essential in the latter version. Certain questions specifically target audio elements in the video, such as character dialogue, vocal commands to infotainment systems, and ambient sounds. This addition incorporates linguistic content that links information across both visual and auditory channels. The process for obtaining answers follows the MAIA guidelines. For the newly crafted set of 300 questions, data was gathered using a questionnaire given to four volunteers. These individuals are native Italian speakers, each born in Italy, with at least a high school diploma. They interacted with 100 videos and responded to the corresponding 300 questions, allowing for the collection of independent responses and four distinct versions of each answer. The structured approach remains unchanged: the videos serve as the multimodal information source, the questions specify what information needs extraction, and the human-generated answers provide the necessary linguistic reference for developing subsequent benchmark components.

Caption-Foil Generation

Building upon the human-annotated responses, the construction of *MAIA-AV* involves the automated generation of *caption–foil* pairs for the Visual Statement Verification (VSV) task. This process follows the methodology established in *MAIA*: each question-answer pair is first converted into a declarative true statement (*True Statement*, TS) and then modified to create a minimally contrastive false statement (*False Statement*, FS). Consequently, the caption and foil share maximum linguistic structure, differing mainly in question-relevant information. While the original benchmark collects eight human responses per question to yield eight TS–FS pairs, *MAIA-AV* adopts a scaled-down approach: four independent responses per question yield four captions and four corresponding foils. With three questions per video (spatial, temporal, and causal), each video comprises twelve caption–foil pairs. The automated generation uses `GPT-4o-2024-08-06`, matching the exact model version used in the original *MAIA* pipeline. Generation progresses in two distinct stages. In the first stage, the model receives only the question and a single human answer. Instead of creating novel content, the prompt directs the model to synthesize the question–answer pair into a single declarative statement, preserving the original semantic content. The instructions mandate minimal alterations to annotators' phrasing while requiring the conversion of first-person expressions into an impersonal stance. As a result, phrases like "I think that" or "I believe that" are removed from the caption without discarding any underlying uncertainty. Restricting outputs to concise declarative sentences leverages natural annotator variance while avoiding reliance on the model for factual core generation. This initial phase employs the prompt used for True Statement generation in *MAIA*:

```
Sei un assistente che crea affermazioni in italiano sui video.
Data una domanda Q e una risposta A relative a un video, devi creare
un'affermazione S basata su A.
Mentre generi S, cerca di non alterare le parole che compongono A.
Se A include verbi o frasi in prima persona (ad es., 'penso', 'credo'),
riformula S in modo impersonale, evitando la prima persona.
L'affermazione deve essere una frase breve e dichiarativa.[1]
```

The instructions include few-shot examples demonstrating question–answer fusion, with specific coverage of negative responses, non-deducible information, and expressions of uncertainty. Appended to the prompt, the target question and annotator response appear in `Q:` and `A:` format, conditioning the model to output solely the statement `S:`. The second stage processes the generated caption to yield the corresponding foil. Unlike the initial phase, this prompt adapts to the target question category. The core methodology

[1] English translation: *You are an assistant that creates statements in Italian about videos. Given a question Q and an answer A concerning a video, you must create a statement S based on A. While generating S, try not to alter the words composing A. If A includes first-person verbs or phrases (e.g., 'I think,' 'I believe'), rephrase S to be impersonal, avoiding a first-person perspective. The statement should be a concise, declarative sentence.*

relies on constructing *minimal pairs*: the model modifies only the information necessary to invalidate the caption, avoiding extraneous additions and preserving sentence structure. Consequently, distinguishing between caption and foil tests a model's ability to ground claims in audiovisual content rather than relying on surface linguistic cues. This design adheres to *MAIA*, directing GPT-4o to alter exclusively elements relevant to the specified semantic domain. Across all categories, the system prompt assigns a uniform role:

`Sei un assistente progettato per creare foil a partire dalle caption.`[2]

Subsequent instructions vary by targeted semantic dimension. For **spatial** questions, foil generation alters strictly the entity's position or location:

```
Data una caption in italiano (C) riguardante la posizione o la loca-
lizzazione di qualcuno o qualcosa, il tuo compito è creare il suo
foil (F) modificando solo l'informazione spaziale.
Non aggiungere altre informazioni rispetto a quanto dichiarato in
C.
```
[3]

An accompanying example relocates a woman standing in a poppy field to a school setting. Preserving the subject and surrounding text isolates spatial information as the sole determinant of truth value. For **temporal** questions, instructions dictate altering the temporal relationships within the caption:

```
Data una caption in italiano (C) riguardante gli eventi e quando av-
vengono, il tuo compito è creare il suo foil (F) modificando solo
l'informazione temporale.
Non aggiungere altre informazioni rispetto a quanto dichiarato in
C.
```
[4]

[2] English translation: *You are an assistant designed to create foils based on captions.*

[3] English translation: *Given an Italian caption (C) regarding the position or location of someone or something, your task is to create its foil (F) by changing only the spatial information. Don't add other information with respect to what is stated in C.*

[4] English translation: *Given an Italian caption (C) regarding events and when they happen, your task is to create its foil (F) by changing only the temporal information. Don't add other information with respect to what is stated in C.*

An illustrative example converts the chronological order of two events into an alternative sequence, preserving core semantic content while modifying event timing or order. For **causal** questions, foil construction alters the cause-and-effect relationship expressed in the caption:

```
Data una caption in italiano (C) riguardante le cause o gli effetti
degli eventi, il tuo compito è creare il suo foil (F) modificando
le cause o l'effetto dell'evento principale.
Non aggiungere altre informazioni rispetto a quanto dichiarato in
C.
```
[5]

The provided example retains the principal action, such as a person beginning to run, while altering solely the underlying cause. This focuses the contrast on the causal link rather than arbitrary scene modifications. In the final task setup, option ordering is randomized, preventing positional bias and ensuring performance metrics reflect semantic comprehension rather than structural pattern exploitation.

Implementation Details The *caption–foil* generation pipeline is implemented in `caption_foil.py`. The script relies on `pandas` for tabular data processing and the OpenAI Python client for model inference. Both caption and foil generation utilize `gpt-4o-2024-08-06`, with authentication managed via the `OPENAI_API_KEY` environment variable. Questions are loaded from `data/vsv/question_classification.csv`. Only the video identifier, principal dimension, and question text are required for this stage:

```python
questions = pd.read_csv(
    QUESTIONS_FILE,
    sep=";",
    encoding="utf-8-sig",
)
questions = questions[
```

[5] English translation: *Given an Italian caption (C) regarding the causes or the effects of events, your task is to create its foil (F) by changing the causes or the effect of the main event. Don't add other information with respect to what is stated in C.*

```python
    ["video_name", "principal_dimension", "question_text"]
].copy()
```

Prior to processing, the script verifies column availability, discards missing values, normalizes category labels, and confirms the presence of only the three expected macroareas. Video identifiers are standardized using `normalize_video_name()`, which extracts the numeric identifier and converts it to the format used across the dataset:

```python
def normalize_video_name(value):
    value = str(value).strip()
    match = re.search(r"(\d+)", value)
    if not match:
        raise ValueError(
            f"Impossibile ricavare il numero del video da: {value}"
        )

    return f"video{int(match.group(1)):03d}.mp4"
```

Additionally, the script ensures that each video is associated with exactly one spatial, one temporal, and one causal question. Duplicate video–macroarea pairs are rejected prior to generation. Human responses are automatically loaded from all CSV files within `data/vsv/raw_human_answers`. The file count is validated against the expected number of annotators:

```python
files = sorted(RAW_ANSWERS_DIR.glob("*.csv"))
if len(files) != EXPECTED_ANNOTATORS:
    raise ValueError(
        f"Attesi {EXPECTED_ANNOTATORS} file di annotatori in "
        f"{RAW_ANSWERS_DIR}, trovati {len(files)}"
    )
```

Each file must contain the columns `titolo_video`, `macro-area`, and `risposta`. For backward compatibility, the alternative column name `risposte` is also accepted and automatically renamed. Annotator identifiers are extracted directly from the filenames:

```
if "risposta" not in df.columns and "risposte" in df.columns:
    df = df.rename(columns={"risposte": "risposta"})
df = df[["titolo_video", "macroarea", "risposta"]].copy()
df["video_name"] = df["titolo_video"].map(
    normalize_video_name
)
df["macroarea"] = (
    df["macroarea"]
    .astype(str)
    .str.lower()
    .str.strip()
)
df["annotator"] = file.stem.removeprefix("risposte_")
```

Questions and responses are subsequently aligned via the shared `video_name` and `macroarea` fields. The merge is constrained as *many-to-one*, matching multiple annotator responses to a single question:

```
data = answers.merge(
    questions,
    on=["video_name", "macroarea"],
    how="left",
    validate="many_to_one",
)
```

Additional consistency checks validate question keys against annotator responses. Missing answers or orphan responses interrupt execution prior to invoking any API requests. Responses corresponding to the same video and macroarea are grouped into a common pool. The pool identifier is formed by concatenating the video identifier with the corresponding macroarea:

```
data["video_id"] = (
    data["video_name"].str.removesuffix(".mp4")
```

```
)
data["pool_id"] = (
    data["video_id"] + "_" + data["macroarea"]
)
```

Each pool must contain exactly four responses. Individual entries are assigned a sequential position and a unique identifier:

```
data["pool_item"] = (
    data.groupby("pool_id").cumcount() + 1
)
data["id"] = (
    data["pool_id"]
    + "_"
    + data["pool_item"].astype(str).str.zfill(2)
)
```

API calls are centralized within `call_openai()`, which serves both caption and foil generation functions:

```
response = client.chat.completions.create(
    model=MODEL,
    messages=messages,
    max_tokens=max_tokens,
)
return response.choices[0].message.content.strip()
```

The function executes up to five attempts for recoverable API failures. Rate-limit errors trigger a 30-second sleep interval, whereas generic API errors pause execution for 10 seconds. Authentication errors terminate execution immediately:

```
except RateLimitError:
    if attempt == retries - 1:
        raise
    time.sleep(30)
```

```python
except AuthenticationError:
    raise RuntimeError(
        "Errore di autenticazione: controlla OPENAI_API_KEY."
    )
except APIError:
    if attempt == retries - 1:
        raise
    time.sleep(10)
```

For each annotated response, caption and foil generation proceed sequentially. The former accepts the question and human answer as input, whereas the latter accepts the generated caption along with its associated macroarea:

```python
caption = generate_caption(
    row["question_text"],
    row["risposta"],
)
foil = generate_foil(
    caption,
    row["macroarea"],
)
```

Outputs are post-processed via `clean_generation()`, which strips optional `S:` or `F:` prefixes returned by the model:

```python
def clean_generation(text, prefix):
    text = text.strip()
    if text.startswith(f"{prefix}: "):
        text = text[3:]
    elif text.startswith(f"{prefix}:"):
        text = text[2:]

    return text.strip()
```

To accommodate long-running executions, the script implements a checkpointing mechanism. Cached entries are reused only if their question, annotator, and human answer match the current source data and both generated fields are non-null:

```
checkpoint_is_valid = (
    saved is not None
    and str(saved["question"]) == str(row["question_text"])
    and str(saved["annotator"]) == str(row["annotator"])
    and str(saved["human_answer"]) == str(row["risposta"])
    and pd.notna(saved["caption"])
    and pd.notna(saved["foil"])
)
```

Newly generated entries are appended incrementally to `caption_foil_checkpoint.csv`:

```
pd.DataFrame(generated_rows).to_csv(
    CHECKPOINT_FILE,
    index=False,
    encoding="utf-8-sig",
)
```

Following generation, caption and foil positions are randomized independently within each question category. A fixed random seed guarantees reproducibility while maintaining a balanced distribution of options across positions:

```
rng = random.Random(RANDOM_SEED)
for category, indices in (
    df.groupby("question_category").groups.items()
):
    indices = list(indices)
    n = len(indices)
    labels = [0] * (n // 2) + [1] * (n // 2)

    if n % 2:
        labels.append(rng.randint(0, 1))
```

```python

    rng.shuffle(labels)
```

The resulting `target` variable indicates the position of the correct caption: a value of 0 places the caption in `answer1`, whereas a value of 1 assigns it to `answer2`:

```python
df["answer1"] = df.apply(
    lambda row:
    row["caption"]
    if row["target"] == 0
    else row["foil"],
    axis=1,
)
df["answer2"] = df.apply(
    lambda row:
    row["foil"]
    if row["target"] == 0
    else row["caption"],
    axis=1,
)
```

The final dataset is saved to `data/vsv/caption-foil/caption_foil.csv`. Alongside the randomized options and target labels, the output retains all metadata required to trace each pair back to its originating video, question, annotator, and human response:

```python
columns = [
    "id",
    "video_id",
    "video_name",
    "pool_id",
    "pool_item",
    "question_category",
    "question",
```

```
    "annotator",
    "human_answer",
    "caption",
    "foil",
    "answer1",
    "answer2",
    "target",
]
final_df[columns].to_csv(
    OUTPUT_FILE,
    index=False,
    encoding="utf-8-sig",
)
```

Caption–Foil Quality Validation

The automatic generation of a foil does not, by itself, guarantee that the resulting sentence constitutes a valid negative counterpart of the corresponding caption. Although generation is constrained through category-specific prompts, the model may introduce changes that are semantically unrelated to the targeted reasoning phenomenon, alter more information than necessary, or produce a sentence that remains compatible with the original caption. For this reason, the generated *caption–foil* pairs undergo an additional semi-automatic validation stage before being included in the final VSV evaluation set. The validation procedure follows the two-step strategy adopted in *MAIA*, adapted here to the three reasoning dimensions considered in *MAIA-AV*. The first stage is a *structural check*, whose purpose is not simply to determine whether the caption and foil differ, but to verify that their difference concerns the semantic dimension the corresponding question is meant to evaluate. This distinction is fundamental for preserving the diagnostic nature of the benchmark. If, for instance, a temporal foil modified the identity of an entity rather than the ordering of two events, a model could distinguish the two statements through entity recognition alone, without any temporal reasoning. Similarly, a causal pair would stop providing a controlled test of causal understanding if the foil simultaneously changed the cause, the participants, and the location of the event. Structural verification is what keeps the contrast introduced during foil generation aligned with the competency each question is meant to probe. For **causal** instances, validation requires the foil to modify the cause, the effect, or the causal relation expressed by the caption, while preserving the remaining content as much as possible. For **temporal** instances, the contrast must concern temporal information, including event ordering, before–after relations, or the moment at which an event occurs. The **spatial** category calls for a slightly broader criterion. Unlike the original *MAIA* taxonomy, where spatial reasoning is split into Total and Partial subcategories, the spatial questions in *MAIA-AV* combine phenomena from both settings. The validation prompt therefore accepts changes involving locations, positions, directions, and relative spatial relations, including relations that hold only during a particular phase of the video. This broader formulation avoids forcing the new dataset into a distinction its annotation scheme does not retain, while keeping the core requirement that the foil differ from the caption specifically in its spatial content. Pairs that satisfy this first criterion move on to a second validation step based on *Natural Language Inference* (NLI), where the caption is treated as the premise and the foil as the hypothesis. The model assigns one of three standard NLI labels: *Entailment*, *Contradiction*, or *Neutral*. The expected relation is *Contradiction*, since the foil is designed to offer an alternative statement that should not hold under the same interpretation as the caption. This step addresses a different source of error than the structural check. A foil may correctly manipulate the intended semantic dimension while still failing to create a sufficiently distinct alternative. Conversely, a sentence may differ substantially from the caption yet fail the structural test because the modification does not isolate the phenomenon under evaluation. The NLI result is nevertheless treated as a diagnostic signal rather than an infallible automatic decision, particularly for causal and, to a lesser extent, spatial statements. Two alternative causes, for example, may be logically compatible in isolation even though only one is intended to describe the event represented in the dataset. In such cases a general-purpose NLI model may classify the relation as *Neutral* rather than *Contradiction*. A similar pattern was observed during the original *MAIA* validation, where qualitative inspection showed that many pairs labelled as neutral were nevertheless valid caption–foil contrasts. For this reason, only pairs that pass the structural check and receive the *Contradiction* label are accepted automatically. Structurally valid pairs classified as *Neutral* are marked for manual review, while *Entailment* cases are given higher review priority, since they provide stronger evidence that the foil may remain semantically compatible with the caption. Category-sensitive structural verification and category-independent semantic inference thus provide two complementary safeguards. The former preserves the correspondence between each pair and the reasoning ability it is meant to test. The latter checks whether the resulting negative statement establishes an adequate semantic opposition with the corresponding positive statement. Importantly, neither stage is used to silently discard uncertain data. Every instance flagged as structurally invalid, entailed, neutral, or affected by execution errors is isolated in a dedicated review set, manually inspected, and, whenever the validation reveals an actual problem, corrected so that the intended semantic contrast is restored. The corrected pair is then run back through the same validation procedure. Human verification at this final stage keeps automatic classifier errors from propagating into the benchmark, while the pipeline as a whole remains predominantly automated.

Automatic Validation and Manual Revision

The automatic validation procedure was used as a quality-control step to identify caption–foil pairs that required further inspection. The structural check evaluated whether the foil modified the semantic dimension associated with the corresponding category, while the NLI stage assessed the relation between caption and foil in terms of *Entailment*, *Contradiction*, or *Neutral*. The output of these automatic checks was not treated as a definitive decision on the validity of each pair. In particular, some instances labelled as *FAIL_STRUCTURAL*, *Neutral*, or *Entailment* could still represent acceptable contrasts when considered in relation to the original question, the human answer, and the intended semantic category. This is especially relevant for cases in which the contrast is expressed through the answer-bearing element itself, such as a change of location in spatial questions or a change in the participant occupying a specific temporal position. Similarly, in causal questions, two alternative explanations may be classified as *Neutral* by an NLI model even when only one of them is supported by the video. For this reason, the complete caption–foil file was subsequently revised manually. Each automatically flagged instance was inspected together with its question, human answer, semantic category, caption, and foil. The automatic labels were used only to prioritize potentially problematic cases, not to determine whether a pair had to be discarded or modified. When a caption–foil pair was considered semantically valid and sufficiently consistent with the intended contrast, it was retained without changes. Whenever the manual inspection revealed an actual issue, such as a foil targeting the wrong semantic dimension, an ambiguous or insufficiently discriminative contrast, a formulation that did not adequately reflect the original human answer, or a mismatch between the question and the generated statements, the pair was manually corrected instead. The final dataset thus reflects a combination of automatic quality-control signals and human verification. Modifications were introduced only when they were considered necessary after direct inspection of the corresponding instance.

Implementation Details The validation procedure is implemented as a separate post-processing stage operating on the output of the caption–foil generation pipeline. Its input is the complete dataset stored in `data/vsv/caption-foil/caption_foil.csv`. Validation is performed on the canonical `caption` and `foil` fields rather than on the randomized `answer1` and `answer2` columns used by the final VSV task, since option randomization affects only presentation order and must not interfere with the semantic relation being assessed. The relevant paths and expected semantic categories are defined explicitly:

```
INPUT_PATH = Path(
    "data/vsv/caption-foil/caption_foil.csv"
)
OUTPUT_DIR = Path(
    "data/vsv/caption-foil/validation"
)

STRUCTURAL_OUTPUT = (
    OUTPUT_DIR /
    "caption_foil_structural_validation.csv"
```

```
)

FINAL_OUTPUT = (
    OUTPUT_DIR /
    "caption_foil_validation.csv"
)

VALID_CATEGORIES = {
    "causale",
    "temporale",
    "spaziale",
}
```

Before any API request is issued, the script verifies the availability of the fields required for validation and checks that all category labels belong to the three supported macroareas:

```
REQUIRED_COLUMNS = {
    "id",
    "question_category",
    "caption",
    "foil",
}

missing_columns = (
    REQUIRED_COLUMNS - set(df.columns)
)

if missing_columns:
    raise ValueError(
        "Mancano le seguenti colonne nel CSV: "
        + ", ".join(sorted(missing_columns))
    )
```

The first validation stage selects a category-specific prompt according to the value stored in `question_category`. All three prompts share the same decision format but define validity according to different semantic constraints. Causal pairs, for instance, explicitly require the modification to concern causal information:

```
if category == "causale":
    return (
        "Given an Italian caption (C) dealing "
        "with causal relations between events "
        "and its foil (F), your task is to assess "
        "whether F is a valid causal foil of C.\n\n"

        "To be valid, F must modify the cause, "
        "the effect, or the causal relation "
        "expressed in C while preserving the "
        "remaining relevant content as much "
        "as possible.\n"

        "If F is a valid causal foil, output "
        "'correct'. Otherwise output "
        "'not correct'.\n\n"

        f"C: {caption}\n"
        f"F: {foil}\n"
        "Output:"
    )
```

The temporal prompt follows the same structure but constrains the acceptable difference to event ordering or other temporal properties:

```
if category == "temporale":
    return (
        "To be valid, F must modify the temporal "
        "information expressed in C, such as "
```

```
        "the ordering of events, before/after "
        "relations, the moment at which an event "
        "occurs, or another temporally relevant "
        "relation.\n"
    )
```

For spatial instances, the prompt is deliberately formulated to cover both stable locations and time-dependent spatial configurations:

```
if category == "spaziale":
    return (
        "The spatial category considered here "
        "includes both locations and spatial "
        "relations, including cases in which "
        "spatial information refers to a specific "
        "moment or phase of the video.\n\n"

        "To be valid, F must modify spatial "
        "information expressed in C, such as "
        "the location, position, direction, or "
        "spatial relation of a person, object, "
        "or event.\n"
    )
```

The model is constrained to produce only one of two labels, `correct` or `not correct`. Outputs are normalized before being stored, to remove minor formatting variations:

```
def normalize_structural_label(text):
    value = clean_text(text).lower()

    if value.startswith("output:"):
        value = value[len("output:"):].strip()

    value = value.rstrip(".").strip()
```

```

    if value == "not correct":
        return "not correct"

    if value == "correct":
        return "correct"

    return "invalid"
```

The structural result is saved for every caption–foil pair in the `structural_eval` field. Incremental serialization is performed after each processed instance, so that completed evaluations survive an interruption:

```
df.at[index, "structural_eval"] = result

df.to_csv(
    STRUCTURAL_OUTPUT,
    index=False,
)
```

The second stage applies only to pairs that pass structural validation. Caption and foil are mapped respectively to the NLI premise `S1` and hypothesis `S2`:

```
prompt = (
    "Your task is to determine the natural "
    "language inference (NLI) relationship "
    "between S1 and S2.\n\n"

    "The possible labels are:\n"
    "- Entailment: S2 logically follows "
    "from S1.\n"
    "- Contradiction: S2 contradicts S1.\n"
    "- Neutral: S1 and S2 are related, "
    "but S2 neither follows from nor "
```

```
    "contradicts S1.\n\n"

    f"S1: {caption}\n"
    f"S2: {foil}\n"
    "Output:"
)
```

The resulting output is normalized to one of the three expected NLI labels:

```
if value == "contradiction":
    return "Contradiction"

if value == "neutral":
    return "Neutral"

if value == "entailment":
    return "Entailment"
```

The two validation signals are then combined into an explicit status variable, with automatic acceptance limited to pairs that are structurally valid and classified as contradictions:

```
def get_validation_status(
    structural_eval,
    nli_eval,
):
    if structural_eval != "correct":
        return "FAIL_STRUCTURAL"

    if nli_eval == "Contradiction":
        return "PASS"

    if nli_eval == "Neutral":
        return "REVIEW"

```

```python
    if nli_eval == "Entailment":
        return "REVIEW_HIGH_PRIORITY"

    return "ERROR"
```

Pairs that fail the structural check do not proceed automatically to semantic acceptance and are marked as FAIL_STRUCTURAL. Structurally valid pairs classified as Neutral are assigned REVIEW, while Entailment produces REVIEW_HIGH_PRIORITY. This hierarchy keeps uncertain model judgements from being treated as definitive dataset errors. At the end of the procedure, all non-passing cases are extracted into a dedicated manual-review file:

```python
review_mask = df[
    "validation_status"
].isin(
    [
        "REVIEW",
        "REVIEW_HIGH_PRIORITY",
        "FAIL_STRUCTURAL",
        "ERROR",
    ]
)

review_df = df.loc[
    review_mask
].copy()

review_df.to_csv(
    OUTPUT_DIR /
    "caption_foil_manual_review.csv",
    index=False,
)
```

This file serves as the interface between automatic and human validation. Each flagged instance is manually inspected by comparing its caption, foil, question, and semantic category. Cases where the automatic validator is overly conservative but the contrast is in fact appropriate are retained unchanged. When inspection instead confirms that the foil modifies the wrong semantic property, remains compatible with the caption, introduces unrelated information, or otherwise fails to provide the intended minimal contrast, the foil is corrected by hand. Corrected instances are then re-evaluated through the same structural and NLI checks before being incorporated into the final dataset. The complete validation output retains the original metadata alongside `structural_eval`, `nli_eval`, and `validation_status`. This preserves full traceability from each final VSV item back to its generated caption, foil, question, annotator response, automatic validation decisions, and any manual correction.

3. Study Design and Experimental Setup

3.1 Overview of the Evaluated Models

Qwen Family

The Qwen models considered in this work share an omnimodal architecture designed to process heterogeneous information within a unified computational framework. Both Qwen2.5-Omni and Qwen3-Omni accept textual, visual, and acoustic inputs, with modality-specific encoders transforming raw signals into representations compatible with a shared language-model backbone. Multimodal integration takes place after perceptual encoding, and particular attention is paid throughout the family to the temporal coordination of audio and video [24, 25].

Qwen2.5-Omni-3B

Qwen2.5-Omni introduces the *Thinker-Talker* architecture, separating multimodal understanding from speech generation. The *Thinker* processes the input modalities and produces textual representations and responses, while the *Talker* takes the Thinker's internal representations and generates speech from them. Perceptual understanding and spoken generation thus remain functionally distinct components of an otherwise end-to-end pipeline [24]. Text is processed through the Qwen tokenizer. Audio is resampled to 16 kHz and converted into a 128-channel mel-spectrogram, with the encoder producing representations at a temporal resolution of roughly 40 ms. Visual information is handled by a Vision Transformer derived from Qwen2.5-VL and shared across image and video inputs; frames are sampled dynamically to preserve relevant visual content while matching the temporal granularity of the acoustic stream [24]. The modality-specific encoders are limited to perceptual extraction: their output is passed to the *Thinker*, where multimodal information is processed within the shared Transformer. Both audio and visual encoders additionally support block-wise processing, allowing long multimodal sequences to be handled incrementally [24]. Multimodal alignment relies on *Time-aligned Multimodal Rotary Position Embedding* (TMRoPE), which represents positional information along temporal, vertical, and horizontal dimensions. Text is assigned identical positional values across all three; images retain a constant temporal coordinate while preserving spatial position along height and width; audio receives temporal positions tracking its progression through the acoustic stream [24]. Video combines both axes: each frame keeps its two-dimensional spatial layout while carrying a temporal coordinate tied to its position in the clip, on a scale coordinated with the acoustic representation. Audio and video can thus be placed within a common temporal reference system, which is what allows the *Thinker* to relate acoustic and visual evidence occurring at corresponding moments. Encoded representations from both streams are arranged within a single multimodal sequence and processed jointly by the language model's attention mechanism, so integration occurs before response generation rather than through a late combination of independent modality-specific predictions [24]. The *Talker* is a dual-track autoregressive Transformer decoder, conditioned on the high-dimensional representations it receives from the *Thinker*; the resulting discrete speech units are converted into a waveform through a sliding-window diffusion Transformer [24]. The technical report describes this organization at the level of the Qwen2.5-Omni family as a whole, without a fusion mechanism specific to the 3B variant [24].

Qwen3-Omni-30B-A3B-Instruct

Qwen3-Omni keeps the Thinker-Talker organization of its predecessor while redesigning several components. It handles text, images, audio, and video, and supports both textual and spoken generation; in the 30B-A3B configuration, both the *Thinker* and the *Talker* adopt a Mixture-of-Experts architecture, increasing overall capacity while limiting the parameters activated at each inference step [25]. The *Thinker* still carries out multimodal understanding and text generation by combining representations from dedicated perceptual encoders. The *Talker*, however, is more loosely coupled to it than in Qwen2.5-Omni: rather than depending directly on the Thinker's high-level textual hidden states, it conditions speech generation on textual context together with high-dimensional visual and acoustic representations [25]. Audio is processed by the *Audio Transformer* (AuT). Raw audio is resampled to 16 kHz and represented as a 128-channel mel-spectrogram; convolutional downsampling brings the resulting representations to roughly 12.5 Hz, or about 80 ms per acoustic frame. AuT uses attention windows of different sizes to support both global audio understanding and block-wise processing [25]. Visual input is processed by an encoder derived from Qwen3-VL and initialized from SigLIP2-So400m, shared across images and video; as in Qwen2.5-Omni, frames are sampled dynamically to preserve visual content while matching the temporal granularity of the audio stream [25]. TMRoPE is retained for multimodal alignment, again along temporal, vertical, and horizontal dimensions: text keeps identical coordinates across all three, images hold a constant temporal coordinate while preserving spatial position, and audio now receives absolute temporal identifiers at roughly 80 ms resolution [25]. Video frames are assigned temporal identifiers taken from their actual timestamps, on the same scale used for audio, so acoustic and visual representations share an absolute temporal coordinate system while spatial information is still carried separately through height and width [25]. This replaces the fixed temporal chunking of Qwen2.5-Omni with alignment based on absolute temporal identifiers, producing a finer correspondence between acoustic and visual events and removing the dependence on a predefined chunk size [25]. The aligned representations are then processed by the Mixture-of-Experts *Thinker*, so multimodal integration occurs within the shared Transformer computation after perceptual encoding and temporal alignment; the pre-training strategy reinforces this from the outset, drawing on unimodal and cross-modal data spanning image-text, audio-text, video-text, video-audio, and video-audio-text combinations [25]. Speech generation is likewise redesigned. The *Talker* predicts a multi-codebook acoustic representation based on residual vector quantization: the first codebook is generated autoregressively, while a Multi-Token Prediction module estimates the remaining residual codebooks. The final waveform is produced by *Code2Wav*, a causal convolutional renderer built for incremental synthesis [25].

Gemma Family

The Gemma models considered in this work implement multimodality through two distinct strategies. Gemma 3n E4B relies on dedicated perceptual encoders that convert acoustic and visual signals into representations processed by the language-model backbone. Gemma 4 12B instead adopts a unified, encoder-free architecture, mapping raw perceptual information directly into the decoder's embedding space through lightweight projection modules [18, 6].

Gemma 3n E4B IT

Gemma 3n accepts text, images, audio, and video and generates textual responses; separate perceptual encoders convert acoustic and visual signals into token representations for the central language model [18]. The E4B configuration is built on *MatFormer* (Matryoshka Transformer), which nests a smaller E2B computational path within the larger E4B structure; both are trained jointly, allowing the model to operate under different computational budgets without requiring separate weights [18]. Gemma 3n also introduces *Per-Layer Embeddings*, storing a substantial portion of the embedding parameters outside accelerator memory. As a result, E4B keeps only around four billion core parameters resident on the accelerator despite a larger overall parameter count ,a gain in computational efficiency rather than a change to how modalities are integrated [18]. Audio is processed by an encoder based on the Universal Speech Model, producing roughly one acoustic token every 160 ms (about six tokens per second) that feed directly into the language model [18]. Visual input is handled by MobileNet-V5-300M, supporting resolutions of 256×256, 512×512, and 768×768 pixels across both image and video content [18]. A *Multi-Scale Fusion VLM adapter* processes the resulting visual representations before they reach the language model; its role is limited to combining features at different scales and does not amount to a general audiovisual fusion mechanism [18]. Multimodal integration remains centered on the language-model backbone: audio and visual encoders produce modality-specific representations that become available to the Transformer during generation, and the available documentation does not describe a shared temporal alignment mechanism between acoustic and visual signals comparable to TMRoPE in the Qwen models [18]. Gemma 3n further employs *KV Cache Sharing*, letting keys and values from intermediate attention layers be reused by subsequent ones, which reduces redundant computation during the prefill stage, a benefit that is particularly relevant for the long sequences produced by audio and video input [18].

Gemma 4 12B IT

Gemma 4 is a natively multimodal family built on a decoder-only Transformer, and the 12B configuration departs from the rest of the family architecturally. Rather than relying on perceptual encoders, it is trained from scratch under a unified encoder-free paradigm, with visual and acoustic input mapped directly into the language model's embedding space through lightweight projection modules [6]. This sets it apart from the other Gemma 4 configurations: the smaller E2B and E4B models retain distinct vision and audio encoders, and even the larger variants keep dedicated visual processing. Gemma 4 12B alone drops both the vision and the audio encoder, replacing them with a direct projection of raw perceptual information [6]. For visual input, the model operates on RGB patches of $48 \times 48 \times 3$ values. A large linear projection of roughly 35 million parameters takes the place of the Vision Transformer used elsewhere in the family, mapping the resulting representations directly into the language model's embedding space [6]. Spatial information is preserved by adding two-dimensional coordinate-based positional embeddings to the projected patches, followed by a final LayerNorm before the visual representations enter the backbone, enough to keep spatial structure available without a separate visual encoder [6]. The audio pathway follows the same principle. Raw audio, sampled at 16 kHz, is divided into 40 ms segments of 640 values each; the USM-based Conformer encoder used by the other Gemma 4 variants is removed entirely, and each 640-dimensional segment is projected directly into the language model's embedding space [6]. No additional positional encoding is introduced for these acoustic representations, since their sequential order already preserves the temporal structure of the audio stream. Visual and acoustic information thus enter the same embedding space through modality-specific linear projections rather than through independent high-capacity encoders [6]. Multimodal integration accordingly takes place directly inside the decoder-only Transformer: the projection modules establish representational compatibility between raw perceptual input and the language model's embedding space, and the Transformer itself carries out the subsequent interaction between perceptual and linguistic information, narrowing the separation between perceptual encoding and language processing found in more conventional multimodal systems [6]. The 12B model is dense and combines local and global self-attention, with five local attention blocks per global block. Local layers use RoPE, while global layers use p-RoPE with $p = 0.25$; the architecture also reuses keys as values in global attention layers, reducing the memory footprint of the global KV

3.2 Preliminary Analysis as a Diagnostic Tool for Multimodal Assessment

The preliminary analysis aims to determine which information relevant to the subsequent resolution of *MAIA-AV* questions can be autonomously recovered by each model from the visual content of the video alone. The analysis is accordingly conducted in a *question-independent* setting: the model receives solely the video input, with no access to the questions, corresponding answers, audio streams, or transcriptions. The output is a structured and temporally grounded representation of the visual content, which is only later compared with the informational requirements of individual questions. This separation distinguishes the model's capacity to extract visual information from its ability to retrieve relevant evidence when guided by a linguistic query. While the preliminary analysis does not directly measure performance on the VSV task, its purpose is to characterize what the model can represent prior to the introduction of a specific question-answering objective. Isolating these two abilities addresses a key limitation of evaluations based solely on final accuracy. A correct prediction does not guarantee that the model has correctly identified relevant entities, reconstructed events, or established the relations necessary to support its output. Conversely, an incorrect answer fails to reveal which stage of the process caused the failure. The preliminary analysis thus introduces an intermediate diagnostic layer between the raw video content and the final task outcome. Methodologically, this approach relates to *VILMA* [10], specifically regarding its distinction between proficiency tests and main tests. In *VILMA*, proficiency tests evaluate fundamental capabilities deemed necessary for resolving complex temporal phenomena via a zero-shot caption–foil paradigm. While our method does not replicate this specific evaluation protocol, it adopts the same underlying principle: complex performance must be interpreted relative to the prerequisite capabilities that underpin it. In *MAIA-AV*, these prerequisites are examined through a structured representation of the visual content. For each video, four multimodal and omnimodal models, subsequently evaluated on the VSV task, independently analyze temporally localized video segments. Outputs are initially kept separate to enable cross-representation comparison without prematurely merging predictions. Inter-model agreement is interpreted as an indicator of **prediction stability** rather than absolute ground truth, since multiple models may converge on an identical yet unsupported interpretation. The representation is constructed progressively across three levels: **entities**, **events**, and **relations between events**. This hierarchy separates directly observable perceptual information from information that requires temporal integration or higher-level inference. The **entity level** describes elements identified within the visual content and their observable properties, detailing the objects, individuals, and other scene components detected by the model together with their corresponding temporal intervals of presence. The **event level** captures actions, interactions, and state changes tied to specific temporal intervals. Because overlapping video segments may yield duplicate descriptions of the same event, extracted outputs are consolidated to eliminate redundancy while preserving genuinely recurrent events. Temporal anchoring maintains a direct link between each event and the video segment providing its supporting evidence. The **relation level** organizes extracted events according to temporal and inferential dependencies. Temporal relations encompass configurations such as *before*, *after*, *during*, and *overlaps*, supporting event reordering and the tracking of changes across the video timeline. Causal relations are treated independently of temporal succession: simple temporal precedence does not by itself establish a cause-and-effect relationship. The analysis accordingly differentiates causal information directly supported by visual evidence from relations that require further inference over events, states, or intentions, keeping observable content separate from information introduced by the model's own inferential process. The distinction between representational levels matters because questions assigned to the same nominal category can demand substantially different types and volumes of evidence. Identifying an object's location within a single frame, for instance, requires fundamentally different operations than determining its position following a sequence of movements. Similarly, recognizing an isolated event differs from reconstructing a sequence of multiple events or inferring causal links between them. To capture these distinctions systematically, questions are analyzed along three operational dimensions: **spatial**, **temporal**, and **causal**. A unified three-level scale is applied across each dimension. Level 0 denotes cases where the required information can be directly retrieved. Level 1 involves integrating an explicit relation or a limited number of evidence units. Level 2 demands broader reconstruction across multiple events, temporal phases, state changes, or implicit relations. These criteria are summarized in Table 3.1. This classification reflects the informational requirements of the questions rather than their linguistic complexity, accounting for properties such as the temporal distribution of evidence, the number of relevant evidence units, the presence of inter-event relations, and whether the required information is explicit or implicit. Once constructed independently, the visual representation can be compared against the question requirements to determine whether the entities, events, and relations necessary to answer a question are already captured within the vision-only representation. The evaluation therefore does not depend directly on the model's capacity to generate the correct answer. Instead, it assesses whether the supporting information is present within its visual representation. This comparison is particularly relevant for *MAIA-AV*, since it establishes a visual baseline prior to introducing additional modalities. Prior findings [4] showed that, in the original *MAIA* setting, incorporating acoustic information yielded no systematic performance gains, while including transcriptions led to a slight yet statistically robust decline in accuracy, consistent with a dataset originally designed around visual content rather than information explicitly conveyed via audio. The preliminary analysis thus establishes *what can be seen* prior to evaluating the contribution of *what can be heard*. It clarifies whether the visual channel inherently contains the required evidence and highlights categories where visual information alone proves insufficient, so that the incremental contributions of audio and transcriptions can be assessed against an explicit visual baseline rather than solely through shifts in downstream task accuracy.

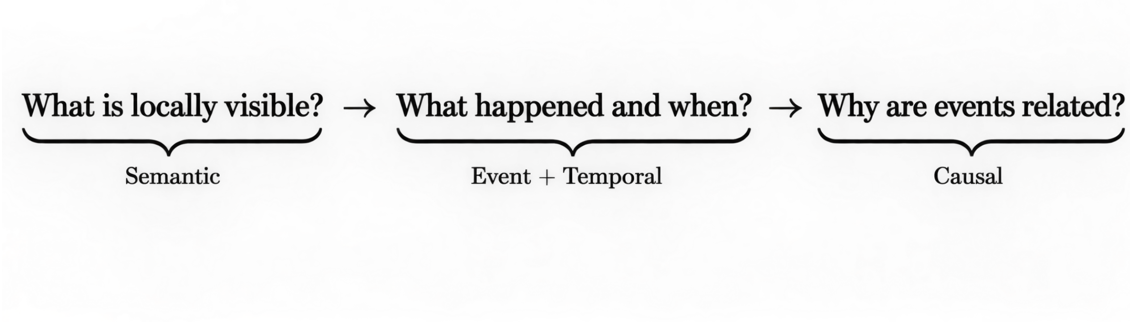

Figure 3.1: Conceptual schema of the preliminary analysis.

Overall, the preliminary analysis transforms *MAIA-AV* from a purely result-driven evaluation into a diagnostic, competence-oriented framework. Final performance can thus be analyzed across specific sub-capabilities, including entity recognition, event identification, temporal integration, spatial reconstruction, and causal inference, disentangling evidence availability failures from errors in downstream selection or integration during the VSV task.

Category	Level 0	Level 1	Level 2
Spatial (S)	Direct retrieval of an entity's location or spatial property from a single frame.	Explicit spatial relation or event-conditioned localization involving at most two evidence units.	Spatial reconstruction across multiple phases, including movement, relocation, or state change.
Temporal (T)	Recognition of a single event or state without temporal comparison.	Temporal localization or pairwise ordering between two explicit events.	Multi-event temporal reconstruction, including duration, repetition, onset, or state evolution.
Causal (C)	Retrieval of an explicitly stated cause without inference.	Explicit cause-and-effect relation involving at most two evidence units.	Implicit or multi-event causal reconstruction requiring contextual or intentional inference.

Table 3.1: Complexity levels for spatial, temporal, and causal questions.

Video Preprocessing: Temporal Segmentation and Frame Sampling

Prior to the entity extraction phase, each video is preprocessed by dividing it into four-second temporal windows with a stride of three seconds. Consecutive spans share one second of visual content. This overlap reduces the risk that an action or state change occurring near the boundary between two segments is only partially observed, while maintaining shared context across consecutive observations without altering the chronological order of the video. For each segment, the `start_time` and `end_time` are preserved, keeping each segment associated with its original temporal position in the video. This information is maintained throughout all subsequent stages of the pipeline, allowing observations to be reconnected with their exact position in the video. The windowing process serves exclusively to provide temporal indexing and structure visual evidence, leaving content and semantic interpretation to subsequent phases. Redundant information is deliberately retained at this stage to prioritize continuity of evidence, with redundancy resolution deferred to the output consolidation step. The preprocessing phase ultimately yields an ordered sequence of temporally localized visual units that serve as the shared foundation for all subsequent stages of analysis.

Implementation Details The preprocessing stage is implemented in `video_preprocessing.py` to convert each source video into a set of temporally indexed visual representations. The script first retrieves core metadata via `ffprobe`, including duration, frame rate, spatial resolution, codec, and audio stream presence. Although audio metadata is recorded, the acoustic signal is excluded from visual representation construction. Frames are extracted using OpenCV by seeking directly to target temporal positions, and each is saved as a JPEG image with its original timestamp embedded in both the filename and the JSON manifest, preserving temporal provenance independently of downstream processing. A core output of the script is the dense visual stream, sampled by default at 4 FPS:

```
parser.add_argument("--dense-fps", type=float, default=4.0)
```

Extracted frames are saved in the `dense_frames` directory for each video. Simultaneously, the script generates 32 uniform global frames and performs shot-boundary detection via PySceneDetect. These boundaries define variable-length segments with a target duration of 4 seconds, ranging from 2 to 6 seconds, and a default overlap of 0.5 seconds, extracting eight frames per segment. The preprocessing output is serialized into `metadata.json`, `frames.json`, `shots.json`, and `segments.json`. For the subsequent preliminary analysis, semantic extraction scripts operate directly on the dense stream rather than shot-based segments. Utility functions locate each video's `dense_frames` directory, parse timestamps from filenames, and sort frames chronologically, so that the model receives temporally ordered visual observations instead of isolated or semantically segmented images. The inference temporal windows are configured as follows:

```
parser.add_argument("--window-seconds", type=float, default=4.0)
parser.add_argument("--stride-seconds", type=float, default=3.0)
```

A 4-second window with a 3-second stride yields a 1-second overlap between consecutive observations. The pipeline selects dense frames within these nominal boundaries, and if frame counts exceed the allowed maximum, frames are sampled uniformly across the window. At 4 FPS, a 4-second interval contains 16 frames, remaining well below the default 32-frame ceiling. Duplicate consecutive windows are automatically omitted. Each temporal unit is encapsulated in a `TemporalWindow`, mapping a unique segment ID to its start time, end time, and frame paths. Recorded boundaries reflect the timestamps of the first and last selected frames rather than nominal window limits, with millisecond precision maintained to ensure exact alignment with the source video. The transition to semantic extraction relies strictly on *ordered dense frames*. For each window, the model receives selected RGB images alongside a prompt specifying the video ID, temporal interval, and frame-to-timestamp map, with no questions, answers, transcriptions, or acoustic signals provided. Temporal segmentation thus serves purely as an indexing mechanism for visual evidence, deferring semantic interpretation to the subsequent extraction stage.

Semantic Representation Extraction

Building upon the defined temporal windows, the pipeline proceeds with semantic representation extraction. This stage generates a structured semantic description for each window, processing every window independently. The extraction procedure is repeated across all four evaluated models. Although the models employ distinct architectures and inference modalities, each receives identical inputs and is constrained to produce a standardized representation adhering to a common schema, as shown in Table 3.2. Consequently, performance differences can be attributed directly to model capabilities rather than annotation formatting. For each window, the representation spans seven semantic categories:

entities : Operating at the most fundamental descriptive level, each entity receives a local identifier, a concise label, visually observable attributes, and its role within the scene when discernible. The representation tracks the temporal occurrence of the entity and the specific frames providing visual evidence. These local identifiers enable consistent references across other descriptive categories within the same segment.

actions : Actions represent visually observable occurrences, specifying the performing agent and target object whenever feasible.

events : Events denote broader semantic units linked to one or more participants across a given temporal span. This distinction separates the detection of an isolated action from the broader event context in which it occurs.

spatial_relations : Spatial configurations between entities or scene elements are defined using a subject-relation-object structure. Permissible relations include configurations such as *in front of*, *behind*, *above*, *below*, *inside*, *outside*, *left of*, *right of*, *near*, and *on*. These spatial descriptors are temporally localized, meaning a spatial configuration observed in one segment is not assumed to hold throughout the video.

state_changes : State changes capture observable transformations affecting an entity. By contrasting a prior state with a subsequent state, this structure represents temporal variations within a segment. This level is essential when answering a question requires tracking changes in an entity's properties, position, or condition over time.

temporal_relations : The model describes temporal links such as *before*, *after*, *overlaps*, *during*, *starts*, *finishes*, and *simultaneous*. These relations, however, remain strictly localized to the current window: global video temporal structure is not reconstructed at this stage, nor are event correspondences across different segments systematically resolved.

causal_hypotheses : Causal candidate relations are explicitly separated from directly observed facts. The framework requires causal hypotheses to be flagged as inferred evidence, preventing temporal precedence from being automatically interpreted as a cause-and-effect relationship, and keeping a clear boundary between observed evidence and model inference.

The distinction between observed and inferred evidence is preserved through the *evidence_type* field, accompanied by a confidence score between 0 and 1. Each element also includes an *evidence_frames* list identifying supporting frames. During output normalization, frame references are retained only if they fall within the processed window, and generated temporal intervals are similarly bounded by segment limits, keeping semantic representations strictly aligned with their supporting visual evidence. Categories may remain empty if a model detects insufficient evidence within a segment. At the segment level, the pipeline preserves *segment_id*, *start_time*, *end_time*, and *input_frames*. At this initial stage, a unified global video description is not yet constructed. Because consecutive windows overlap, identical entities, actions, or events may be extracted repeatedly across adjacent segments. This redundancy is intentionally preserved during extraction to maintain a direct mapping between each prediction and its supporting local evidence, leaving entity reconciliation and global temporal aggregation to subsequent pipeline stages.

Field	Information represented
entities	Identified entities, observable attributes, roles, and temporal occurrence.
actions	Local actions, their actors, and involved objects.
events	Temporally extended local events and their participants.
spatial_relations	Spatial configurations between entities or scene elements.
state_changes	Observable transitions between an entity's previous and subsequent states.
temporal_relations	Temporal relations between events identified within the same local context.
causal_hypotheses	Candidate causal relations explicitly marked as inferred evidence.

Table 3.2: Semantic information extracted from each temporal window.

Implementation Details Entity and semantic analysis is implemented through a unified extraction pipeline shared across all evaluated models. Model-specific wrapper scripts load the corresponding architectures and format selected input frames, while temporal framing, prompt generation, schema definitions, validation logic, and serialization are centralized in `utils.py`, so that every model is evaluated against an identical semantic protocol regardless of underlying architectural differences. The common semantic space is defined across seven categories:

```
SEMANTIC_CATEGORIES = (
    "entities",
    "actions",
    "events",
    "spatial_relations",
    "state_changes",
    "temporal_relations",
    "causal_hypotheses",
)
```

The categories illustrated in Table 3.2 specify the structured information required from each temporal window. For each temporal window, the pipeline constructs a standardized extraction prompt via `build_semantic_prompt`, which supplies the video ID, window boundaries, and an explicit frame-to-timestamp map alongside the category definitions:

```python
frame_map = "\n".join(
    f'- frame {index}: "{path.name}", '
    f'timestamp={get_timestamp(path):.3f}s'
    for index, path in enumerate(
        window.frame_paths,
        start=1,
    )
)
```

Rather than eliciting free-form descriptions, the prompt[1] enforces strict visual grounding: every extracted element must reference supporting frames, temporal boundaries, a confidence score, and an `evidence_type`. Directly visible details are classified as `observed`, whereas inferential claims are labeled as `inferred`, and the prompt explicitly instructs the model not to treat simple temporal succession as causality:

```
    """Non inventare dettagli non visibili.
    Usa evidence_type="observed"
    per fatti direttamente visibili.
    Usa evidence_type="inferred"
    soltanto per inferenze inevitabili
    o ipotesi causali.
    Per ogni elemento indica confidence
    tra 0 e 1 e i nomi esatti dei frame
    che lo supportano.
    Non confondere una successione
    temporale con una relazione causale.
```

[1]English translation: *"Do not invent details that are not visible. Use evidence_type="observed" for directly visible facts. Use evidence_type="inferred" only for unavoidable inferences or causal hypotheses. For each element, indicate confidence between 0 and 1 and the exact frame names that support it. Do not confuse temporal succession with a causal relationship. If a category contains no elements, return an empty list. Return exclusively a valid JSON object."*

```
    Se una categoria non contiene elementi,
    restituisci una lista vuota.
    Restituisci esclusivamente
    un oggetto JSON valido."""
```

Models are constrained to output structured JSON objects containing standardized metadata, including `start_time`, `end_time`, `evidence_frames`, `evidence_type`, and `confidence`. Entity identifiers generated within a window can be cross-referenced across actions, events, and relations in the same segment. The main execution loop is encapsulated in `process_video_with_inf` After dense frames are partitioned into temporal windows, each window is processed sequentially:

```
for window_index, window in enumerate(
    windows,
    start=1,
):
    prompt = build_semantic_prompt(
        video_directory.name,
        window,
    )
    response = inferencer(
        window.frame_paths,
        prompt,
    )

    parsed = parse_model_json(response)

    segment = normalize_semantic_output(
        parsed,
        window,
    )

    segments.append(segment)
```

The `inferencer` component is the sole model-dependent step in this loop, accepting frame paths and prompts and returning textual output. All downstream parsing and validation operations remain identical across architectures. To manage output formatting inconsistencies from generative models, the parsing stage strips Markdown wrappers and extracts the primary JSON object. If standard decoding fails, the pipeline attempts recovery using `json_repair` before marking the response invalid:

```
try:
    parsed = json.loads(candidate)
except json.JSONDecodeError as first_error:
    try:
        from json_repair import repair_json
        parsed = json.loads(
            repair_json(candidate)
        )
    except Exception as repair_error:
        raise SemanticOutputError(
            f"JSON non valido: {first_error}"
        ) from repair_error
```

Structural constraints are subsequently enforced by `normalize_semantic_output`, which clips temporal boundaries to window limits, restricts confidence scores to the interval $[0, 1]$, and filters out invalid frame references. Evidence types are validated against `observed`, `inferred`, or `uncertain`, with causal hypotheses automatically reclassified as `inferred` if initially marked as `observed`:

```
confidence = _safe_float(
    item.get("confidence"),
    0.0,
)
item["confidence"] = round(
    max(0.0, min(1.0, confidence)),
```

```
    4,
)
evidence_type = str(
    item.get(
        "evidence_type",
        "observed",
    )
).lower()
if evidence_type not in {
    "observed",
    "inferred",
    "uncertain",
}:
    evidence_type = "uncertain"
if (
    category == "causal_hypotheses"
    and evidence_type == "observed"
):
    evidence_type = "inferred"
```

Frame-level grounding validation retains only filenames present in the input temporal window:

```
valid_evidence = set(
    window.evidence_frames
)
item["evidence_frames"] = [
    str(name)
    for name in evidence_frames
    if str(name) in valid_evidence
]
```

Normalized segments are enriched with temporal metadata from the preprocessing stage:

```
normalized = {
    "segment_id": window.segment_id,
    "start_time": window.start_time,
    "end_time": window.end_time,
    "input_frames": window.evidence_frames,
}
```

Outputs are saved for each video in a `_semantic.json` file alongside model metadata and window configuration settings. Window failures are logged in an `errors` field containing segment identifiers, temporal bounds, frame lists, and error descriptions, preserving optional raw responses for diagnostic review. This phase ultimately transforms temporally indexed visual evidence into structured local representations. Cross-window deduplication is deliberately omitted at this step to preserve local grounding, leaving element consolidation and full video reconstruction to subsequent analysis stages.

Event Analysis

Spatial Relations

Spatial information captures entity configurations within the scene and tracks positional changes across localized visual observations. Unlike temporal relations, spatial features are extracted directly during the semantic representation phase, while visual windows are still accessible. Spatial relations encode explicit configurations between entities, or between an entity and a scene element. These include positional, proximity, containment, and orientation relations such as *left of*, *right of*, *above*, *below*, *inside*, *outside*, *in front of*, *behind*, *near*, and *on*. Because these descriptors are tied to localized segments, a single entity pair may exhibit distinct spatial arrangements across different temporal intervals. Displacements and state transitions also drive spatial variation. A positional change is represented by combining the spatial relations observed at different timestamps with the corresponding action and any recorded `state_changes`, distinguishing static scene layouts from dynamic rearrangements. Event-dependent localization further evaluates spatial configurations relative to an event interval, before, during, or after, isolating how a prior action alters an entity's final position. Methodologically, co-occurrence does not imply a spatial relation. The presence of two entities within the same temporal segment provides insufficient grounds to assume proximity, orientation, containment, or contact: a relation is retained only when explicitly captured in the semantic analysis. Spatial evidence currently remains bound to these local semantic descriptions. The event extraction stage draws on `spatial_relations` for event reconstruction but does not construct a global spatial graph.

Temporal Relations

Temporal information is formalized during event extraction, where local semantic descriptions are consolidated into a chronological timeline covering the full video. Each consolidated event retains a `start_time` and an `end_time` defining its temporal boundary and, implicitly, its duration. Relations between events are modeled explicitly through pairs of `event_id` references. Supported relation types are *before*, *after*, *overlaps*, *during*, and *simultaneous*, each linking two existing events in the consolidated representation together with an associated confidence score. *Before* and *after* express temporal precedence and sequence, *during* denotes temporal containment, *overlaps* indicates partial interval sharing, and *simultaneous* marks concurrent occurrences. This structure captures chronological succession, repetition, containment, and overlap across the video timeline while keeping distinct occurrences separate. Generation order within the JSON output does not itself constitute temporal evidence: dependencies derive strictly from segment timestamps and relational metadata rather than from the model's output sequence. The temporal schema adopts the following structure:

```
"temporal_relations": [
{
"first_event": "E0001",
"relation": "before|after|overlaps|during|simultaneous",
"second_event": "E0002",
"confidence": 0.0
}
```

```
]
```

This consolidated event structure also serves as input for the subsequent causal analysis, though chronological organization is kept distinct from causal inference: temporal precedence does not by itself establish a cause-and-effect relationship.

Implementation Details The event analysis is implemented as a second-stage transformation over the structured semantic representations produced by the previous phase. No visual input is processed again at this stage. For each model, the corresponding semantic output is loaded and used as the only source of information for event reconstruction. The event analyzer can reorganize and consolidate previously extracted information, but it cannot introduce evidence obtained from a new inspection of the video, preserving the separation between perceptual extraction and subsequent structural reasoning. The input therefore consists of the temporally indexed semantic segments generated for the same model. Each segment retains its identifier and temporal boundaries together with the semantic categories extracted from the corresponding visual window. The information made available to the event analysis includes entities, actions, local events, spatial relations, state changes, temporal relations, and causal hypotheses:

```
keys = (
"segment_id",
"start_time",
"end_time",
"entities",
"actions",
"events",
"spatial_relations",
"state_changes",
"temporal_relations",
"causal_hypotheses",
)
```

These fields are considered jointly because an event may not be represented exclusively in the local `events` field. An action or an observable state transition, for instance, may provide relevant information for reconstructing an event even when the local semantic description does not encode it as an independent event. The event analysis therefore operates over the complete local semantic representation rather than simply concatenating the previously extracted event lists. Before inference, the semantic segments are arranged according to their temporal position and serialized into a structured textual input. The model is explicitly instructed to treat this representation as a closed source of evidence, with the common prompt constraining the reconstruction process through a set of rules designed to prevent additional perceptual or commonsense information from being introduced:

```

* usa esclusivamente le informazioni presenti
  nell’analisi semantica
* non inventare azioni, oggetti, intenzioni,
  emozioni o cause mancanti
* unisci soltanto i duplicati causati dalla
  sovrapposizione delle finestre
* mantieni separati eventi realmente distinti
  o ripetuti
* ricava i tempi soltanto dai segmenti forniti
```

A central function of this phase is the consolidation of local descriptions produced by overlapping temporal windows. Since the semantic extraction uses 4-second windows with a 3-second stride, adjacent segments share part of the same visual evidence, and the same physical event can consequently appear in multiple semantic outputs. During event reconstruction, descriptions referring to the same occurrence are merged into a single event, an operation intended to remove redundancy introduced by the preprocessing strategy rather than repetitions that genuinely occur in the video. The prompt explicitly enforces this distinction. Events with similar descriptions are not automatically treated as duplicates: two occurrences are merged only when their temporal context and supporting segments indicate that they represent the same event observed through overlapping windows. Repeated actions occurring at different moments, by contrast, remain represented as distinct events, so that repetition is preserved as a meaningful temporal property of the video. The consolidated representation follows a common JSON schema. Each reconstructed event contains a unique identifier, a concise description, temporal boundaries, the participants involved, the semantic segments supporting the event, the type of evidence, and a confidence value:

```
{
"event_id": "E0001",
"description": "string",
"start_time": 0.0,
"end_time": 0.0,
"participants": ["entity or participant"],
"evidence_segments": ["segment_0001"],
"evidence_type":
"observed|inferred|uncertain",
"confidence": 0.0
}
```

The `event_id` field provides a stable reference for subsequent relational analyses. Events are assigned progressive identifiers such as `E0001`, `E0002`, and `E0003`, associated with the consolidated chronological representation rather than inherited from the local semantic outputs, so that events can be referred to consistently once multiple local descriptions have been merged. Temporal localization is inherited from the semantic analysis, since the event extractor does not estimate event boundaries by accessing the original frames. Instead, `start_time` and `end_time` must remain compatible with the temporal intervals of the semantic segments supporting the event. When the same occurrence is represented across multiple overlapping windows, its consolidated interval can reflect the temporal evidence distributed across those segments. The `evidence_segments` field preserves provenance after consolidation. An event extracted from a single semantic window may contain only one segment identifier, whereas an event observed across overlapping windows may be associated with several:

```
"evidence_segments": [
"segment_0004",
```

```
"segment_0005"
]
```

Consolidation removes duplicated event descriptions without discarding the evidence from which they originated, keeping the resulting event representation traceable to the local observations produced in the previous phase, even though it is more compact than the semantic representation itself. Participants are derived from entities or participants already represented in the semantic analysis and are not generated as independent new entities at this stage, so that the event representation preserves semantic continuity between entity extraction and event reconstruction. The epistemic status of each event is represented through `evidence_type`. The analysis distinguishes three values:

```
"observed"
"inferred"
"uncertain"
```

`Observed` is used for events supported by information previously represented as directly visible. `Inferred` is reserved for information that already requires an inferential interpretation of the available semantic evidence. `Uncertain` represents cases in which the available evidence does not support a stronger classification. The event extraction stage is not intended to transform directly observed information into unsupported inference. Confidence is represented numerically in the $[0, 1]$ interval and subsequently normalized to ensure a consistent representation. Normalization also verifies the structural validity of the generated output before serialization. In addition to individual events, the same inference produces an initial global temporal organization. Temporal relations connect events through their newly assigned identifiers rather than through their textual descriptions:

```
"temporal_relations": [
{
"first_event": "E0001",
"relation":
"before|after|overlaps|during|simultaneous",
"second_event": "E0002",
```

```
"confidence": 0.0
}
]
```

The allowed relations distinguish precedence, succession, containment, partial temporal overlap, and simultaneity, providing an explicit representation of the temporal structure that emerges once local events have been consolidated. Temporal relations cannot be derived solely from the order in which events are written in the generated JSON. The prompt explicitly requires temporal organization to depend on the intervals and relations available in the semantic representation: the order in which the text is generated is not treated as temporal evidence, and temporal succession is likewise kept distinct from causal interpretation. The model output is parsed as structured JSON rather than retained as free-form text. Responses are cleaned of possible Markdown delimiters, and the JSON object is extracted before decoding. The same robust parsing strategy used in the semantic stage allows malformed but recoverable outputs to be repaired when possible:

```
text = (
text.replace("‘json", "")
        .replace("‘", "")
.strip()
)

start = text.find("{")
end = text.rfind("}")

if start < 0 or end < start:
raise ValueError(
"La risposta non contiene un oggetto JSON"
)

payload = json.loads(
text[start:end + 1]
```

```
)
```

This phase primarily changes the granularity of the representation. The semantic analysis describes evidence locally within individual overlapping windows, while the event analysis reorganizes those observations into a coherent event-based representation of the complete video, removing redundancies caused by temporal overlap, preserving genuinely repeated occurrences, assigning stable identifiers, maintaining temporal and evidential provenance, and establishing the first explicit relations among events, all without any new inspection of the visual input.

Causal Inference

Causal inference operates exclusively on the consolidated event representation, independently of perceptual extraction. This phase receives no visual frames or raw multimodal input, relying solely on previously reconstructed events and their established `temporal_relations`. Isolating causal reasoning in this way prevents the recovery of omitted visual details or the insertion of unobserved events. Before processing, the event representation is pruned to its essential fields: `event_id`, description, temporal interval, participants, supporting semantic segments, `evidence_type`, and confidence. The analysis considers four types of causal dependency:

- `causes`: one event directly produces the outcome represented by another.
- `enables`: one event establishes the conditions required for another to occur.
- `motivates`: the available evidence supports an intentional or motivational link between events.
- `prevents`: one event inhibits or counteracts another.

Each entry links a `cause_event` to an `effect_event`, optionally supplemented by `supporting_events` where additional events are needed to substantiate the inference. No new events are introduced during this phase. The resulting structure follows this schema:

```json
"causal_relations": [
  {
    "cause_event": "E0001",
    "relation": "causes|enables|motivates|prevents",
    "effect_event": "E0002",
    "evidence_type": "direct|inferred|uncertain",
    "supporting_events": ["E0003"],
    "explanation": "string",
    "confidence": 0.0
  }
]
```

The epistemic status of each relation is indicated by `evidence_type`. Dependencies grounded in explicit physical interactions or state transitions are labeled `direct`. Connections that require contextual interpretation are labeled `inferred`, while partially supported links receive `uncertain`. Motivational dependencies require explicit support within the event representation rather than speculative plausibility. Temporal succession does not imply causation: a chronological link such as *before* or *after* remains insufficient on its own to establish a causal relation. Dependencies are introduced only when the semantic content explicitly supports causation, enabling, motivation, or prevention. Keeping these stages separate ensures that temporal ordering is never mistaken for causal evidence, preserving the distinction between extracted facts and inferential reasoning.

Implementation Details The causal inference stage is a text-only processing step applied to the consolidated event representation produced in the preceding phase. For each video, the pipeline loads the corresponding `_events.json` file generated by the model. No raw video frames, audio signals, transcriptions, or downstream text prompts are supplied, so the causal analyzer operates exclusively on a closed representation of previously extracted visual information. Prior to inference, the event file is pruned by the `compact` function, which retains only the fields required for causal reasoning:

```python
def compact(payload):
    return {
        "id_video": payload.get("id_video"),
        "events": [
            {
                "event_id": event.get("event_id"),
                "description": event.get("description"),
                "start_time": event.get("start_time"),
                "end_time": event.get("end_time"),
                "participants": event.get("participants", []),
                "evidence_segments": event.get("evidence_segments", [])
    ,
                "evidence_type": event.get("evidence_type"),
                "confidence": event.get("confidence"),
            }
            for event in payload.get("events", [])
        ],
        "temporal_relations": payload.get("temporal_relations", []),
    }
```

This filtering isolates causal reasoning from the original perceptual features, requiring the model to derive dependencies strictly from the consolidated events and their temporal layout. The allowable relation types and evidence categories are bounded explicitly in code:

```python
ALLOWED_RELATIONS = {
    "causes",
    "enables",
    "motivates",
    "prevents",
}
ALLOWED_EVIDENCE = {
    "direct",
```

```
    "inferred",
    "uncertain",
}
```

These definitions separate direct cause-and-effect links (`causes`) from enabling conditions (`enables`), intentional drivers (`motivates`), and inhibitory actions (`prevents`). Inference is guided by a standardized prompt that instructs the model to rely solely on the provided events and temporal relations, explicitly forbidding it from treating simple chronological succession as causal evidence:

```

* usa esclusivamente gli eventi e le relazioni
temporali presenti nell'input
* non usare domande, caption, foil o altre
informazioni esterne
* la semplice successione temporale NON implica causalita'
* crea una relazione soltanto quando il contenuto
degli eventi supporta una dipendenza causa-effetto,
una condizione abilitante, una motivazione oppure
una prevenzione
* cause_event ed effect_event devono essere ID
di eventi presenti nell'input
* non inventare eventi, oggetti, intenzioni o cause
non ricavabili dalla rappresentazione fornita
* se non esiste evidenza sufficiente, restituisci
causal_relations come lista vuota
```

Within this schema, relationships are classified by epistemic confidence: `direct` for explicit physical or state-change interactions, `inferred` for contextual or intentional deductions, and `uncertain` for weakly supported cues. Output formatting enforces the same structured JSON schema used elsewhere in the pipeline:

```
{
"causal_relations": [
```

```
{
    "cause_event": "E0001",
    "relation": "causes|enables|motivates|prevents",
    "effect_event": "E0002",
    "evidence_type": "direct|inferred|uncertain",
    "supporting_events": ["E0003"],
    "explanation": "string",
    "confidence": 0.0
}
]
}
```

The fields `cause_event` and `effect_event` reference existing event IDs, while `supporting_events` optionally logs contextually relevant secondary events. Output parsing strips Markdown delimiters and decodes the outer JSON object. If formatting errors occur, `json_repair` attempts structural recovery, and if parsing fails entirely, a secondary generation call requests strictly formatted JSON without altering the input prompt:

```
text = text.replace("json", "").replace("", "").strip()
start = text.find("{")
end = text.rfind("}")
candidate = text[start : end + 1]
try:
    return json.loads(candidate)
except json.JSONDecodeError:
    from json_repair import repair_json
    return json.loads(repair_json(candidate))
```

A normalization step then applies a sequence of structural checks before the relations are accepted. It first collects the set of valid event identifiers parsed from the input:

```
valid_ids = {
    event.get("event_id")
    for event in payload.get("events", [])
```

```
    if event.get("event_id")
}
```

Candidate relations that reference an unknown event, or that link an event to itself, are discarded:

```
if (
    cause not in valid_ids
    or effect not in valid_ids
    or cause == effect
):
    continue
```

Relations using an unrecognized type are dropped, while evidence labels outside the allowed set default to `inferred`:

```
if rel not in ALLOWED_RELATIONS:
    continue
if evidence not in ALLOWED_EVIDENCE:
    evidence = "inferred"
```

Relations sharing the same cause, relation, and effect are then deduplicated, keeping only the first occurrence of each triplet:

```
key = (cause, rel, effect)
if key in seen:
    continue
seen.add(key)
```

Supporting-event references are filtered in the same way, excluding invalid identifiers as well as the cause and effect events themselves:

```
supporting = [
    event_id
    for event_id in relation.get("supporting_events", [])
    if event_id in valid_ids and event_id not in {cause, effect}
```

```
]
```

Finally, confidence scores are clipped to the $[0, 1]$ interval:

```
try:
    confidence = float(relation.get("confidence", 0.0))
except (TypeError, ValueError):
    confidence = 0.0
confidence = max(0.0, min(1.0, confidence))
```

Final outputs are serialized per video into `_causal.json` files containing metadata, model parameters, and the resulting relations. Unresolvable errors yield empty relation arrays rather than speculative fallbacks.

NLI-Based Visual Proficiency Assessment

The final stage of the preliminary analysis integrates the information produced at the semantic, event, temporal, spatial, and causal levels into a consolidated representation of the visual evidence recovered by each model. The procedure operates exclusively on the outputs generated during the preceding phases, preserving the separation between visual analysis and question-dependent evaluation without performing any further inspection of the video. The integrated representation combines complementary information produced at different levels of the pipeline. Entity descriptions, spatial relations, and state changes originate from the semantic analysis. Consolidated events provide a global event-level representation together with their participants, temporal boundaries, supporting segments, evidence type, and confidence. Temporal relations connect these events through their identifiers, while causal relations encode the dependencies established during causal inference. These components are treated not as independent observations, but as progressively structured views of the same visual evidence. Finalization does not amount to a simple concatenation of all intermediate outputs. Redundancies introduced by overlapping temporal windows have already been resolved during event consolidation, where multiple descriptions of the same occurrence are merged while genuinely repeated events remain distinct, and duplicated or structurally invalid causal relations have already been removed during causal normalization. The epistemic status assigned during the previous stages is preserved as well: information classified as `observed`, `inferred`, or `uncertain` remains distinguishable in the consolidated representation, together with the corresponding confidence values. Uncertain information is not automatically promoted to positive evidence, nor are conflicting descriptions resolved through an additional interpretation introduced at this stage. The final representation thus captures the entities, events, spatial configurations, temporal organization, and causal dependencies recovered by the model, without expanding the evidence beyond what the visual analysis actually produced. Crucially, **finalization does not constitute an answer to any VSV question**. Its purpose is to fix a representation of what the model has recovered from the visual modality before the linguistic content of the evaluation item is introduced. Questions, captions, and foils therefore play no role in constructing the representation, and once it has been finalized, its content remains fixed throughout the evaluation of the corresponding question: the question may provide context for interpreting the linguistic alternatives, but it cannot trigger the extraction of new entities, events, relations, or visual evidence. The proficiency assessment itself is formulated as a Natural Language Inference task. The finalized visual representation acts as the **premise**, while the caption and the corresponding foil are evaluated independently as alternative hypotheses. GPT-4o is employed exclusively as an NLI judge, not to answer the original question, generate a new description of the video, or select directly between the two alternatives. Its role is restricted to determining the logical relation between the evidence contained in the finalized representation and each hypothesis. For both caption and foil, the judge assigns one of three labels: `entailment`, `contradiction`, or `neutral`. `Entailment` indicates that the hypothesis is supported by the available visual representation. `Contradiction` indicates that it is incompatible with the represented evidence. `Neutral` is used when the available information is insufficient to establish either relation, so that missing evidence is not interpreted as contradiction and absence of information is not treated as negative evidence. The caption and foil are judged separately. A caption–foil pair receives `PASS` only when the finalized representation entails the caption and contradicts the foil. Every other combination produces `NOT PASS`. This rule adopts a deliberately conservative criterion. Entailment of the caption alone is insufficient, since the same representation must also provide enough evidence to reject the minimally contrastive foil. Contradiction of the foil, in turn, cannot compensate for a caption classified as `neutral`. A model is therefore considered proficient on a given pair only when its visual representation supports both sides of the required distinction. The procedure is repeated independently for the four caption–foil pairs associated with each question. Question-level proficiency requires consistency across the complete set: a question receives `PASS` only when all four pairs satisfy the pair-level criterion, and is classified as `NOT PASS` if at least one pair fails. This strict aggregation reduces the influence of individual linguistic formulations and requires the recovered visual evidence to remain sufficient across multiple, minimally different realizations of the same underlying item. For each pair, the procedure records the classification assigned to the caption, the classification assigned to the foil, and the resulting binary decision, which are then aggregated into the final proficiency label associated with the question. The resulting measure is accordingly distinct from conventional question-answering accuracy: rather than evaluating whether the model can generate or select the correct answer after being exposed to the question, it evaluates whether the information independently recovered from the visual channel is sufficiently complete and discriminative to support the correct statement while rejecting its minimally contrastive alternative, connecting question-independent visual analysis to the competence-oriented evaluation objective of the preliminary analysis.

Implementation Details The final evaluation is implemented in `final_evaluation.py`, which operates on the finalized representations produced for the four evaluated models. These representations are stored as JSON files in `data/preliminar_analysis/final_results` and constitute the fixed visual premises used throughout the proficiency assessment. The script expects one finalization file for each model and verifies their presence before starting the evaluation:

```
EXPECTED_MODELS = (
"gemma",
"gemma-4-12B",
"qwen",
```

```
"qwen-30B",
)

FINAL_DIR = Path(
"data/preliminar_analysis/final_results"
)

final_files = [
FINAL_DIR / f"{model}.json"
for model in EXPECTED_MODELS
]
```

Finalized representations are loaded independently for each model and indexed by normalized video identifiers. The loading procedure accepts different top-level JSON organizations, including lists of video representations, dictionaries containing `videos` or `results`, individual video objects, and dictionaries directly indexed by video ID, allowing the subsequent NLI procedure to retrieve the complete representation associated with a video without modifying its content. Question metadata and caption–foil pairs are loaded separately from `question_classification.csv` and `caption_foil.csv`. Video identifiers and question categories are normalized before the two sources are joined on the video and principal question dimension. The question is therefore introduced only after the finalized visual representation has been loaded, and is retained as contextual information for the NLI judge rather than incorporated into the premise itself. The NLI evaluation uses `gpt-4o-2024-08-06` with deterministic generation:

```
MODEL = "gpt-4o-2024-08-06"
risposta = client.chat.completions.parse(
model=MODEL,
temperature=0,
messages=[
{
"role": "system",
```

```
"content":
f"{PROMPT}\n\n{REGOLE[categoria]}",
},
{
"role": "user",
"content": contenuto,
},
],
response_format=RispostaNLI,
)
```

Each request contains three distinct fields: the question, the finalized representation used as the premise, and one hypothesis. The caption and foil are never evaluated jointly. Instead, two independent calls are performed using the same premise and question context:

```
nli_caption = classifica(
premessa,
row["domanda"],
row["categoria"],
row["caption"],
)

nli_foil = classifica(
premessa,
row["domanda"],
row["categoria"],
row["foil"],
)
```

The system prompt explicitly restricts the judge to the information contained in the premise. External knowledge, completion of missing information, and plausibility-based reasoning are excluded, and incomplete, weak, or ambiguous evidence must be classified as `neutro`. Additional constraints are introduced according to the question category: spatial relations cannot be inferred from simple co-occurrence, temporal relations cannot be inferred from the textual order of events, and causal relations cannot be established from temporal succession alone. The model response is constrained through a Pydantic schema rather than parsed from unrestricted natural-language output:

```
class RispostaNLI(BaseModel):
etichetta: Literal[
"implicazione",
"contraddizione",
"neutro",
]
```

This design restricts each inference to one of the three admissible NLI labels, preventing the judge from generating explanations or free-form answers. A missing or structurally invalid classification raises an error rather than being converted into an evaluation decision. Pair-level proficiency is subsequently computed through a deterministic rule implemented outside GPT-4o:

```
def valuta_coppia(caption, foil):
return (
"PASS"
if caption == "implicazione"
and foil == "contraddizione"
else "NOT PASS"
)
```

GPT-4o's role is thus limited to assigning the two NLI labels: it does not directly decide whether a proficiency test is passed. The binary outcome is derived programmatically, receiving `PASS` only when the finalized representation entails the caption and contradicts the corresponding foil. All other label combinations result in `NOT PASS`. Results are written incrementally to a model-specific checkpoint in the `nli` subdirectory. Before evaluating a pair, the script checks whether its identifier has already been processed, allowing interrupted executions to resume without repeating completed API calls. Each stored row contains the model, pair identifier, video, category, question, caption, foil, the two NLI classifications, and the resulting pair-level outcome. Once all model-specific evaluations have been completed, pair-level records are concatenated into `nli_risultati.csv`. Question-level proficiency is then computed by grouping results according to model, video, category, and question:

```
df = risultati.groupby(
[
"modello",
"video_id",
"categoria",
"domanda",
]
).agg(
totale_coppie=("esito", "size"),
coppie_pass=("pass", "sum"),
).reset_index()

df["proficiency"] = [
"PASS" if passate == totale else "NOT PASS"
for passate, totale in zip(
df["coppie_pass"],
df["totale_coppie"],
)
]
```

The aggregation adopts the same conservative principle used at pair level: proficiency is assigned only when every caption–foil pair associated with the question passes. Since the experimental data contain four pairs for each question, this corresponds to the required $4/4$ criterion, and a single `NOT PASS` is sufficient to classify the question as `NOT PASS`. The script additionally stores the number of evaluated pairs, the number of successful pairs, and their ratio in `question_proficiency.csv`. The implementation keeps the responsibilities of the two components distinct: the finalized JSON representation remains unchanged and functions solely as the visual premise, GPT-4o performs constrained NLI classification over each caption and foil, and the final proficiency decision is produced by deterministic aggregation rules rather than by an additional generative judgment.

3.3 Core Reasoning Assessment Framework

Experimental Logic and Hypotheses

The experimental design of MAIA-AV goes beyond a simple comparison of model accuracy across conditions. Its primary objective is to analyze the role played by individual modalities in the decision process and verify whether correct answers are genuinely derived from the available evidence. Experiments are structured around a **Modality-Ablation Design**: the test item is held constant while the input modalities are systematically varied, so that shifts in prediction can be observed as a specific information source is added or removed. This control enables **paired comparisons**, attributing behavioral variance more precisely to the selected modality configuration rather than to differences between items. The design further distinguishes between **Modality Sufficiency** and **Modality Necessity**. A modality is sufficient when it independently contains all the information required to solve the task, though sufficiency does not imply that the modality is uniquely useful, nor does it preclude further benefit from additional sources. Conversely, supplementing a sufficient modality with another does not guarantee a performance gain: the added input may be redundant, introduce irrelevant noise, or even interfere with a previously correct prediction. Our experimental setup thus shifts the focus from overall model capability toward how effectively a system integrates the available modalities, and whether distinct modality preferences emerge. This perspective motivates the concepts of **Modality Dominance**, **Modality Gain, Repair, Damage**, and **Cross-Modal Interaction**, formally defined as evaluation metrics in subsequent sections. This framework is rooted in the principle of **controlled information removal** established in *Lost in Transcription* [4], where the underlying task remains fixed while the information accessible to the model is systematically varied, making it possible to evaluate whether an added modality compensates for the absence of another or provides an advantage when partial evidence is already available. That earlier study focused on assessing acoustic contributions within a benchmark built around visual evidence, asking whether the acoustic or textual signal could compensate for absent visual evidence and how this contribution shifted as visual evidence was added. MAIA-AV extends this paradigm through a more articulated manipulation of input modalities. The framework begins with four conditions that establish how much relevant information is retrievable from a single source at a time: complete absence of input, and the isolated acoustic, visual, and transcribed modalities. Two multimodal configurations are then introduced: video with native audio, the full audiovisual clip, and video with audio transcriptions, muted video paired with text. In these multimodal settings, the question extends beyond whether audio merely assists a primarily visual decision. The goal instead is to understand how the availability and representation of information across different modalities reshape the model's overall reasoning, turning the core question from a simple *does audio help* into a broader investigation of how distinct information sources jointly contribute to multimodal reasoning.

Visual Statement Verification

All experimental conditions adopt the same task formulation, **Visual Statement Verification** (VSV). Each decision is evaluated on a pair consisting of a caption, a correct description, and a foil, a minimally contrastive false description. The two alternatives are presented to the model as a binary choice, and the system must return exclusively 0 or 1. This constrained formulation minimizes the confounds associated with open-ended text generation and enables direct comparison across models and input configurations, since the same experimental unit can be observed under different modality configurations without any change to prompt structure or evaluation criteria. This task structure supports two analytical levels: **Item-Level Decisions**, corresponding to single caption-foil choices, and **Aggregate Accuracy**, which evaluates sets of items linked to the same underlying conceptual category to assess whether the model maintains consistent behavior across different formulations. Both metrics are formally defined in subsequent sections.

Domanda	
Dove cammina la bambina alla fine della scena?	
Caption	**Foil**
Alla fine della scena, la bambina cammina **su un tappeto**.	Alla fine della scena, la bambina cammina **su un sentiero**.
Question	
Where does the little girl walk at the end of the scene?	
Caption	**Foil**
At the end of the scene, the little girl walks **on a carpet**.	At the end of the scene, the little girl walks **on a path**.

Table 3.3: Example of a minimally contrastive caption–foil pair, reported in Italian with its English translation.

Experimental Setup

The protocol is operationalized through a standardized pipeline, with model-specific adapters covering only the interface differences between architectures. This pipeline centralizes item preparation, the definition of experimental conditions, prompt construction, input management, and output storage, so that the same evaluation logic applies uniformly across models. Inputs consist of caption-foil pairs, original videos, muted versions without audio, separate audio files, and precomputed transcripts. The pre-established order of alternatives is preserved for every item, and the target label retains the information needed to verify decision accuracy. For each item, the raw response generated by the model is preserved alongside the binary label subsequently extracted from it, allowing valid responses to be distinguished from unparsable output and any execution errors to be documented. A lack of support for a given input type is not treated as an incorrect response. When an architecture cannot process a specific modality, the corresponding configuration is marked as unavailable, and the model is excluded only from comparisons requiring that condition.

The protocol should therefore be read as a model-by-mode matrix, in which the availability of conditions varies across architectures.

Cross-Architecture Input Standardization The experimental protocol remains uniform across model families. Input delivery is adapted to each architecture's native multimodal interface, including the different processors and formats used by Qwen and Gemma, ensuring an identical information payload across architectures without altering the experimental design.

Video Processing Visual content follows a standardized sampling policy: each video is represented using 32 frames evenly distributed across its duration, keeping the amount of visual information comparable across architectures while accommodating the preprocessing requirements of each processor. Visual-only conditions use a version of the video with audio removed, while audiovisual configurations pair the visual stream with the acoustic track specified by the protocol. Sampling is therefore a shared experimental choice, while differences among the visual encoders themselves are addressed separately in the architecture-specific descriptions.

Audio and Transcription Processing The preparation of the acoustic component is tailored to meet the specifications of each experimental condition. In the audio-only scenario, the model is provided with a preprocessed audio file corresponding to the associated video. Conversely, in the audio-visual condition, the original, unmuted video is supplied. When necessary, the model-specific adapter extracts the acoustic stream, converts it into the required format for the corresponding audio frontend, and applies resampling if a 16 kHz signal is mandated. Transcriptions adhere to a distinct methodological framework. They are not generated during the primary experiment, nor do they serve as an intermediate output of the model under evaluation. Instead, they are precomputed inputs maintained consistently across relevant conditions and integrated directly into the textual context. This methodological distinction is crucial, as transcription captures the recognized linguistic content, while the audio signal retains additional acoustic properties. By comparing these two forms of representation, one can study not only the presence of information but also the manner in which it becomes accessible to the model.

Inference Protocol Inference parameters are configured to minimize stochastic variability. Because the task requires a binary classification choice rather than open-ended generation, deterministic decoding is enforced through greedy search, setting `temperature = 0` or `do_sample=False` depending on the model interface, and the maximum output length is strictly constrained to a minimal token budget, limiting non-compliant formatting and keeping outputs consistent across models and conditions. Inference runs in `bfloat16` precision where supported, across execution backends including vLLM and Hugging Face Transformers. These infrastructure choices preserve the underlying experimental logic, while hardware-dependent parameters, such as GPU allocation and framework-specific workarounds, are detailed in the technical appendix for reproducibility.

Shared Evaluation Infrastructure A small set of components is shared unchanged across every experimental condition, and is described here once rather than repeated for each mode below. Media resources are resolved for each item from its video identifier:

```
def paths(row):
    stem = Path(row["video_name"]).stem
    return {
        "audio": AUDIO_DIR / f"{stem}.mp3",
        "mute": MUTE_DIR / f"{stem}.mp4",
        "video": VIDEO_DIR / f"{stem}.mp4",
    }
```

Each condition draws on a different subset of these paths, or on none at all, as detailed below. Prompts are built centrally, before any request reaches a model adapter:

```
def prompt(row, mode, transcript=""):
    return PROMPTS[mode].format(
        answer1=row["answer1"],
        answer2=row["answer2"],
        transcript=transcript,
```

```
    )
```

The `transcript` argument is populated only for the two conditions that require it:

```
transcript = (
    transcripts.get(row["video_name"], "")
    if mode in (
        "only_transcription",
        "transcript_video",
    )
    else ""
)
```

Precomputed transcriptions are loaded once from a dedicated file and indexed by video name:

```
TRANSCRIPTIONS = Path(
    "data/input/transcription/transcriptions.csv"
)

def transcriptions():
    with TRANSCRIPTIONS.open(
        encoding="utf-8-sig",
        newline=""
    ) as f:
        return {
            r["video_name"]: r["transcription"]
            for r in csv.DictReader(f, delimiter=";")
        }
```

Which modalities reach the model in a given condition is determined by two flags, defined once for the whole protocol:

```
use_video = mode in {
    "only_video",
```

```
    "video_audio",
    "transcript_video",
}

use_audio = mode in {
    "only_audio",
    "video_audio",
}
```

`video_audio` is the only mode in which both flags are active simultaneously, providing a direct, implementation-level distinction between the joint audiovisual configuration and the unimodal ablations. Every condition delegates inference to the corresponding model adapter through the same call:

```
raw = infer(
    mode,
    row,
    prompt(row, mode, transcript),
    paths(row),
)
pred = prediction(raw)
```

Regardless of internal multimodal interface, every model is required to return a binary decision. The generated text is normalized through a shared parser that extracts the first isolated occurrence of `0` or `1`:

```
def prediction(text):
    m = re.search(r"(?<!\d)[01](?!\d)", str(text))
    return int(m.group()) if m else None
```

If neither label can be recovered, the output is classified as `invalid` rather than mapped automatically to one of the two alternatives. Valid predictions are compared against the stored target through the same rule in every condition:

```
result = (
```

```
    "correct" if pred == int(row["target"])
    else "wrong" if pred is not None
    else "invalid"
)
```

Sampling is disabled throughout, so that inference is deterministic and variation between predictions reflects the information available in each condition rather than sampling noise. For every item, the pipeline logs a fixed set of fields, including item and video metadata, model identifier, experimental condition, raw output, parsed prediction, and correctness status:

```
FIELDS = [
    "id", "video_id", "video_name", "pool_id",
    "pool_item", "question_category", "question",
    "caption", "foil", "answer1", "answer2",
    "target", "model", "mode", "raw_model_output",
    "predicted_label", "is_correct", "result", "error",
]
```

Results are written to a mode-specific directory with a separate file per model. Existing item identifiers are cached before execution, so that interrupted runs resume without re-computing completed instances, and outputs are flushed immediately after writing to avoid data loss. The system instruction is likewise shared across the entire pipeline, and is not repeated for each condition below:

```
Sei un assistente binario per un task di Visual Statement Verification.
Rispondi esclusivamente con 0 oppure 1.
```
[2]

Model-specific code is accordingly limited to weight loading, chat template formatting, and generation execution. Prompt construction, decision formatting, output parsing, correctness evaluation, error handling, and serialization are identical across models, so that performance differences reflect model capability rather than inconsistencies in the exper-

[2] English translation: *You are a binary assistant for a Visual Statement Verification task. Respond exclusively with either 0 or 1.*

imental protocol. What follows for each condition is therefore limited to what changes: the modality flags in effect, the media or text actually attached to the request, and the condition-specific instruction.

Zero-input Condition

The no-input condition removes perceptual information entirely, providing the model exclusively with the textual caption-foil pair required to complete the task. No video frames, audio signal, or transcription is supplied. Rather than assessing audiovisual comprehension, its function is strictly diagnostic: it establishes the extent to which decisions can be derived from the linguistic plausibility of the options, structural regularities in the dataset, or systematic parametric biases of the model. Performance well above chance in this condition is a strong indicator of linguistic shortcuts operating independently of genuine content understanding.

Implementation Details Neither flag introduced above is active in this condition, and no transcript is retrieved: the model receives only the two textual alternatives, with the original question, video frames, audio, and transcription entirely omitted. The condition-specific prompt is:

```
Basandoti esclusivamente sulle due alternative testuali, scegli.[3]

0: {answer1}
1: {answer2}

Rispondi esclusivamente con 0 oppure 1.
```

where `answer1` and `answer2` are the two members of the caption-foil pair. Their numerical ordering carries no semantic meaning, since the dataset's target label independently indicates whether the correct caption occupies position `0` or `1`, so the model performs a binary selection between competing linguistic realizations rather than open-ended generation. Disabling sampling is especially important here, since it removes output variance unrelated to the underlying linguistic representations. Results are written to a dedicated

[3]English translation: *Based exclusively on the two textual alternatives, choose.*

no_input directory, one file per model, following the shared logging and resumability scheme described above.

Audio-only Condition

In the audio-only condition, the model receives exclusively the raw acoustic signal, with no visual frames or transcription provided. This isolates which caption-foil pairs can be resolved from spoken content, environmental sound, music, or prosody, and comparing performance here against other configurations reveals whether acoustic information retains independent utility once combined with vision, or is instead diminished or overshadowed by a dominant modality.

Implementation Details Only the `use_audio` flag is active: the model receives the audio resolved via `paths(row)["audio"]` together with the two textual alternatives, while the original question is kept only as dataset metadata and no transcript is retrieved. The condition-specific prompt is:

```
Ascolta esclusivamente l'audio fornito.
Scegli quale delle due descrizioni è corretta sulla base delle
informazioni ricavabili esclusivamente dall'audio.
0: {answer1}
1: {answer2}
Rispondi esclusivamente con 0 oppure 1.
```
[4]

How this resource reaches the model is architecture-dependent: models that accept audio files directly receive it through their native processor, while architectures requiring raw waveforms decode the signal into the format expected by their inference backend. Either way, this concerns only input formatting and introduces no textual or visual information beyond what the shared infrastructure already provides. Results are written to a dedicated `only_audio` directory, one file per model. This condition differs from the no-input

[4]English translation: *Listen exclusively to the provided audio. Choose which of the two descriptions is correct based on the information available exclusively from the audio. 0: {answer1}. 1: {answer2}. Respond exclusively with either 0 or 1.*

baseline solely in the availability of the acoustic signal, so the comparison between them isolates whether speech, environmental sound, music, or prosody carry enough information to discriminate captions from foils, independently of visual evidence or its textual transcription.

Video-only Condition

In the video-only condition, the model receives exclusively the visual stream, with the audio track removed. This is the primary baseline for evaluating which decisions can be sustained by visual evidence alone, and it plays a central role in every paired comparison that follows: introducing audio or transcription afterward measures the marginal contribution of that source relative to a baseline where visual context is already available.

Implementation Details Only the `use_video` flag is active. To guarantee that no acoustic information reaches the model, this condition uses the muted video resolved via `paths(row)["mute"]` together with the two textual alternatives and no transcript. The condition-specific prompt is:

```
Osserva il video fornito.
Scegli quale delle due descrizioni è corretta sulla base
esclusivamente delle informazioni visive disponibili.

0: {answer1}
1: {answer2}

Rispondi esclusivamente con 0 oppure 1.[5]
```

The video is represented with the standard budget of 32 frames, sampled at uniformly spaced indices across its full duration rather than from a single local interval:

```
N_FRAMES = 32

frame_indices = np.linspace(
```

[5] English translation: *Observe the provided video. Choose which of the two descriptions is correct based exclusively on the available visual information. 0: {answer1}. 1: {answer2}. Respond exclusively with either 0 or 1.*

```
    0,
    total_frames - 1,
    min(N_FRAMES, total_frames),
    dtype=int,
)
```

Videos with fewer than 32 frames retain all available frames. Extracted frames are converted to RGB before being passed to the multimodal processor, and depending on the native interface of the architecture, the resulting sequence is represented either as a video input or as an ordered sequence of images, a difference confined to the model-specific adapter that does not alter the visual evidence available in the experiment. Results are written to a dedicated `only_video` directory, one file per model. Comparison with the no-input condition quantifies the contribution of visual grounding on its own, while comparison with the audiovisual conditions establishes the reference point needed to determine whether additional modalities improve, preserve, or interfere with visually grounded predictions.

Transcription-only Condition

In the transcription-only condition, the model receives exclusively the textual transcription of the audio track. This isolates spoken linguistic content from non-verbal acoustic features such as prosody, speaker characteristics, background noise, and music, evaluating performance driven strictly by explicit verbal information.

Implementation Details Neither modality flag is active for this condition. The model receives the transcript retrieved via `transcriptions()` together with the two textual alternatives. No video frames, video representation, or raw audio signal are provided, and the original question is kept only as dataset metadata. The condition-specific prompt is:

```
Leggi esclusivamente la trascrizione dell’audio fornita.
Scegli quale delle due descrizioni è corretta sulla base
esclusivamente delle informazioni testuali disponibili nella
trascrizione.
```

```
Trascrizione dell'audio: {transcript}
0: {answer1}
1: {answer2}
Rispondi esclusivamente con 0 oppure 1.[6]
```

This condition differs from the no-input setting in a single controlled respect: the caption-foil pair is now accompanied by the textual representation of the audio track, so the comparison between them isolates whether speech-derived linguistic information contributes to the decision even without direct perceptual access to the original video or acoustic signal. Because no multimodal resource is attached, the transcription is tokenized as ordinary text alongside the task instruction and the two alternatives, without ever passing through an audio encoder. This is central to the condition's design: it evaluates access to the semantic content extracted from audio rather than to the acoustic modality itself, so prosody, speaker characteristics, environmental sound, and other non-textual properties remain unavailable unless they were explicitly captured in the transcription. Results are written to a dedicated `only_transcription` directory, one file per model. Comparison with the no-input condition measures the contribution of the transcription itself, comparison with the audio-only condition distinguishes the semantic information preserved by textualization from what is available in the original acoustic signal, and comparison with the conditions that include visual evidence tests whether the same linguistic information remains useful once combined with the primary visual channel.

The Audio-Visual (AV) Condition

The audiovisual condition represents the natural configuration of information available to the model, comprising both the visual sequence and the native acoustic signal from the original video, provided as a single integrated stream rather than as the separately combined channels of the video-plus-transcription condition described below. This is the core configuration for testing whether the simultaneous presence of multiple modalities

[6] English translation: *Read exclusively the provided audio transcription. Choose which of the two descriptions is correct based exclusively on the textual information available in the transcription. Audio transcription: {transcript}. 0: {answer1}. 1: {answer2}. Respond exclusively with either 0 or 1.*

yields an advantage over unimodal conditions or instead leads to *unimodal collapse*. Its value extends beyond aggregate performance, since it also allows item-level observation of whether the acoustic channel repairs errors made in the video-only condition or introduces new incorrect decisions.

Implementation Details Both `use_video` and `use_audio` are active here, the only condition in which both are true simultaneously. The model receives the original, non-muted video resolved via `paths(row)["video"]`, ensuring both modalities originate from the same source stimulus, together with the two textual alternatives and no transcript. The condition-specific prompt is:

```
Osserva il video e ascolta il relativo audio.
Scegli quale delle due descrizioni è corretta sulla base
dell’insieme delle informazioni visive e acustiche disponibili.
0: {answer1}
1: {answer2}
Rispondi esclusivamente con 0 oppure 1.
```
[7]

When an architecture requires the acoustic stream as a separate input, the waveform is extracted directly from the same original file used for the visual channel, rather than from an independent recording, preserving the correspondence between the two modalities at the stimulus level:

```python
def extract_audio(video):
    out = (
        Path(tempfile.gettempdir())
        / "maia_av_audio_cache"
        / f"{video.stem}.wav"
    )

```

[7] English translation: *Observe the video and listen to its audio. Choose which of the two descriptions is correct based on the combined visual and acoustic information available. 0: {answer1}. 1: {answer2}. Respond exclusively with either 0 or 1.*

```python
    out.parent.mkdir(exist_ok=True)

    if not out.exists():
        subprocess.run(
            [
                "ffmpeg",
                "-loglevel", "error",
                "-y",
                "-i", str(video),
                "-vn",
                "-ac", "1",
                "-ar", "16000",
                str(out),
            ],
            check=True,
        )

    return out
```

producing a mono waveform at 16 kHz. The video itself follows the same 32-frame uniform sampling used in the video-only condition, with indices additionally deduplicated:

```python
N_FRAMES = 32

frame_indices = np.unique(
    np.linspace(
        0,
        total_frames - 1,
        min(N_FRAMES, total_frames),
        dtype=int,
    )
)
```

Both streams are then exposed to the inference backend as explicit multimodal inputs where the adapter requires it, or recovered directly from the original video by adapters offering native audiovisual processing, a difference confined to the model-specific adapter that does not affect the evidence available during inference:

```
multi_modal_data = {}

if video is not None:
    multi_modal_data["video"] = video

if audio is not None:
    multi_modal_data["audio"] = (
        audio,
        AUDIO_SAMPLE_RATE,
    )

request["multi_modal_data"] = multi_modal_data
```

Results are written to a dedicated `video_audio` directory, one file per model. This condition is best understood not as an independent task but as the joint-modality counterpart of the two unimodal ablations: comparison with video-only provides a controlled measure of whether adding raw acoustic information improves, preserves, or interferes with decisions already supported by vision, while comparison with the transcription-based multimodal condition below distinguishes the contribution of native acoustic processing from that of the linguistic information preserved in a transcription.

The Visual-Language (VL) Configuration

The video-plus-transcription condition combines the visual component with a textual representation of speech. This is the second multimodal configuration, allowing direct audiovisual integration to be compared against textualized linguistic information: comparison with the audiovisual condition tests whether the informative content of the acoustic track is exploited more effectively once converted to text, while comparison with video-only clarifies the benefit of audio-derived linguistic information when visual context is already available.

Implementation Details Only `use_video` is active, since this condition does not expose the raw acoustic signal. The model receives the muted video resolved via `paths(row)["mute"]`, together with the transcript retrieved via `transcriptions()` and the two textual alternatives. The condition-specific prompt combines the visual instruction with the transcription:

```
Osserva il video fornito e leggi la trascrizione dell'audio.
Scegli quale delle due descrizioni è corretta sulla base
dell'insieme delle informazioni visive e testuali disponibili.

Trascrizione dell'audio: {transcript}

0: {answer1}
1: {answer2}

Rispondi esclusivamente con 0 oppure 1.[8]
```

The video follows the same 32-frame uniform sampling described for the video-only condition. No audio extraction takes place here. Since only the transcription of the acoustic channel is exposed to the model, never the raw signal itself, the muted video is sufficient and the native waveform is never decoded. Results are written to a dedicated `transcript_video` directory, one file per model. Because this condition and the audiovisual condition differ only in how the acoustic content reaches the model, native waveform versus transcription, while the visual channel and the underlying decision remain identical, their comparison isolates the effect of textualization itself on top of an already visually grounded prediction.

[8]English translation: *Observe the provided video and read the audio transcription. Choose which of the two descriptions is correct based on the combined visual and textual information available. Audio transcription: {transcript}. 0: {answer1}. 1: {answer2}. Respond exclusively with either 0 or 1.*

Metrics Selection and Aggregation Strategy

Linguistic and Structural Bias Assessment under Zero-Input

The no-input condition calls for different analytical treatment than the other experimental settings. Since neither visual nor acoustic information is available, performance in this condition cannot be interpreted as evidence of multimodal understanding, and is therefore unsuitable for direct inclusion in standard performance comparisons. Instead, dedicated metrics are used to quantify **linguistic and structural shortcuts**, that is, whether the model can exploit linguistic regularities, structural properties of the alternatives, or systematic biases in the dataset to solve the task without perceptual evidence. Let N denote the number of evaluated items. For each item i, let $y_i \in \{0,1\}$ be the target label and $\hat{y}_i \in \{0,1\}$ the prediction produced by the model.

Raw Accuracy The first measure is *Raw Accuracy*, the proportion of items for which the predicted label matches the target:

$$\mathrm{Acc}_{\mathrm{NoInput}} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}\left(\hat{y}_i = y_i\right), \tag{3.1}$$

where $\mathbb{I}(\cdot)$ is the indicator function, equal to 1 when the condition holds and 0 otherwise. Since each item requires a binary choice between two alternatives, a model selecting uniformly at random has expected accuracy

$$\mathbb{E}\left[\mathrm{Acc}_{\mathrm{random}}\right] = \frac{1}{2} = 0.5. \tag{3.2}$$

An accuracy close to 0.5 is therefore compatible with random guessing, while a substantially higher value indicates that the model identifies the target more often than chance would predict despite the absence of perceptual evidence, suggesting that some property of the alternatives' linguistic form may be predictive of the correct answer. Raw Accuracy should not be read as a measure of language comprehension: the model is not required to understand the underlying video event, nor does it have access to the evidence needed for grounded reasoning. The metric only captures its ability to recover the target from the two textual alternatives available in this condition.

Language-Only Advantage To express the deviation from the random baseline more directly, we compute the *Language-Only Advantage* (LOA), obtained by subtracting the expected chance level from the observed no-input accuracy:

$$\mathrm{LOA} = \mathrm{Acc}_{\mathrm{NoInput}} - 0.5. \tag{3.3}$$

LOA $= 0$ indicates performance exactly at the random baseline, positive values indicate performance above chance, and negative values indicate performance below chance. The main advantage of this transformation is interpretability: rather than reporting the raw accuracy value, LOA directly expresses the amount of performance exceeding what chance would predict, and can be used as a compact indicator of the potential contribution of linguistic or structural shortcuts. A positive LOA does not identify the nature of the shortcut. It only establishes that the textual alternatives contain sufficient regularities for the model to perform above chance, so additional diagnostics are required to determine which properties may contribute to this effect.

Matthews Correlation Coefficient Accuracy alone can conceal degenerate decision strategies, since a model with a systematic preference for one output label may still obtain non-negligible accuracy when predictions are unevenly distributed across the two positions. For this reason, the analysis also includes the *Matthews Correlation Coefficient* (MCC). For a binary task, predictions can be summarized through the four entries of the confusion matrix, true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN):

$$\mathrm{MCC} = \frac{\mathrm{TP} \cdot \mathrm{TN} - \mathrm{FP} \cdot \mathrm{FN}}{\sqrt{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})(\mathrm{TN}+\mathrm{FP})(\mathrm{TN}+\mathrm{FN})}}. \tag{3.4}$$

The numerator compares concordant and discordant predictions, correct decisions contributing through $\mathrm{TP} \cdot \mathrm{TN}$ and the two error types through $\mathrm{FP} \cdot \mathrm{FN}$, while the denominator normalizes this difference against the marginal distributions of targets and predictions. MCC ranges between -1 and 1. Values close to 1 indicate strong agreement between predictions and targets, values close to 0 indicate little or no association, and negative values indicate a systematic inverse association. MCC complements Raw Accuracy by weighing all four components of the confusion matrix rather than only the diagonal, distinguishing genuine target-sensitive performance from apparently successful behavior produced by strongly unbalanced prediction strategies.

Position Bias *Position Bias* measures directly whether the model systematically favors one candidate position independently of its content. Let

$$p_0 = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left(\hat{y}_i = 0\right) \tag{3.5}$$

denote the empirical probability of selecting the first alternative, with $p_1 = 1 - p_0$ the probability of selecting the second. Under perfectly balanced behavior, $p_0 = p_1 = 0.5$, so we define the magnitude of the positional deviation as

$$\mathrm{PB} = \left| p_0 - 0.5 \right|, \tag{3.6}$$

and the preferred position as

$$\mathrm{PreferredPosition} = \underset{j \in \{0,1\}}{\operatorname{argmax}}\, p_j. \tag{3.7}$$

A Position Bias of 0 indicates perfectly balanced selection, and the maximum possible value, 0.5, is reached when the model always selects the same position. This diagnostic matters because above-chance Raw Accuracy does not by itself demonstrate the exploitation of meaningful linguistic information: part of the observed result may instead reflect a systematic preference for the first or second alternative, which Position Bias measures directly.

Lexical Attraction *Lexical Attraction* evaluates whether the model preferentially selects the alternative that is lexically closer to the original question. For each item i, let q_i denote the question and a_{i0}, a_{i1} the two alternatives, each represented as a TF–IDF vector. The cosine similarity between the question and alternative j is

$$s_{ij} = \cos\left(\mathbf{v}_{q_i}, \mathbf{v}_{a_{ij}}\right) = \frac{\mathbf{v}_{q_i}^{\top} \mathbf{v}_{a_{ij}}}{\left\|\mathbf{v}_{q_i}\right\|_2 \left\|\mathbf{v}_{a_{ij}}\right\|_2}, \qquad j \in \{0,1\}, \tag{3.8}$$

and the lexically closest alternative is

$$\ell_i = \underset{j \in \{0,1\}}{\operatorname{argmax}}\, s_{ij}. \tag{3.9}$$

The *Lexical Attraction Rate* measures how often the model selects this alternative:

$$\mathrm{LAR} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\left(\hat{y}_i = \ell_i\right). \tag{3.10}$$

A high value indicates a preference for the candidate sharing greater lexical overlap with the question, though this behavioral tendency alone does not establish that lexical overlap is a dataset bias. To evaluate the structural relationship between lexical similarity and the target, we separately compute the *Lexical Target Alignment*:

$$\mathrm{LTA} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\left(y_i = \ell_i\right), \tag{3.11}$$

the frequency with which the correct alternative is also the one with the highest lexical similarity to the question. The two measures capture different phenomena, LAR reflecting model behavior and LTA reflecting dataset structure:

$$\mathrm{LAR} \rightarrow \text{model behavior}, \tag{3.12}$$

$$\mathrm{LTA} \rightarrow \text{dataset structure}. \tag{3.13}$$

A high LAR is only informative as a shortcut if lexical similarity is genuinely correlated with the target:

$$\mathrm{LAR} \gg 0.5 \quad \text{and} \quad \mathrm{LTA} \gg 0.5 \tag{3.14}$$

jointly provide evidence for both the availability of a lexical shortcut in the dataset and the model's tendency to exploit it.

Length Bias A parallel analysis applies to the length of the two alternatives. Let $L(a_{ij})$ denote the length of alternative j for item i, and let

$$h_i = \operatorname*{argmax}_{j\in\{0,1\}} L(a_{ij}) \tag{3.15}$$

identify the longer candidate. The model's preference for longer alternatives is the *Length Attraction Rate*:

$$\mathrm{LenAR} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\left(\hat{y}_i = h_i\right), \tag{3.16}$$

a behavioral property of the model, with values well above 0.5 indicating a tendency to select the longer alternative. As with lexical similarity, a separate measure establishes whether this heuristic is actually rewarded by the dataset, the *Length Target Alignment*:

$$\text{LenTA} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}\left(y_i = h_i\right), \tag{3.17}$$

the frequency with which the correct alternative is also the longer one. The distinction again separates model behavior from dataset structure:

$$\text{LenAR} \to \text{preference of the model for longer alternatives,} \tag{3.18}$$

$$\text{LenTA} \to \text{association between alternative length and the target.} \tag{3.19}$$

A model may prefer longer alternatives even when length carries no information about correctness, just as the dataset may contain a length-correctness correlation that a given model fails to exploit. Considering both quantities together keeps these two cases from being conflated.

No-Input Category Breakdown The main no-input diagnostics are additionally computed separately for each reasoning category. Let c denote a category and

$$I_c = \{i \mid \text{category}(i) = c\} \tag{3.20}$$

the subset of items belonging to it, with $N_c = |I_c|$. Category-specific Raw Accuracy and Language-Only Advantage follow directly:

$$\text{Acc}_c = \frac{1}{N_c}\sum_{i\in I_c}\mathbb{I}\left(\hat{y}_i = y_i\right), \qquad \text{LOA}_c = \text{Acc}_c - 0.5, \tag{3.21}$$

and MCC and Position Bias are computed analogously by restricting the confusion matrix and prediction frequencies to I_c, with

$$p_{0,c} = \frac{1}{N_c}\sum_{i\in I_c}\mathbb{I}\left(\hat{y}_i = 0\right), \qquad \text{PB}_c = \left|p_{0,c} - 0.5\right|. \tag{3.22}$$

This decomposition makes it possible to determine whether no-input solvability varies systematically across reasoning categories, without treating a higher category-level accuracy as evidence that the model is more competent at the corresponding reasoning process: since no perceptual evidence is available in this condition, such differences instead indicate that some groups of items expose stronger linguistic or structural regularities than others. This distinction matters directly when no-input results are later compared against performance under perceptual conditions, since differences observed between spatial, temporal, and causal questions should not be attributed automatically to multimodal reasoning ability when part of the variation may already be present before any perceptual modality is introduced. The category breakdown therefore provides the baseline needed to separate genuine grounding effects from differences rooted in the linguistic structure of the task.

Item-Aligned Modality Influence Metrics

Let c denote an experimental condition and $C_i^{(c)} \in \{0,1\}$ indicate whether the model answers item i correctly under condition c:

$$C_i^{(c)} = \mathbb{I}\left(\hat{y}_i^{(c)} = y_i\right), \tag{3.23}$$

where y_i is the target label and $\hat{y}_i^{(c)}$ the prediction obtained under c. For every comparison between a baseline condition b and a condition n obtained by adding a new modality, only items available in both are retained, giving the paired evaluation set

$$\mathscr{I}_{b,n} = \mathscr{I}_b \cap \mathscr{I}_n, \qquad N_{b,n} = \left|\mathscr{I}_{b,n}\right|. \tag{3.24}$$

All modality-contribution metrics are computed on this common subset, which preserves item-level comparability and keeps differences in data availability from affecting the estimated contribution of the added modality.

Conditional Modality Gain The *Conditional Modality Gain* measures the change in performance produced by adding a new source of information to a fixed baseline, for the two primary comparisons

$$\text{Video Only} \rightarrow \text{Audio + Video} \qquad \text{and} \qquad \text{Video Only} \rightarrow \text{Video + Transcription.} \tag{3.25}$$

Let b denote the baseline condition and n the condition after the additional modality has been introduced. Accuracy on the paired subset is

$$\mathrm{Acc}_b = \frac{1}{N_{b,n}} \sum_{i \in \mathscr{I}_{b,n}} C_i^{(b)}, \qquad \mathrm{Acc}_n = \frac{1}{N_{b,n}} \sum_{i \in \mathscr{I}_{b,n}} C_i^{(n)}, \tag{3.26}$$

and the Conditional Modality Gain is their difference, which, since both accuracies are computed on the same items, can equivalently be written as an item-level average:

$$\Delta_{b \to n} = \mathrm{Acc}_n - \mathrm{Acc}_b = \frac{1}{N_{b,n}} \sum_{i \in \mathscr{I}_{b,n}} \left(C_i^{(n)} - C_i^{(b)} \right). \tag{3.27}$$

This second formulation makes the paired nature of the metric explicit. Each item contributes $+1$ when the additional modality turns an incorrect response into a correct one, -1 when it turns a correct response into an incorrect one, and 0 when correctness is unchanged, so $\Delta_{b \to n} > 0$ indicates a positive net contribution of the added modality and $\Delta_{b \to n} < 0$ an overall detrimental effect. For the two main comparisons, the corresponding gains are

$$\Delta_{\mathrm{Audio|Video}} = \mathrm{Acc}_{\mathrm{Audio+Video}} - \mathrm{Acc}_{\mathrm{VideoOnly}}, \qquad \Delta_{\mathrm{Transcription|Video}} = \mathrm{Acc}_{\mathrm{Video+Transcription}} - \mathrm{Acc}_{\mathrm{VideoOnly}}. \tag{3.28}$$

These quantities measure the marginal contribution of acoustic or transcribed information when visual evidence is already available, directly extending the controlled information-addition principle adopted in *Lost in Transcription*.

Repair Rate Aggregate gain does not reveal how individual decisions change. The *Repair Rate* isolates items answered incorrectly in the baseline condition and measures how often the additional modality converts them into correct predictions:

$$N_{\mathrm{repair}} = \sum_{i \in \mathscr{I}_{b,n}} \mathbb{I}\left(C_i^{(b)} = 0 \wedge C_i^{(n)} = 1 \right), \qquad N_{\mathrm{base\ error}} = \sum_{i \in \mathscr{I}_{b,n}} \mathbb{I}\left(C_i^{(b)} = 0 \right), \tag{3.29}$$

$$\mathrm{RepairRate}_{b \to n} = \frac{N_{\mathrm{repair}}}{N_{\mathrm{base\ error}}} = P\left(C^{(n)} = 1 \mid C^{(b)} = 0 \right). \tag{3.30}$$

The metric conditions exclusively on baseline failures, measuring the corrective contribution of the newly available modality rather than its effect on the complete item set. For audio added to visual evidence,

$$\mathrm{RepairRate}_{A|V} = P\left(C^{(AV)} = 1 \mid C^{(V)} = 0 \right), \tag{3.31}$$

where a high value indicates that acoustic evidence frequently resolves decisions that cannot be solved from visual evidence alone.

Damage Rate The *Damage Rate* captures the opposite transition, the proportion of baseline-correct items that become incorrect once the additional modality is introduced:

$$N_{\text{damage}} = \sum_{i \in \mathscr{I}_{b,n}} \mathbb{I}\left(C_i^{(b)} = 1 \wedge C_i^{(n)} = 0\right), \qquad N_{\text{base correct}} = \sum_{i \in \mathscr{I}_{b,n}} \mathbb{I}\left(C_i^{(b)} = 1\right), \quad (3.32)$$

$$\text{DamageRate}_{b \to n} = \frac{N_{\text{damage}}}{N_{\text{base correct}}} = P\left(C^{(n)} = 0 \mid C^{(b)} = 1\right). \quad (3.33)$$

For the addition of audio to video,

$$\text{DamageRate}_{A|V} = P\left(C^{(AV)} = 0 \mid C^{(V)} = 1\right), \quad (3.34)$$

a transition that provides a direct behavioral measure of cross-modal interference: the additional source does not merely fail to contribute useful information, it actively overturns a previously correct decision. Conditional Modality Gain alone cannot distinguish repair from damage. When $N_{\text{repair}} = N_{\text{damage}}$, the net gain $\Delta_{b \to n} = 0$ even though a substantial fraction of individual predictions has changed, a situation fundamentally different from $N_{\text{repair}} = N_{\text{damage}} = 0$, where the additional modality produces no change in correctness at all. Paired decisions are accordingly partitioned into four mutually exclusive transition classes:

$$\begin{aligned}
N_{\text{stable correct}} &= \sum_i \mathbb{I}\left(C_i^{(b)} = 1 \wedge C_i^{(n)} = 1\right), \\
N_{\text{repair}} &= \sum_i \mathbb{I}\left(C_i^{(b)} = 0 \wedge C_i^{(n)} = 1\right), \\
N_{\text{damage}} &= \sum_i \mathbb{I}\left(C_i^{(b)} = 1 \wedge C_i^{(n)} = 0\right), \\
N_{\text{stable wrong}} &= \sum_i \mathbb{I}\left(C_i^{(b)} = 0 \wedge C_i^{(n)} = 0\right),
\end{aligned} \quad (3.35)$$

satisfying

$$N_{b,n} = N_{\text{stable correct}} + N_{\text{repair}} + N_{\text{damage}} + N_{\text{stable wrong}}, \quad (3.36)$$

the item-level decomposition underlying the aggregate gain.

Modality Interdependence Analysis

Factorial Modality Interaction Analysis The joint contribution of audio and video is analyzed through a 2×2 factorial design, with $A \in \{0,1\}$ indicating the availability of the acoustic modality and $V \in \{0,1\}$ the availability of the visual modality:

$$\begin{array}{c|cc} & V=0 & V=1 \\ \hline A=0 & \text{No Input} & \text{Video Only} \\ A=1 & \text{Audio Only} & \text{Audio + Video} \end{array} \tag{3.37}$$

The *No Input* condition is included solely to complete the factorial structure, representing the cell $(A,V)=(0,0)$ in which neither perceptual modality is present, without implying that it constitutes a multimodal comprehension condition. Let $Y_i \in \{0,1\}$ indicate whether item i is answered correctly. Correctness probability is modeled through logistic regression,

$$\operatorname{logit}[P(Y_i=1)] = \beta_0 + \beta_A A_i + \beta_V V_i + \beta_{AV}(A_i V_i), \qquad \operatorname{logit}(p) = \log\left(\frac{p}{1-p}\right). \tag{3.38}$$

Writing $p_{av} = P(Y=1 \mid A=a, V=v)$ for the four cell probabilities, the coefficients admit a direct interpretation:

$$\beta_0 = \operatorname{logit}(p_{00}), \qquad \beta_A = \operatorname{logit}(p_{10}) - \operatorname{logit}(p_{00}), \qquad \beta_V = \operatorname{logit}(p_{01}) - \operatorname{logit}(p_{00}), \tag{3.39}$$

$$\begin{aligned} \beta_{AV} &= \operatorname{logit}(p_{11}) - \operatorname{logit}(p_{10}) - \operatorname{logit}(p_{01}) + \operatorname{logit}(p_{00}) \\ &= [\operatorname{logit}(p_{11}) - \operatorname{logit}(p_{01})] - [\operatorname{logit}(p_{10}) - \operatorname{logit}(p_{00})]. \end{aligned} \tag{3.40}$$

The interaction thus measures whether the effect of audio changes as a function of visual availability. Under the null hypothesis $H_0 : \beta_{AV} = 0$, the joint audiovisual configuration is compatible with additive modality effects on the logit scale, while $\beta_{AV} > 0$ indicates a joint effect stronger than the separate modality contributions predict and $\beta_{AV} < 0$ a joint effect weaker than that additive expectation. On the odds-ratio scale, $\mathrm{OR}_{AV} = \exp(\beta_{AV})$, so that $\mathrm{OR}_{AV} > 1$ corresponds to positive interaction and $\mathrm{OR}_{AV} < 1$ to negative interaction. For the estimated coefficient $\hat{\beta}_{AV}$, the Wald statistic

$$z = \frac{\hat{\beta}_{AV}}{\mathrm{SE}\left(\hat{\beta}_{AV}\right)} \tag{3.41}$$

yields the two-sided p-value $p = 2\left[1-\Phi(|z|)\right]$, where $\Phi(\cdot)$ is the cumulative distribution function of the standard normal distribution. The 95% confidence interval for the interaction coefficient is

$$\mathrm{CI}_{95\%}\left(\beta_{AV}\right) = \hat{\beta}_{AV} \pm 1.96\,\mathrm{SE}\left(\hat{\beta}_{AV}\right), \tag{3.42}$$

and its odds-ratio counterpart is obtained through exponentiation,

$$\mathrm{CI}_{95\%}\left(\mathrm{OR}_{AV}\right) = \left[\exp\left(\hat{\beta}_{AV} - 1.96\,\mathrm{SE}\left(\hat{\beta}_{AV}\right)\right), \exp\left(\hat{\beta}_{AV} + 1.96\,\mathrm{SE}\left(\hat{\beta}_{AV}\right)\right)\right]. \tag{3.43}$$

Standard errors are estimated using a cluster-robust covariance matrix at the `pool_id` level, letting observations within the same pool be statistically dependent while treating different pools as independent clusters. Only items observed in all four factorial cells are retained, restricting the analysis to

$$\mathscr{I}_{2\times 2} = \mathscr{I}_{00} \cap \mathscr{I}_{10} \cap \mathscr{I}_{01} \cap \mathscr{I}_{11}, \tag{3.44}$$

a complete-case restriction that preserves the paired structure required for a valid comparison of modality effects.

Joint-Modality Evaluation

Joint-Modality Accuracy Aggregate audiovisual accuracy does not distinguish items that genuinely require multiple modalities from items already solvable from a single source. The *Joint-Modality Accuracy* restricts evaluation to items classified as requiring a specific combination of modalities. Let $J_i^{AV} \in \{0,1\}$ indicate whether item i requires both audio and visual information, with $\mathscr{J}_{AV} = \left\{i \mid J_i^{AV} = 1\right\}$. Audio–Video Joint-Modality Accuracy is then

$$\mathrm{JMA}_{AV} = \frac{\sum_{i\in\mathscr{J}_{AV}} C_i^{(AV)}}{|\mathscr{J}_{AV}|}. \tag{3.45}$$

For items requiring visual information together with an audio-derived transcription, the analogous definitions $J_i^{VT} \in \{0,1\}$ and $\mathscr{J}_{VT} = \left\{i \mid J_i^{VT} = 1\right\}$ give

$$\mathrm{JMA}_{VT} = \frac{\sum_{i\in\mathscr{J}_{VT}} C_i^{(VT)}}{|\mathscr{J}_{VT}|}. \tag{3.46}$$

This restriction changes the object being measured. General audiovisual accuracy is $\mathrm{Acc}_{AV} = P\left(C^{(AV)} = 1\right)$, whereas Joint-Modality Accuracy is $\mathrm{JMA}_{AV} = P\left(C^{(AV)} = 1 \mid J^{AV} = 1\right)$.

High performance on the first can arise from items solvable through a single modality, while high performance on the second requires successful predictions specifically within the subset for which joint information has been identified as necessary, making it a more selective estimate of multimodal integration.

Alignment Sensitivity

Alignment Sensitivity measures whether model performance depends on the correct correspondence between information sources. Let a denote the correctly aligned multimodal condition and m the corresponding misaligned condition, with evaluation restricted to $\mathscr{I}_{a,m} = \mathscr{I}_a \cap \mathscr{I}_m$. Aligned and misaligned accuracies are

$$\mathrm{Acc}_{\mathrm{aligned}} = \frac{1}{|\mathscr{I}_{a,m}|} \sum_{i \in \mathscr{I}_{a,m}} C_i^{(a)}, \qquad \mathrm{Acc}_{\mathrm{misaligned}} = \frac{1}{|\mathscr{I}_{a,m}|} \sum_{i \in \mathscr{I}_{a,m}} C_i^{(m)}, \tag{3.47}$$

and Alignment Sensitivity is their difference which, since both conditions are evaluated on the same items, can equivalently be written item by item:

$$\mathrm{AS} = \mathrm{Acc}_{\mathrm{aligned}} - \mathrm{Acc}_{\mathrm{misaligned}} = \frac{1}{|\mathscr{I}_{a,m}|} \sum_{i \in \mathscr{I}_{a,m}} \left(C_i^{(a)} - C_i^{(m)} \right). \tag{3.48}$$

A positive value, $\mathrm{AS} > 0$, indicates that correct cross-modal correspondence improves performance relative to mismatched inputs, while a value near zero indicates limited sensitivity to the relationship between the modalities. For audiovisual inputs,

$$\mathrm{AS}_{AV} = \mathrm{Acc}_{AV}^{\mathrm{aligned}} - \mathrm{Acc}_{AV}^{\mathrm{misaligned}}, \tag{3.49}$$

and for video and transcription,

$$\mathrm{AS}_{VT} = \mathrm{Acc}_{VT}^{\mathrm{aligned}} - \mathrm{Acc}_{VT}^{\mathrm{misaligned}}. \tag{3.50}$$

This metric enters the main experimental results only when the corresponding misaligned conditions are part of the executed protocol. If misaligned configurations are supported only by the evaluation infrastructure, AS remains a formally defined extension rather than an observed main-experiment metric.

Stratified Behavioral Analysis

Independent Visual-Evidence Estimation The preliminary analysis provides an estimate of visual evidence availability that is independent of the VSV predictions produced in the main experiments.

For each item i, let R_i^V denote the structured representation extracted exclusively from the visual stream, obtained from the analysis of entities, events, and inferential relations, and let Q_i^* denote the information required to resolve the corresponding question. A Natural Language Inference stage compares R_i^V with Q_i^* and produces an evidence-availability variable

$$E_i^V = f_{\mathrm{NLI}}\left(R_i^V, Q_i^*\right) \in \{0,1\}, \tag{3.51}$$

where $E_i^V = 1$ indicates that the visual representation contains sufficient information according to the adopted NLI criterion, and $E_i^V = 0$ indicates that the required information cannot be established from it.

This variable is not defined from VSV correctness, $E_i^V \neq C_i^{(V)}$: the two quantities sit at distinct analytical levels. E_i^V estimates whether the required information is available in the independently extracted visual representation, while $C_i^{(V)}$ records whether the model actually produces the correct VSV decision under the video-only condition. Evidence availability, in short, is not the same thing as observed task success.

Mapping Preliminary and Experimental Outcomes Preliminary and main-experiment outputs are joined through their shared item identifiers. For each item i, the resulting analytical representation contains at least the triplet

$$\left(E_i^V, C_i^{(N)}, C_i^{(m)}\right), \tag{3.52}$$

where E_i^V is the independently estimated visual-evidence availability, $C_i^{(N)}$ correctness in the *No Input* condition, and $C_i^{(m)}$ correctness under experimental modality configuration m. The complete modality vector can be written as

$$\mathbf{C}_i = \left(C_i^{(N)}, C_i^{(A)}, C_i^{(V)}, C_i^{(AV)}, C_i^{(T)}, C_i^{(VT)}\right), \tag{3.53}$$

with N, A, V, AV, T, and VT denoting *No Input*, *Audio Only*, *Video Only*, *Audio + Video*, *Transcription Only*, and *Video + Transcription*, respectively.

The preliminary taxonomy assigns each item to an independently defined stratum $S_i \in \{S_0, S_1, \ldots, S_K\}$, not derived from main-experiment correctness, $S_i \not\equiv g(\mathbf{C}_i)$, but acting as an external stratification variable. Performance under condition m can therefore be computed separately for each class S_k,

$$\mathrm{Acc}_{m,S_k} = \frac{\sum_i \mathbb{I}(S_i = S_k) C_i^{(m)}}{\sum_i \mathbb{I}(S_i = S_k)}, \tag{3.54}$$

with the corresponding difference between two modality conditions b and n within the same stratum given by

$$\Delta_{b \to n}^{(S_k)} = \mathrm{Acc}_{n,S_k} - \mathrm{Acc}_{b,S_k}. \tag{3.55}$$

This stratification lets experimental behavior be interpreted jointly with independently estimated evidence availability. The pair $\left(E_i^V = 0, C_i^{(V)} = 1\right)$ identifies an item solved correctly under the video-only condition despite the absence of sufficient evidence in the independently extracted visual representation, while $\left(E_i^V = 1, C_i^{(V)} = 0\right)$ identifies a failure despite independently estimated visual evidence being available. The additional No-Input variable further separates perceptual from non-perceptual success. $\left(E_i^V = 0, C_i^{(N)} = 1\right)$ is compatible with a decision supported by linguistic or structural regularities rather than by visual grounding, while $\left(E_i^V = 1, C_i^{(N)} = 0, C_i^{(V)} = 1\right)$ is consistent with a performance gain that emerges only once visual evidence becomes available. The preliminary classes $S_0, S_1, \ldots, S_K$ thus provide an independent stratification layer for the modality-ablation results. Their role is analytical rather than evaluative: they allow observed VSV decisions to be interpreted against the information independently recoverable from the visual stream, without defining the classes from the same predictions that are subsequently analyzed against them.

3.4 Attention-Based Modality Analysis

The attention analysis complements the behavioural experiments described above. Rather than asking only whether a modality's presence affects performance, it investigates how the model internally allocates attention across the available sources of information during generation. The design follows directly from the findings discussed earlier for *Lost in Transcription* [4]: the availability of multiple modalities does not by itself guarantee their effective integration, and behavioural comparisons alone cannot show how a model distributes its internal processing when several sources are available at once. A model may receive both video and audio while attending almost entirely to the visual representations, or a transcription may attract substantial decoder attention even where its addition barely changes task accuracy. For this reason, the attention analysis provides an internal counterpart to the behavioural modality-ablation experiments. The behavioural analysis addresses the question:

> Does the availability of a given modality change model performance?

The attention analysis instead addresses the complementary question:

> When multiple modalities are available, how is attention distributed across them during answer generation?

The distinction matters for questions of modality dominance and unimodal collapse. Strong performance under a multimodal configuration does not demonstrate that every available modality contributes meaningfully to the decision, since a model can reach a correct prediction by relying mainly on a dominant source while largely ignoring the rest. Comparing unimodal and multimodal conditions exposes this behaviour at the output level, but says nothing about whether the internal computation reflects the same imbalance. Tracing how much attention is directed toward visual, acoustic, and textual representations during generation provides that missing evidence, showing whether the dominance observed behaviourally is also reflected in the model's internal allocation. The comparison between the *Audio + Video* and *Video + Transcription* conditions is especially informative here. Both settings preserve the visual channel but carry the information associated with the audio track in different representational forms, native acoustic signal in one case, language in the other. Comparing the two allows us to ask whether linguistic reformulation makes audio-derived information more accessible to the language decoder, and whether this in turn changes the attention assigned to the video. The question connects directly to the observations made in *Lost in Transcription*. If transcription-based input attracts substantially more attention than raw audio, this would help explain why multimodal models behave differently once acoustic information is converted into text. If instead the decoder keeps concentrating attention on visual representations in both conditions, that would point to persistent visual dominance regardless of how the acoustic content is represented. The analysis is thus meant to connect three complementary levels: input availability, which measures whether adding or removing a modality changes task performance, semantic accessibility, which evaluates whether the information required by the question can already be recovered from the visual stream alone, and internal allocation, which investigates how the model distributes attention across the available sources while producing the answer.

$$\begin{aligned} \text{Modality Availability} &\rightarrow \text{Behavioural Contribution} \\ \text{Visual Content} &\rightarrow \text{Semantic Accessibility} \\ \text{Internal Attention} &\rightarrow \text{Modality Allocation} \end{aligned} \tag{3.56}$$

When examined together, the three levels provide a more comprehensive understanding of multimodal integration than considering each one individually. They help differentiate between a modality that is accessible but receives minimal attention, one that receives significant attention but offers little behavioral improvement, and one where both internal focus and performance shifts indicate a true multimodal contribution. Attention, in this context, is not viewed as a direct indicator of causal influence. A high level of attention doesn't necessarily mean a modality determines the final outcome, just as a low level doesn't imply its irrelevance. Instead, attention functions as an indicator of how resources are internally distributed, while causal insights continue to emerge from manipulating modalities in controlled settings and observing the resulting performance advantages. The analysis of attention serves as a supplement to the behavioral ablation framework, rather than a substitute for it.

Setup	Input	Description
No Input	None	The model receives no multimodal information. The setup isolates how far the answer can be inferred from linguistic priors or dataset biases alone.
Audio Only	Audio	The model receives only the raw audio track, while the visual content is removed. It measures the information recoverable from the acoustic modality in isolation.
Video Only	Video	The model receives only the visual content of the video, without audio. It measures how much task-relevant information the visual modality alone provides.
Audio + Video	Audio + Video	The complete audiovisual signal is provided to the model. It represents the fully multimodal condition, assessing the contribution of joint audio-visual processing.
Transcription + Video	Transcription + Video	The visual content is combined with a textual transcription of the audio track. It tests whether linguistic information extracted from audio is exploited more effectively when given explicitly as text.

Table 3.4: Overview of the experimental setups used to evaluate the contribution of different input modalities.

4. Results and Discussions

4.1 Preliminary Analysis Results

Expected Behaviour under the Preliminary Condition

Before examining the experimental outcomes, it is useful to lay out qualitative expectations for model performance under the frame-only condition. This provides a methodological anchor for interpreting distinctions across question categories, rather than deriving explanations retrospectively from the results themselves. For spatial questions, a relatively high presence of visual evidence is expected, given the inherent visibility of entity locations, movements, object configurations, and spatial relationships within the visual stream. Complexity may increase, however, when this information must be reconstructed across multiple temporal intervals, as when tracking an entity through a sequence of movements or deriving a spatial configuration that results from several successive actions. Temporal questions are expected to display a more varied profile. Visually distinctive actions and event sequences can often be pinpointed through timestamp data and organized by the temporal structure produced during Event Analysis, but not all temporal information is necessarily recoverable from visual input alone. Some questions may involve events whose interpretation depends on linguistic context, or on relations between moments that remain ambiguous from visual information alone. The adequacy of the resulting representation therefore hinges on the quality of chronological reconstruction as much as on the inherent nature of the events involved. Causal questions are anticipated to be the most demanding under the visual-only condition. The visual stream can capture event succession, interactions, and observable state changes, but the underlying reasons for these events often depend on spoken dialogue or remain simply indistinguishable from static imagery. Motivations, intentions, and explanatory relationships are less likely to be directly supported by visual evidence alone. This expectation also underlines a methodological distinction maintained throughout the analysis: temporal succession is not treated as evidence of causation. These expectations are formed independently of the empirical results and serve only as a qualitative reference. No claim is made about the magnitude of category-level differences or about which category will ultimately score highest. Proficiency results are then compared with downstream Visual Statement Verification (VSV) performance, in order to separate evidence availability from task execution. Visual proficiency asks whether the independently constructed representation contains enough information to distinguish a caption from its foil. VSV performance asks whether the model produces the correct response once the question and the condition-specific inputs are directly available. Together, the two measures offer a diagnostic view that plain task accuracy cannot provide on its own. A Pass proficiency result paired with a correct VSV response indicates that the relevant visual evidence was both represented and successfully used downstream. A Pass result paired with an incorrect response instead points to a failure occurring after the evidence was already available, in selection, integration, or later reasoning. When both proficiency and VSV performance are negative, the most plausible explanation is a perceptual or representational shortfall, though this comparison alone cannot pinpoint the exact locus of failure. The most diagnostically informative configuration is the opposite one: proficiency Not Pass paired with a correct downstream answer. Here, the model selects the correct alternative despite a visual representation that does not meet the proficiency criterion, which may reflect partial cues, alternative evidence, parametric priors, shortcut strategies, or simple guessing. The comparison alone cannot isolate which of these mechanisms is at play, but it does establish that correctness and evidence availability are not the same thing, a distinction the sections below return to repeatedly.

Entity, Event, and Inference Recovery

As discussed in Section 3.2, the preliminary analysis asks whether the visual content of a video alone allows a model to answer a question correctly. It also serves a diagnostic role of its own, verifying that the newly constructed questions do not inherit the same limitation as the original MAIA benchmark [22], namely a reliance on visual content so strong that it prevents a genuine assessment of cross-modal integration. The analysis targets an earlier stage of the process than VSV itself, the semantic construction of what has been observed. This makes it possible to check whether a model can independently extract the required information from video frames, separating knowledge that is genuinely grounded in visual evidence from responses shaped by linguistic priors. Table 4.1 reports the aggregate scores for the three models currently available. Scores for Qwen-3B are not reported

Model	**Entity**	**Event**	**Inference**	**Comprehension**
Gemma	27.87	23.13	16.48	20.46
Gemma-4-12B	33.77	30.07	21.48	25.90
Qwen-30B	**34.47**	**31.53**	**23.95**	**27.75**

Table 4.1: Average preliminary-analysis scores across entity recovery, event recovery, inference recovery, and overall comprehension.

here, since the preliminary analysis for this model was not yet available at the time of the present evaluation. Entity-level information is the most accessible, since the model only needs to map a visual element onto its semantic representation, establishing referential grounding between an observed entity and the lexical concept used to describe it. The two larger models confirm a relatively direct grounding mechanism between the visual signal and semantic entities, with average entity-recovery scores of 34.47 for Qwen-30B and 33.77 for Gemma-4-12B, suggesting a stronger ability to associate visual patterns with the corresponding conceptual categories. The smaller Gemma model shows the same internal ordering at lower absolute values, 27.87 for entities against 23.13 for events and 16.48 for inference, indicating that even the most accessible component is far from fully recovered. The decline from entity to inference is present in every configuration and reflects increasing representational demands. Event recovery requires entities to be embedded into dynamic structures involving actions, participants, and state transitions, a predicate-argument composition that goes beyond simple referential grounding. Inference recovery adds a further layer in which the model must derive relations or interpretations that need not correspond to any directly observable perceptual pattern, extending beyond lexicalization into temporal ordering, causal interpretation, or relational dependencies between multiple observed states. Qwen-30B moves from 34.47 for entities to 31.53 for events and 23.95 for inferences. Gemma-4-12B follows the same pattern, from 33.77 to 30.07

to 21.48. Gemma shows the sharpest proportional drop, from 27.87 to 23.13 to 16.48. Model capacity therefore appears to raise the amount of recoverable information at every level without removing the structural difficulty tied to increasingly abstract representation. Correctly identifying a participant does not imply that the model also represents the event that participant is involved in, and correct event identification does not guarantee that the relation required by the question can subsequently be inferred.

Category-Level Differences

Splitting the results by reasoning dimension gives a finer view of what the visual stream actually supports. Tables 4.2, 4.3, and 4.4 report the average scores for each model.

Dimension	Entity	Event	Inference	Comprehension
Causal	39.00	30.80	19.50	25.79
Spatial	29.10	26.10	22.60	25.00
Temporal	15.50	12.50	7.35	10.60

Table 4.2: Gemma preliminary-analysis scores across causal, spatial, and temporal questions.

Dimension	Entity	Event	Inference	Comprehension
Causal	45.30	37.60	25.70	31.75
Spatial	35.90	33.20	28.25	30.90
Temporal	20.10	19.40	10.50	15.05

Table 4.3: Gemma-4-12B preliminary-analysis scores across causal, spatial, and temporal questions.

The category breakdown only partly matches the qualitative expectations set out above. Spatial information is indeed comparatively accessible, particularly for Qwen-30B, but temporal questions produce the lowest scores across every model, while causal questions show relatively strong entity and event recovery despite the expected difficulty of causal inference itself. Each pattern is examined in turn below.

Dimension	Entity	Event	Inference	Comprehension
Causal	43.60	36.50	25.50	30.95
Spatial	**41.50**	**40.50**	**37.05**	**38.85**
Temporal	18.30	17.60	9.30	13.45

Table 4.4: Qwen-30B preliminary-analysis scores across causal, spatial, and temporal questions.

Spatial Information

Spatial questions are the strongest category for Qwen-30B, which reaches an entity score of 41.50, an event score of 40.50, and an inference score of 37.05, for an overall comprehension of 38.85. The relatively small drop across these three levels is informative in itself. Once the relevant entities are recovered, Qwen-30B tends to preserve enough relational information to reconstruct the spatial configuration the question asks about, suggesting that spatial relations can often be supported by local, directly observable evidence. Gemma-4-12B follows the same pattern at lower values, 35.90 for entities, 33.20 for events, and 28.25 for inferences. Gemma reaches 29.10, 26.10, and 22.60 respectively. The category as a whole appears to benefit from the perceptual accessibility of its target relations: locations, relative positions, trajectories, and object configurations can frequently be identified within a restricted temporal interval, keeping the semantic relation the question asks about closely aligned with perceptual structure. Linguistically, spatial questions require associating lexicalized spatial concepts, location, direction, containment, proximity, movement, with geometric or relational properties in the scene, which depends on correctly identifying both the referents and the relation connecting them. Qwen-30B's stronger spatial scores indicate that its visual representation preserves these relational structures more effectively than those produced by the other two models. Even so, spatial grounding is not uniformly successful: overall comprehension stays below 40 even for Qwen-30B, and the presence of a visible object does not guarantee recovery of the specific relation the question targets, particularly when that relation depends on tracking an entity across different moments of the video.

Temporal Information

Temporal questions show the clearest degradation of any category, for every model evaluated. Gemma reaches an overall comprehension of 10.60, Gemma-4-12B 15.05, and Qwen-30B 13.45. The drop is not confined to the inference level. Gemma's entity and event scores are already low, 15.50 and 12.50, Gemma-4-12B reaches 20.10 and 19.40, and Qwen-30B 18.30 and 17.60. Inference scores fall further still, to 7.35, 10.50, and 9.30 respectively. Temporal difficulty, in other words, does not emerge only at the final reasoning step. The preliminary representations frequently fail to recover one or more of the entities and events needed to establish the temporal relation in the first place. A temporal question typically depends on at least two event anchors, both of which must be represented with sufficient specificity and then positioned relative to one another. If one event is missing from the representation, the temporal relation cannot be reconstructed even when the other event was correctly recognized. This pattern surfaces directly in the qualitative annotations: the model often identifies the relevant participants but fails to recover one of the required actions, or fails to anchor it to a sufficiently precise temporal interval, leaving the representation unable to support relations such as *before*, *after*, *first*, or *second*. Temporal interpretation, from a linguistic standpoint, needs more than lexical grounding. Events must be represented as structured semantic units and then ordered within a shared temporal frame, and the low inference scores show that this ordering step is rarely supported when the underlying event representations are themselves incomplete. This is consistent with the original MAIA findings, where temporal phenomena remained particularly difficult under visual-only evaluation [22], and with the broader motivation behind benchmarks such as VILMA, which treat temporal grounding as a capability that cannot be reduced to static visual recognition.

Causal Information

The causal category produces a more differentiated pattern than the initial expectations suggested. Causal questions do not obtain the lowest scores. Gemma and Gemma-4-12B in fact reach their highest comprehension values here, 25.79 and 31.75 respectively, and

Qwen-30B reaches 30.95, only slightly below its spatial score. This should not be read as evidence that causal inference is visually easy. The internal score distribution tells a different story: Gemma-4-12B reaches an entity score of 45.30 and an event score of 37.60, while its inference score drops to 25.70. Qwen-30B shows a similar profile, 43.60, 36.50, and 25.50. Gemma shows the steepest proportional fall, from 39.00 for entities to 30.80 for events and 19.50 for inference. The visual stream, in short, often supplies enough to identify the participants and observable actions in a causal scenario, but the difficulty increases sharply once the representation must encode *why* an event occurs rather than simply *what* occurs. This gap matters linguistically because causal interpretation is a relation between propositions, or between event representations, not a property of a single event. A model can correctly lexicalize an action and identify its participants without representing the explanatory dependency linking that action to an antecedent state, an intention, or a motivation, especially when that dependency was conveyed through spoken explanation rather than anything visually explicit. In such cases the visual representation may contain the observable consequence while offering no support for the corresponding causal interpretation. The results therefore separate *causal event availability* from *causal relation recovery*: the comparatively high entity and event scores show that causal questions frequently concern visually observable situations, while the much lower inference scores show that the dependency needed to actually answer the question stays largely out of reach.

Qualitative Distribution of Comprehension Levels

The continuous proficiency scores are complemented by a qualitative classification into four levels, *insufficient*, *partial*, *sufficient*, and *complete*. Table 4.5 reports the global distribution. The majority of question-specific evidence is classified as insufficient for every model, from 70.0% for Gemma down to 60.7% for Gemma-4-12B and 61.7% for Qwen-30B. Complete representations remain rare even for the strongest configuration, with Qwen-30B reaching 8.7%, ahead of Gemma-4-12B's 6.0% and Gemma's 4.0%. Scale therefore increases the frequency of near-complete representations, but the prevalence of insufficient cases stays substantial even at the top end. This is a meaningful result pre-

Model	Insufficient	Partial	Sufficient	Complete
Gemma	70.0%	21.3%	4.7%	4.0%
Gemma-4-12B	60.7%	25.7%	7.7%	6.0%
Qwen-30B	61.7%	21.0%	8.7%	8.7%

Table 4.5: Distribution of qualitative comprehension levels in the preliminary analysis.

cisely because the representation is built independently of the downstream question, with no explicit guidance toward the entities or events that will later matter. What it measures is spontaneous recoverability of task-relevant evidence, not question-conditioned retrieval. The category breakdown, reported in Table 4.6, sharpens this picture considerably. Tem-

Model	Dimension	Insufficient	Partial	Sufficient	Complete
Gemma	Causal	58%	37%	5%	0%
Gemma	Spatial	66%	15%	7%	12%
Gemma	Temporal	86%	12%	2%	0%
Gemma-4-12B	Causal	48%	37%	14%	1%
Gemma-4-12B	Spatial	55%	22%	7%	16%
Gemma-4-12B	Temporal	79%	18%	2%	1%
Qwen-30B	Causal	51%	34%	14%	1%
Qwen-30B	Spatial	52%	12%	12%	24%
Qwen-30B	Temporal	82%	17%	0%	1%

Table 4.6: Distribution of qualitative comprehension levels by model and principal question dimension.

poral questions carry the highest concentration of insufficient representations across the board, 86% for Gemma, 79% for Gemma-4-12B, 82% for Qwen-30B, and complete temporal representations are almost nonexistent, none for Gemma, only 1% for the other two. Spatial questions look markedly different: Qwen-30B reaches complete comprehension in 24% of spatial cases and sufficient in a further 12%, against 16% and 7% for Gemma-4-12B and 12% and 7% for Gemma. The causal category sits between the two, marked by a comparatively large share of partial representations, 37% for both Gemma and Gemma-4-12B and 34% for Qwen-30B, consistent with the score-level pattern already discussed: the models frequently recover some of the relevant participants and observable actions without ever completing the causal dependency the question actually asks for.

Model Scale and Visual Representation

Scale produces a general improvement in preliminary performance, though its magnitude varies by dimension. Gemma-4-12B consistently outperforms the smaller Gemma model, with overall comprehension rising from 20.46 to 25.90, entity recovery from 27.87 to 33.77, event recovery from 23.13 to 30.07, and inference recovery from 16.48 to 21.48. Qwen-30B reaches the highest global scores, 27.75 overall comprehension against 25.90 for Gemma-4-12B, though the margin at the aggregate level is comparatively narrow. The category breakdown shows why this advantage is not uniform: Qwen-30B's improvement is concentrated in spatial representation, where its comprehension score of 38.85 clearly exceeds Gemma-4-12B's 30.90, while Gemma-4-12B slightly outperforms Qwen-30B on both temporal comprehension, 15.05 against 13.45, and causal comprehension, 31.75 against 30.95. A larger parameter count therefore does not translate into a uniform gain across every form of visual-semantic reasoning. Model family and representation architecture appear to shape which relational structures are preserved most effectively, a pattern the next section returns to when comparing these preliminary results against the main experiment, where model scale again produces markedly uneven effects depending on which kind of evidence is at stake.

4.2 Experimental Results and Discussion

The experimental conditions provide a controlled view of how linguistic, visual, auditory, and transcription-derived information contribute to Visual Statement Verification. Rather than reading multimodal performance only in terms of absolute accuracy, the analysis below treats each condition as evidence about the informational resources available to the model and about its ability to exploit them consistently, and draws throughout on the preliminary analysis in Section 4.1 to separate what a model can perceive from what it can do once perception is paired with a question. The comparison is particularly relevant against the original MAIA results [22] and against the experimental setting introduced in *Lost in Transcription* [4]. MAIA was designed as a visually grounded benchmark, in which the distinction between a true statement and its minimally contrastive foil is expected to depend primarily on information extracted from the video. The original evaluation showed that increasing visual evidence generally improves performance, though gains vary considerably across reasoning categories and temporal phenomena remain particularly difficult. *Lost in Transcription* extended this setting to audio and transcription, finding that raw audio contributed substantially when visual evidence was absent but became almost neutral once dense visual information was available, and that adding textualized audio in the transcription-based setting even produced a small but systematic degradation. The present results paint a different, more heterogeneous picture across model families and scales.

Overall Performance Across Experimental Conditions

Table 4.7 reports item-level accuracy across the five conditions involving perceptual or transcription-derived evidence. All models benefit from the availability of multimodal information, although the relative contribution of each channel varies substantially. Whether

Model	Audio Only	Transcription Only	Video Only	Audio + Video	Video + Transcription
Gemma	74.58%	70.25%	69.83%	**77.17%**	75.08%
Gemma-4-12B	75.92%	69.25%	73.08%	80.50%	**84.75%**
Qwen	73.50%	66.50%	72.50%	**78.92%**	75.50%
Qwen-30B	79.17%	77.00%	73.25%	83.17%	**84.08%**

Table 4.7: Item-level accuracy across unimodal and multimodal experimental conditions. Bold values indicate the best-performing multimodal configuration for each model.

these differences reflect systematic behaviour rather than descriptive fluctuation was tested directly. Each model was evaluated on 1,200 items distributed across 300 pools, and since predictions under different conditions refer to the same underlying items, all comparisons below treat the six conditions as paired observations. An omnibus Cochran's Q test, applied separately to each model, confirms a statistically significant effect of experimental condition in every case, both at item level and at the stricter pool level defined by MAIA's Aggregate Accuracy criterion. The null hypothesis of equal performance across condi-

Model	**Item-level** Q	p	**Pool-level** Q	p
Gemma	194.45	$< .001$	113.63	$< .001$
Gemma-4-12B	361.29	$< .001$	217.78	$< .001$
Qwen	392.89	$< .001$	193.52	$< .001$
Qwen-30B	349.32	$< .001$	195.28	$< .001$

Table 4.8: Cochran's Q tests comparing performance across all six experimental conditions, at item level and at pool level.

tions is therefore rejected for every model configuration. This confirms that modality availability systematically affects predictions and licenses the pairwise comparisons developed throughout the rest of this section, each tested with the exact McNemar test and Holm-corrected for multiple comparisons within each model. A first descriptive observation concerns the relatively limited variability of the video-only condition, ranging from 69.83% for Gemma to 73.25% for Qwen-30B, a spread of only 3.42 points. The two Qwen configurations differ by even less, 72.50% against 73.25%. This compressed range contrasts sharply with the larger differences observed once linguistic or auditory information becomes available, and is compatible with the hypothesis, developed further in Section 4.2, that video-only performance is partly constrained by the perceptual representation itself rather than by the capacity of the language backbone. This interpretation is consistent with the original MAIA analysis, where larger models generally benefited from a 32-frame configuration but continued to show weaknesses in visually demanding categories, particularly temporal duration and other phenomena involving distributed evidence [22]. The present results extend this observation by showing that model capacity has a far clearer effect once an additional linguistic or auditory source becomes available, a pattern examined in detail below.

No-Input Condition and Linguistic Plausibility

The no-input condition isolates the contribution of linguistic information contained in the question and answer alternatives. Since no video, raw audio, or transcription is provided, performance above chance cannot be attributed to perceptual grounding. Three models perform clearly above chance, Qwen-30B at 61.00%, Gemma-4-12B at 59.25%, Gemma at 57.42%, indicating that a non-negligible part of the decision space can be addressed through linguistic plausibility alone. This should not be read as evidence that the questions themselves encode their answers. More plausibly, the model exploits lexical compatibility, semantic plausibility, world knowledge, or distributional regularities learned between a question and its two candidate statements. The no-input condition, in this sense, provides a behavioural estimate of how far linguistic competence can substitute for perceptual evidence. Causal questions show the strongest linguistic-only performance for Gemma, Gemma-4-12B, and Qwen-30B, reaching 68.25%, 68.00%, and 64.75% respectively. Causal interpretation activates knowledge about event structure, selectional preferences, typical causal relations, and commonsense schemas, so a statement describing a conventional relation between an event and its likely cause can receive strong support

Model	**No-Input Accuracy**
Gemma	57.42%
Gemma-4-12B	59.25%
Qwen	50.83%
Qwen-30B	**61.00%**

Table 4.9: Accuracy in the no-input condition. Chance performance is 50%.

Model	**Causal**	**Spatial**	**Temporal**
Gemma	68.25%	55.50%	48.50%
Gemma-4-12B	68.00%	57.75%	52.00%
Qwen	51.25%	50.50%	50.75%
Qwen-30B	64.75%	62.75%	55.50%

Table 4.10: No-input accuracy across the three analyzed reasoning categories.

from the language model independently of any visual evidence. This behavioural pattern is corroborated directly by the preliminary analysis in Section 4.1, where causal inference-recovery scores were comparatively low, 19.50 for Gemma, 25.70 for Gemma-4-12B, 25.50 for Qwen-30B, despite entity and event scores in the same category being among the highest observed. Read together, the two results indicate that causal VSV decisions are particularly susceptible to linguistic and world-knowledge priors, since the visual-only representation typically supplies the participants and observable actions involved in a scenario without encoding the explanatory dependency the question actually asks about. This does not mean causal VSV performance is exclusively language-driven, only that linguistic priors offer a strong alternative source of support whenever the visual representation stops short of the causal relation itself. Temporal questions display the opposite tendency, with no-input accuracy of 48.50% for Gemma, 52.00% for Gemma-4-12B, and 55.50% for Qwen-30B, close to chance throughout. Relative temporal information is far less recoverable from general semantic knowledge, since it typically depends on the actual ordering or duration of events within a specific video rather than on anything a language model could infer from priors alone. Here too the preliminary analysis offers a direct explanation: temporal comprehension scores in Section 4.1 were the lowest of any category, 10.60, 15.05, and 13.45 for the three models. Temporal relations are strongly episode-dependent, so when the preliminary representation fails to recover one of the two event anchors a temporal question depends on, linguistic plausibility offers comparatively little room for compensation. Temporal questions therefore face a double difficulty, weak recovery of the relevant evidence from the video itself, and a linguistic prior too generic to substitute for it, which helps explain why temporal grounding remains one of the most persistent weaknesses across video-language benchmarks, consistent with the original MAIA findings [22]. Qwen represents a special case. Its no-input accuracy of 50.83% looks close to chance, but the model selects the first answer in 99% of its valid predictions. Chance-level accuracy therefore does not correspond to an unbiased decision process here: the model follows an extreme positional preference that happens to yield an approximately balanced overall accuracy only because the correct answer is itself balanced across positions. Accuracy alone cannot distinguish genuine uncertainty from a deterministic shortcut whose success rate happens to approximate chance, which is precisely why the no-input analysis needs to report both linguistic competence and response strategy rather than accuracy in isolation.

Audio-Only Performance

One of the most notable results concerns the audio-only condition. As shown in Table 4.11, raw audio outperforms video-only input for every model. This result matters

Model	Audio Only	Video Only	Difference
Gemma	74.58%	69.83%	+4.75
Gemma-4-12B	75.92%	73.08%	+2.83
Qwen	73.50%	72.50%	+1.00
Qwen-30B	79.17%	73.25%	+5.92

Table 4.11: Comparison between audio-only and video-only item accuracy. Differences are expressed in percentage points.

because MAIA was originally developed as a visually grounded benchmark. Its audio track played no role in constructing the questions and statements, so strong audio-only performance does not show that the benchmark is intrinsically audio-based, only that the acoustic stream carries information strongly correlated with the events and situations it queries. Raw audio can supply several complementary sources of evidence at once, spoken language that explicitly identifies objects, actions, locations, intentions, or causal relations, environmental sounds that signal ongoing events, and acoustic dynamics that offer temporal cues about occurrence or transition. The linguistic component of speech is likely central to this advantage. Once speech is successfully encoded, its semantic representation interfaces naturally with the language backbone responsible for comparing the two candidate statements, narrowing part of the representational gap that separates visual patterns from linguistic propositions. The result is therefore best read as evidence of high *informational accessibility* of the acoustic channel rather than of audio being inherently richer than vision. A visual scene may well contain more information overall while still being harder for the architecture to convert into a representation aligned with the linguistic decision the VSV task requires. The pattern differs markedly from *Lost in Transcription*, where Qwen2.5-Omni-7B obtained 82.70% with 32 visual frames and 82.68% once raw audio was added, a negligible -0.02 point difference, and where raw audio became beneficial mainly once visual evidence was removed entirely, raising black-frame accuracy from 67.40% to 73.19% [4]. The present results reveal a substantially stronger role for audio across the evaluated architectures. This may reflect model-family effects, differences in the reasoning categories considered here, or stronger audio exploitation in more recent omnimodal architectures, and the available results support the behavioural difference without by themselves isolating its architectural cause.

Transcription-Only Performance

Comparing raw audio against its transcription gives a more direct view of the distinction between acoustic and linguistic information.

Model	Audio Only	Transcription Only	Difference
Gemma	74.58%	70.25%	+4.33
Gemma-4-12B	75.92%	69.25%	+6.67
Qwen	73.50%	66.50%	+7.00
Qwen-30B	79.17%	77.00%	+2.17

Table 4.12: Comparison between raw audio and transcription-only accuracy. Positive differences indicate an advantage for raw audio.

Raw audio outperforms transcription for every model, from 7.00 points for Qwen down to 2.17 for Qwen-30B, which shows that the contribution of audio cannot be reduced to a simple speech-to-text transformation. Transcription preserves lexical and propositional content but strips out, or compresses, non-verbal sound, prosody, overlapping speakers, temporal acoustic events, background activity, and other properties of speech delivery, and the transcription pipeline can itself introduce recognition errors or drop uncertain material. The textual representation is, in short, a lossy transformation of the original signal. The gap between the small and large Qwen configurations adds a further, linguistically informative pattern. Qwen reaches 66.50% with transcription alone, Qwen-30B reaches 77.00%, a gap considerably larger than the 0.75-point difference the same two models show in video-only input. Increased model capacity appears especially beneficial once information is already represented in linguistic form, where a larger backbone can exploit lexical semantics, discourse relations, event descriptions, and contextual associations more effectively than a smaller one, so scaling gains look more pronounced in language-mediated reasoning than in isolated visual grounding. This fits the more general distinction between perceptual encoding and linguistic inference: once evidence has been converted to text, what remains is largely semantic selection and integration, whereas visual evidence must first be turned into an appropriate linguistic-semantic representation before the same kind of reasoning can even begin. The present results also contrast with *Lost in Transcription*, where Qwen2.5-VL-7B reached 68.72% in the linguistic-only baseline but only 63.98% with transcription alone, meaning transcription failed to outperform the model's own linguistic prior [4]. The present models instead gain substantially from transcription relative to their no-input baselines, particularly Qwen-30B, suggesting that the ability to convert auxiliary textual evidence into useful task information is not a stable property of an architecture but varies with scale.

Video-Only Performance and the Visual Bottleneck

Video-only accuracy ranges from 69.83% to 73.25%, as already shown in Table 4.7. All four models extract substantial task-relevant evidence from the sampled frames, yet the narrow range across architectures suggests that visual processing constitutes a shared bottleneck rather than a scale-dependent one. Increasing model size from Qwen to Qwen-30B produces only a 0.75-point gain in video-only accuracy, against a 10.50-point gap between the same two models in transcription-only accuracy. Linguistic scale, in other words, primarily improves operations performed once relevant semantic information is already accessible, lexical comparison, semantic compatibility assessment, integration of explicit contextual evidence, and reasoning over propositions, without producing an equivalent improvement in low-level or temporally distributed visual extraction. This distinction becomes considerably sharper once video-only accuracy is set against the preliminary analysis of Section 4.1. Overall visual comprehension there reached only 20.46 for Gemma, 25.90 for Gemma-4-12B, and 27.75 for Qwen-30B, well below the 69.83%, 73.08%, and 73.25% these same models achieve in video-only VSV. The discrepancy is not a contradiction between the two evaluations, but a reflection of the different informational regimes each one operates under. The preliminary analysis constructs a representation with no access to the downstream question, asking whether the required entities, events, and relations are already present in a question-independent description of the video. The VSV condition instead gives the model the question together with two minimally contrastive alternatives, allowing visual processing to become explicitly conditioned on the linguistic target of the task, directing attention toward particular entities, actions, or relations, while the contrastive alternatives themselves narrow the decision space enough that a full, exhaustive semantic representation of the video is not strictly necessary to choose correctly. A low preliminary score therefore shows that the relevant evidence is not reliably present in a model's spontaneous visual representation, not that the same evidence can never be recovered once the question identifies what to look for. The distinction can be summarised as one between *question-independent representation* and *question-conditioned retrieval*, the former measuring what is represented before any task-specific linguistic guidance, the latter measuring what the question itself can pull out of the visual input once it acts as a retrieval constraint. The original MAIA study offers a useful external reference point. With 32 frames, Qwen2.5-VL-7B obtained a pool-based VSV accuracy of 0.44, and Qwen2.5-VL-72B reached 0.54 [22], a clear scaling advantage that nonetheless fell well short of full consistency, with categories such as Spatial Total, Causal, and Implicit Total improving strongly with richer visual evidence while temporal reasoning remained comparatively weak. Direct numerical comparison with the present models must stay cautious, since the current evaluation covers a different category subset and different architectures, but the qualitative pattern is the same: visual evidence improves performance substantially, yet stronger language-model capacity alone does not eliminate the difficulty of grounding temporally and relationally complex information, a difficulty the preliminary analysis locates specifically at the level of what gets extracted from the frames in the first place.

Joint Analysis of Preliminary Representation and Video-Only Performance

To investigate the relationship between the information recovered during the preliminary analysis and downstream VSV performance, a paired comparison was conducted between the question-independent visual representation and the corresponding *video-only* condition.

The analysis was performed at two complementary levels. First, results were aggregated at the video level. For each video, the preliminary scores associated with the causal, spatial, and temporal questions were averaged and compared with the corresponding VSV performance obtained from the same video. Second, the analysis was repeated at the question level, preserving the alignment between each preliminary representation and its corresponding VSV pool.

At the video level, the following preliminary dimensions were considered: entity recovery, event recovery, inference recovery, and overall comprehension. VSV performance was measured both through item-level accuracy and through pool-level Aggregate Accuracy. Since preliminary scores are continuous and bounded, while video-level VSV accuracy is discrete and not normally distributed, Spearman's rank correlation was adopted as the primary association measure.

Video-Level Correlation

Table 4.13 reports the correlation between average preliminary comprehension and downstream *video-only* performance.

The relationship between preliminary comprehension and downstream performance is positive for all models, but generally weak.

Gemma-4-12B presents the clearest association. Average preliminary comprehension correlates significantly with both item-level accuracy and pool-level accuracy. The correlation with item-level performance is

$$\rho = 0.280, \qquad p = .0048,$$

Model	ρ **Item Acc.**	p	ρ **Pool Acc.**	p
Gemma	0.156	.120	0.160	.111
Gemma-4-12B	**0.280**	**.0048**	**0.254**	**.0107**
Qwen-30B	0.113	.262	0.112	.269

Table 4.13: Spearman correlations between average preliminary comprehension and downstream video-only VSV performance at the video level. Item Accuracy is computed over all VSV items associated with each video. Pool Accuracy measures the proportion of fully correct semantic pools within the video.

while the corresponding association with Aggregate Accuracy is

$$\rho = 0.254, \qquad p = .0107.$$

Videos for which Gemma-4-12B constructs a richer question-independent representation therefore tend to be the same videos for which the model achieves stronger downstream VSV performance.

The magnitude of the association remains limited. Preliminary comprehension does not account for most of the variation observed in downstream accuracy. This indicates that the availability of information in the initial visual representation contributes to task success but does not fully determine it.

Gemma and Qwen-30B exhibit the same positive tendency, although their correlations do not reach statistical significance. For Gemma, the correlation between preliminary comprehension and item accuracy is $\rho = 0.156$, while Qwen-30B reaches only $\rho = 0.113$.

The Qwen-30B result is particularly informative. This model obtains the highest average preliminary comprehension among the evaluated systems and also reaches the highest video-only accuracy. Nevertheless, the videos on which Qwen-30B produces its strongest preliminary representations are not systematically the same videos on which it achieves the highest VSV performance.

This weak correspondence suggests that downstream performance cannot be interpreted as a direct consequence of the amount of information spontaneously recovered before question exposure.

Contribution of Entity, Event, and Inference Recovery

The preliminary representation was subsequently decomposed into its three principal dimensions in order to identify which level of semantic information is most strongly associated with downstream VSV performance.

For Gemma-4-12B, event recovery and inference recovery show the strongest correlations with item-level accuracy.

Preliminary Component	ρ	p	**Holm** p
Entity Recovery	0.174	.084	.084
Event Recovery	**0.291**	.0033	**.013**
Inference Recovery	**0.288**	.0036	**.013**
Overall Comprehension	**0.280**	.0048	**.013**

Table 4.14: Spearman correlations between preliminary-analysis components and video-only item accuracy for Gemma-4-12B. Holm-corrected p-values account for multiple comparisons.

Entity recovery alone does not show a statistically robust relationship with downstream accuracy after correction. By contrast, both event recovery and inference recovery remain significant.

This distinction is consistent with the structure of the VSV task. The presence of a visually grounded entity does not necessarily provide enough information to discriminate between a caption and its foil. Correct resolution frequently depends on what the entity is doing, on the relation between multiple events, or on the higher-level inference targeted by the question.

The same pattern is visible at the pool level. Event recovery, inference recovery, and overall comprehension remain significantly associated with Aggregate Accuracy for Gemma-4-12B, while entity recovery provides a weaker contribution.

These results indicate that the diagnostic value of the preliminary representation increases as the extracted information approaches the semantic structure required by the downstream question.

Question-Level Pass and Downstream Success

Video-level aggregation combines causal, spatial, and temporal questions within the same video. To preserve the direct relationship between the preliminary representation and each downstream question, a second analysis was conducted at the question level.

The qualitative preliminary labels were binarized into two conditions. *Sufficient* and *complete* representations were treated as *Pass*, while *partial* and *insufficient* representations were treated as *Not Pass*.

Table 4.15 reports the probability of obtaining a completely correct VSV pool under the two preliminary conditions.

Model	Not Pass	Pass	Odds Ratio	p
Gemma	32.5%	50.0%	2.08	.084
Gemma-4-12B	39.0%	61.0%	**2.44**	**.010**
Qwen-30B	35.9%	67.3%	**3.68**	$< .001$

Table 4.15: Probability of obtaining a fully correct VSV pool as a function of preliminary-analysis proficiency. Pass includes sufficient and complete preliminary representations. Odds ratios quantify the increase in downstream pool success associated with a Pass representation.

The relationship becomes substantially clearer when the analysis is performed at the question level.

For Gemma-4-12B, a Pass preliminary representation is associated with a pool success rate of 61.0%, compared with 39.0% when the representation is classified as Not Pass. The corresponding odds ratio is

$$\mathrm{OR} = 2.44, \qquad p = .010.$$

Qwen-30B exhibits an even stronger effect. Pool accuracy increases from 35.9% in the Not Pass condition to 67.3% in the Pass condition. The corresponding odds ratio is

$$\mathrm{OR} = 3.68, \qquad p < .001.$$

A question for which Qwen-30B constructs a sufficient or complete preliminary representation therefore has more than three times the odds of producing a fully correct downstream pool compared with a question whose preliminary representation is insufficient or partial.

Gemma exhibits the same directional tendency, although the effect does not reach statistical significance at the pool level.

These results demonstrate that the preliminary analysis possesses genuine diagnostic value. When the information required by a question is already present at a sufficient level before question exposure, downstream performance becomes substantially more reliable.

Item-Level Accuracy under Pass and Not Pass Conditions

The same comparison was repeated using item-level VSV accuracy.

Model	**Not Pass**	**Pass**	p
Gemma	68.8%	**80.8%**	.025
Gemma-4-12B	71.1%	**85.4%**	.0027
Qwen-30B	70.3%	**87.5%**	$< .001$

Table 4.16: Video-only item accuracy according to preliminary-analysis proficiency. Pass corresponds to sufficient or complete preliminary representations.

The probability of producing a correct VSV response increases significantly when the preliminary representation satisfies the proficiency criterion. Gemma increases from 68.8% to 80.8%, Gemma-4-12B from 71.1% to 85.4%, and Qwen-30B from 70.3% to 87.5%. The effect is statistically significant for all three models.

The magnitude of the increase is particularly strong for Qwen-30B, where sufficient preliminary evidence is associated with an improvement of more than 17 percentage points.

The Not Pass condition is equally informative. Even when the preliminary analysis indicates that the question-independent representation does not contain sufficient evidence, video-only accuracy remains close to 70% for all models.

This result shows that downstream correctness cannot be equated with complete prior availability of the required information.

A model may still resolve the VSV item after receiving the question because the linguistic input constrains which portions of the visual stream should be inspected. The question can therefore function as a retrieval signal that directs the model toward entities, events, or relations that were not selected during the preliminary question-independent analysis.

Additional mechanisms may also contribute. Partial visual evidence may be sufficient to eliminate one of the alternatives. Linguistic plausibility and parametric world knowledge may support the comparison between minimally contrastive statements. The present analysis does not isolate the relative contribution of these mechanisms, but it demonstrates that successful task execution frequently occurs even when the required evidence was not sufficiently represented beforehand.

Category-Specific Associations

An exploratory category-level analysis was conducted to determine whether the relationship between preliminary comprehension and downstream performance differs across causal, spatial, and temporal questions.

For Gemma, the causal category shows a statistically significant positive correlation:

$$\rho = 0.284, \qquad p = .0042.$$

The effect remains significant after Holm correction:

$$p_{\text{Holm}} = .0127.$$

No statistically significant association emerges for Gemma in the spatial or temporal categories.

For Qwen-30B, spatial questions show a positive association:

$$\rho = 0.215, \qquad p = .032.$$

The effect does not remain significant after correction for the three category-level comparisons:

$$p_{\text{Holm}} = .096.$$

Gemma-4-12B does not show a statistically significant correlation within any single category, despite presenting a significant association at the global video level. This suggests that the relationship between preliminary representation and downstream performance is distributed across different reasoning phenomena rather than being driven by a single category. The category-specific results should therefore be interpreted as exploratory. The strongest evidence concerns the global question-level distinction between Pass and Not Pass representations.

Question-Independent Representation and Question-Conditioned Retrieval

The combined analysis reveals a clear distinction between the availability of visual evidence and successful downstream task execution. A sufficient preliminary representation substantially increases the probability of correct VSV performance. This demonstrates that the quality of the visual representation contributes directly to downstream success. At the same time, the relatively weak video-level correlations and the high VSV accuracy observed under Not Pass conditions show that preliminary proficiency is not a deterministic prerequisite for correct task execution. This distinction follows directly from the informational structure of the two settings. During preliminary analysis, the representation is constructed without access to the question. The model must therefore encode the scene without knowing which entities, events, or relations will subsequently become relevant. During the video-only VSV task, the model receives both the question and the minimally contrastive candidate statements. Linguistic information can guide visual attention toward the specific semantic content required for the decision. The preliminary analysis can therefore be interpreted as a measure of *question-independent evidence availability*, whereas downstream video-only performance reflects a combination of perceptual grounding, question-conditioned retrieval, semantic comparison, and linguistic reasoning. The statistical results support this distinction. Preliminary proficiency predicts downstream success locally, especially when the representation reaches sufficient or complete coverage. However, it explains only a limited proportion of the variation in video-level performance. The relationship between preliminary analysis and VSV accuracy is therefore informative precisely because it is neither absent nor deterministic. Relevant visual evidence improves downstream performance, but correct VSV decisions can also emerge when that evidence was not sufficiently represented before question exposure.

Audio–Visual Integration

Adding raw audio to video improves item-level accuracy for every model. The descrip-

Model	Video Only	Audio + Video	Δ_{audio}
Gemma	69.83%	77.17%	+7.33
Gemma-4-12B	73.08%	80.50%	+7.42
Qwen	72.50%	78.92%	+6.42
Qwen-30B	73.25%	83.17%	+9.92

Table 4.17: Conditional gain produced by adding raw audio to the video-only condition.

Model	Comparison	Base Acc.	New Acc.	Δ	95% CI	Holm p
Gemma	Video $\rightarrow$ Audio+Video	69.83%	77.17%	+7.33	[4.75, 10.00]	$< .001$
Gemma-4-12B	Video $\rightarrow$ Audio+Video	73.08%	80.50%	+7.42	[4.67, 10.25]	$< .001$
Qwen	Video $\rightarrow$ Audio+Video	72.50%	78.92%	+6.42	[3.33, 9.42]	$< .001$
Qwen-30B	Video $\rightarrow$ Audio+Video	73.25%	83.17%	+9.92	[7.25, 12.58]	$< .001$

Table 4.18: Exact McNemar comparisons between the video-only condition and the audio-visual condition. Accuracy differences are in percentage points. Confidence intervals were obtained through cluster bootstrap resampling at the pool level, and reported p-values are Holm-corrected.

tive gains, from 6.42 points for Qwen to 9.92 for Qwen-30B, are all statistically robust: every confidence interval excludes zero and every Holm-corrected test remains significant. Raw audio is therefore not simply ignored once visual evidence is present, a pattern that differs from the 32-frame results in *Lost in Transcription*, where Qwen2.5-Omni-7B showed virtually identical accuracy with and without raw audio [4]. Between 44.75% and 49.84% of video-only errors are corrected once raw audio is introduced, indicating that the acoustic stream frequently carries information capable of resolving insufficient or ambiguous visual evidence, while damage rates stay below 9% for every model. Qwen-30B shows the strongest asymmetry, a Repair Rate of 49.84% against a Damage Rate of only 4.66%, meaning its audio-visual improvement is not the product of generic prediction instability but of the additional modality converting incorrect visual decisions into correct

Model	Repair Rate	Damage Rate
Gemma	44.75%	8.83%
Gemma-4-12B	48.92%	7.87%
Qwen	46.97%	8.97%
Qwen-30B	**49.84%**	**4.66%**

Table 4.19: Repair and damage rates when raw audio is added to video input. Repair Rate measures the proportion of video-only errors corrected by audio. Damage Rate measures the proportion of video-only correct predictions that become incorrect after adding audio.

multimodal ones. The underlying McNemar contingency counts make the direction of this effect explicit: for Qwen-30B, adding raw audio repairs 160 video-only errors while introducing only 41 new ones. Linguistically, this kind of repair is consistent with the acoustic modality supplying a semantic representation that resolves underspecified visual evidence, spoken descriptions lexicalizing events difficult to infer from sparse frames, or environmental sounds disambiguating between competing event interpretations, which the language backbone can then use to judge which candidate statement is better supported. The reciprocal question, whether vision still adds anything once audio is already available, receives a more mixed answer. Vision significantly improves on audio-only performance

Model	Audio Only	Audio+Video	Δ	Holm p
Gemma	74.58%	77.17%	+2.58	.153
Gemma-4-12B	75.92%	80.50%	+4.58	.002
Qwen	73.50%	78.92%	+5.42	$< .001$
Qwen-30B	79.17%	83.17%	+4.00	$< .001$

Table 4.20: Exact McNemar comparisons between audio-only and audio–video conditions.

for Gemma-4-12B, Qwen, and Qwen-30B, but not for Gemma, whose +2.58 point gain does not survive Holm correction ($p = .153$). For this model, raw audio already captures a large share of the evidence that determines the final prediction, and adding vision produces no statistically robust item-level advantage, a pattern compatible with a degree of modality dominance in which the acoustic channel alone accounts for most of what the task requires. The other three models instead show clear, significant benefits from the joint availability of both channels, evidence that vision continues to contribute even once audio is already on the table.

Video and Transcription Integration

The effect of transcription becomes markedly dependent on model scale once visual information is already available. Gemma and Qwen gain 5.25 and 3.00 points respectively,

Model	**Video Only**	**Video + Transcription**	$\Delta_{\text{transcription}}$
Gemma	69.83%	75.08%	+5.25
Gemma-4-12B	73.08%	84.75%	+11.67
Qwen	72.50%	75.50%	+3.00
Qwen-30B	73.25%	84.08%	+10.83

Table 4.21: Conditional gain produced by adding transcription to the video-only condition.

Model	**Comparison**	**Base Acc.**	**New Acc.**	Δ	**95% CI**	**Holm** p
Gemma	Video $\rightarrow$ Video+Transcription	69.83%	75.08%	+5.25	[2.25, 8.25]	$< .001$
Gemma-4-12B	Video $\rightarrow$ Video+Transcription	73.08%	84.75%	+11.67	[8.25, 15.17]	$< .001$
Qwen	Video $\rightarrow$ Video+Transcription	72.50%	75.50%	+3.00	[0.25, 5.75]	.023
Qwen-30B	Video $\rightarrow$ Video+Transcription	73.25%	84.08%	+10.83	[8.17, 13.58]	$< .001$

Table 4.22: Exact McNemar comparisons between the video-only condition and the video-plus-transcription condition.

Gemma-4-12B and Qwen-30B gain 11.67 and 10.83, and every one of these differences remains significant after Holm correction, including Qwen's comparatively small gain ($p = .023$), so even the weakest effect in this comparison cannot be dismissed as noise. The contribution of linguistic auxiliary information clearly increases in the larger configurations. A transcription is already expressed in the representational domain where the language backbone is strongest, but exploiting it well still requires identifying the portions semantically related to the question, suppressing irrelevant discourse, resolving references, and integrating textual evidence with the visual representation, operations the larger models appear considerably better at performing. Increased capacity may therefore add not just richer linguistic knowledge but stronger contextual filtering and semantic alignment between heterogeneous, language-compatible sources of evidence. Unlike

Model	**Transcription Only**	**Video+Transcription**	Δ	**Holm** p
Gemma	70.25%	75.08%	+4.83	.011
Gemma-4-12B	69.25%	84.75%	+15.50	$< .001$
Qwen	66.50%	75.50%	+9.00	$< .001$
Qwen-30B	77.00%	84.08%	+7.08	$< .001$

Table 4.23: Exact McNemar comparisons between transcription-only and video-plus-transcription conditions.

Model	**Repair Rate**	**Damage Rate**
Gemma	43.65%	11.34%
Gemma-4-12B	**64.09%**	7.64%
Qwen	41.21%	11.49%
Qwen-30B	50.78%	**3.75%**

Table 4.24: Repair and damage rates when transcription is added to video input.

the raw-audio comparison of Section 4.2, vision significantly improves on transcription-only performance in every model, and the effect is largest precisely where the video-plus-transcription gain was largest, Gemma-4-12B, where 15.50 additional points correspond to 238 items moving from incorrect to correct against only 52 moving the other way, for a Repair Rate of 64.09% and a Damage Rate of just 7.64%. Qwen-30B shows a similarly stable, asymmetric pattern, a Repair Rate of 50.78% and the lowest Damage Rate observed in any transcription-based comparison, 3.75%. The magnitude and direction of this asymmetry indicate that transcription and visual content carry genuinely complementary information rather than one substituting for the other, since a model relying purely on textualized audio and ignoring the video would show no such systematic, one-sided correction pattern. This behaviour is the opposite of what *Lost in Transcription* reported, where Qwen2.5-VL-7B decreased from 80.67% with 32 frames alone to 78.78% with video and transcription combined, a 1.89 point drop that item-level McNemar analysis found to be systematic, with correct predictions converted into errors more often than incorrect ones were repaired [4]. The current models instead find transcription beneficial rather than interfering, particularly at larger scale, which suggests that transcription-based multimodal integration is not a stable property of the task alone, but depends substantially on model architecture, scale, and the language backbone's ability to regulate auxiliary textual evidence.

Raw Audio and Transcription as Complementary Representations

The purely descriptive ranking between the two multimodal conditions suggests a split by model size. For Gemma and Qwen, raw audio gives the strongest multimodal configuration, 77.17% against 75.08% for Gemma and 78.92% against 75.50% for Qwen. For the larger configurations the ranking flips, 84.75% against 80.50% for Gemma-4-12B and 84.08% against 83.17% for Qwen-30B. Once tested directly, this small-versus-large

Model	**Audio+Video**	**Video+Transcription**	$\Delta_{\mathrm{VT-AV}}$	**Holm** p
Gemma	77.17%	75.08%	-2.08	.153
Gemma-4-12B	80.50%	84.75%	+4.25	.001
Qwen	78.92%	75.50%	-3.42	.009
Qwen-30B	83.17%	84.08%	+0.92	.351

Table 4.25: Direct comparison between native audio–video input and video enriched with transcription. Positive differences favour transcription, negative differences favour raw audio.

story does not hold up cleanly. Gemma's numerical advantage for raw audio and Qwen-30B's numerical advantage for transcription both fail to reach significance ($p = .153$ and $p = .351$), so for these two models the item-level data cannot statistically distinguish between the two multimodal representations. A significant, representation-dependent preference emerges only for the remaining two models, and not in a way that lines up neatly with scale, Gemma-4-12B performs significantly better with transcription-derived linguistic information ($\Delta = +4.25$ points, $p = .001$), while Qwen shows the opposite pattern, native raw audio outperforming transcription by 3.42 points ($p = .009$). The data therefore do not support a general claim that smaller models systematically prefer raw audio and larger models systematically prefer transcription, only that the advantage of one representation over the other is model-dependent, likely reflecting the specific interaction between a model's modality-specific encoders and the linguistic reasoning capacity of its backbone, rather than a property of scale on its own. This also clarifies why transcription should not be treated as a simple stand-in for audio. Raw audio preserves acoustic and temporal information at the cost of requiring cross-modal representation alignment, while transcription discards part of that information in exchange for direct access to lexical and propositional structure, and which of the two wins out for a given model depends on the balance it strikes between perceptual richness and linguistic accessibility, a balance that this analysis shows is architecture-specific rather than governed by parameter count alone.

Aggregate Accuracy and Prediction Consistency

Item-level accuracy does not reveal whether a model behaves consistently across the several minimally contrastive realizations tied to the same underlying question. Aggregate Accuracy, which counts a pool as correct only when every one of its items is answered correctly, gives a stricter estimate of robustness. The gap between item-level and aggregate

Model	Audio Only	Transc. Only	Video Only	Audio + Video	Video + Transc.
Gemma	41.67%	37.67%	34.00%	49.00%	42.67%
Gemma-4-12B	47.00%	36.00%	42.00%	55.33%	**64.00%**
Qwen	40.33%	31.33%	38.33%	50.00%	44.33%
Qwen-30B	55.67%	52.00%	41.33%	60.67%	**63.33%**

Table 4.26: Aggregate Accuracy across experimental conditions. A pool is counted as correct only when the model correctly resolves all minimally contrastive items associated with it.

performance is substantial for every model, confirming that isolated correct predictions do not necessarily correspond to stable semantic discrimination. Multimodal conditions improve consistency as well as raw correctness, Gemma-4-12B rises from 42.00% Aggregate

Accuracy in video-only to 64.00% with video and transcription, Qwen-30B from 41.33% to 63.33%. This connects directly to MAIA's own evaluation philosophy, which treats visually grounded competence as unproven by isolated correct decisions if semantically equivalent reformulations produce inconsistent predictions [22]. The original 32-frame VSV results reached 0.44 pool accuracy for Qwen2.5-VL-7B and 0.54 for Qwen2.5-VL-72B [22], while the present multimodal conditions push above 0.60 for the larger models, though direct comparison remains limited by the different model families and category subsets involved. Repeating the paired significance tests at pool level largely confirms these item-level results while also exposing an important exception in prediction consistency. Adding raw audio to video significantly improves Aggregate Accuracy for every

Model	**Comparison**	**Base AA**	**New AA**	Δ	**Holm** p
Gemma	Video $\rightarrow$ Audio+Video	34.00%	49.00%	+15.00	$< .001$
Gemma	Video $\rightarrow$ Video+Transcription	34.00%	42.67%	+8.67	.019
Gemma-4-12B	Video $\rightarrow$ Audio+Video	42.00%	55.33%	+13.33	$< .001$
Gemma-4-12B	Video $\rightarrow$ Video+Transcription	42.00%	64.00%	+22.00	$< .001$
Qwen	Video $\rightarrow$ Audio+Video	38.33%	50.00%	+11.67	$< .001$
Qwen	Video $\rightarrow$ Video+Transcription	38.33%	44.33%	+6.00	.071
Qwen-30B	Video $\rightarrow$ Audio+Video	41.33%	60.67%	+19.33	$< .001$
Qwen-30B	Video $\rightarrow$ Video+Transcription	41.33%	63.33%	+22.00	$< .001$

Table 4.27: Exact McNemar comparisons at pool level using Aggregate Accuracy.

model. Transcription produces a significant pool-level gain for Gemma, Gemma-4-12B, and Qwen-30B, but not for Qwen, whose item-level improvement from 72.50% to 75.50% (Section 4.2, $p = .023$) does not carry over to its Aggregate Accuracy gain from 38.33% to 44.33% ($p = .071$). For this model, transcription improves local, item-level decisions without providing corrections systematic enough to raise the number of fully consistent pools to a statistically robust degree, an effect detectable in isolated predictions but too weak to survive the stricter, semantic-consistency criterion MAIA is specifically designed to enforce. Gemma-4-12B and Qwen-30B, by contrast, both gain 22 percentage points in Aggregate Accuracy from transcription, remaining highly significant after correction, indicating that for these two models the additional linguistic evidence improves not just how many decisions are correct but how stable those decisions are across minimally different formulations of the same content.

Audio–Video Interaction Effect

Positive multimodal gains do not by themselves imply that audio and video combine independently. A logistic interaction analysis over the 2×2 factorial design of no-input, audio-only, video-only, and audio-visual conditions returns a negative interaction coefficient for every model.

Model	$\beta_{A \times V}$	p**-value**	**Odds Ratio**
Gemma	-0.399	$< .001$	0.671
Gemma-4-12B	-0.355	.001	0.701
Qwen	-0.636	$< .001$	0.529
Qwen-30B	-0.298	.011	0.743

Table 4.28: Audio–video interaction coefficients estimated from the factorial comparison between no-input, audio-only, video-only, and audio–video conditions. Negative coefficients indicate a sub-additive interaction.

All four coefficients are negative and statistically significant, Qwen showing the strongest interaction at $\beta_{A \times V} = -0.636$ and Qwen-30B the weakest at -0.298. A negative coefficient does not mean combining audio and video reduces absolute performance, since audio-visual accuracy is higher than video-only accuracy for every model, as already established by the significant McNemar comparisons in Section 4.2. It means instead that the joint gain is smaller than an additive model of independent modality contributions would predict, pointing to partial informational redundancy or competition between the two channels, some of what audio contributes may already be recoverable from the video, and the model may fail to preserve the full benefit of each source once both compete within a shared representational or attentional space.

Read together, these two results, a significant audio-visual improvement over video alone, and a significant sub-additive interaction, describe two distinct properties of multimodal behaviour rather than contradicting one another. The pairwise tests establish that audio provides useful additional evidence whenever video is available. The interaction coefficient establishes that this additional evidence is not combined with the visual channel as an independent, additive source. Audio and video are complementary, in that both improve performance, but part of their informational content overlaps or is not fully preserved during integration, which is exactly the distinction between *multimodal benefit* and *multimodal integration* that this thesis has argued for throughout. The models clearly benefit from multiple channels. Their interaction coefficients show that those channels are not combined as independent additive sources of evidence.

This result is compatible with the modality-balancing problem already identified in *Lost in Transcription*, where Qwen2.5-Omni dynamically reduced attention to audio as visual information increased, without the attention still allocated to audio necessarily translating into corresponding gains in reasoning performance [4]. The present behavioural results extend this observation across different architectures: additional modalities are used, but their informational contribution stays partly redundant, and their combination falls short of a fully additive effect.

Model Scale and Linguistic Integration

Model scale affects the different input modalities unevenly. Its weakest effect appears in the video-only condition, where Qwen and Qwen-30B differ by only 0.75 points, and its strongest effect appears in transcription-based conditions, where Qwen-30B exceeds Qwen by 10.50 points with transcription alone and by 8.58 points with video plus transcription. The Gemma family shows the same tendency, a 3.25-point advantage for Gemma-4-12B over Gemma in video-only input widening to 9.67 points in video plus transcription.

Scaling, in other words, disproportionately improves the exploitation of linguistically structured contextual evidence. A larger language backbone can encode finer lexical distinctions, maintain longer discourse dependencies, resolve referential relations more effectively, and evaluate semantic compatibility between the question, the transcription, and the candidate statements, all of which matter once acoustic information has already been converted into text. Visual input poses a different problem: before any of that linguistic reasoning can apply, the model has to derive a sufficiently informative semantic representation from perceptual evidence in the first place, and increasing language-model capacity offers no guarantee that the relevant entities, events, spatial relations, or temporal dependencies were correctly extracted to begin with.

This asymmetry is directly supported by the preliminary analysis in Section 4.1, where scale raised entity, event, and inference recovery scores but never removed the underlying entity-event-inference hierarchy, and where the video-only bottleneck discussed in Section 4.2 traced this same limitation back to what the visual representation itself contains, independently of any linguistic reasoning applied afterward. The original MAIA results point in the same direction: Qwen2.5-VL-72B substantially outperformed Qwen2.5-VL-7B in the 32-frame pool-based evaluation, 0.54 against 0.44 [22], yet continued to report weak performance on temporally demanding categories despite this scaling advantage, suggesting that increased overall capacity does not uniformly resolve perceptual grounding limitations.

Taken together, the present results sharpen this picture considerably. Scaling is especially effective once evidence is already available in linguistic form, while gains on isolated visual input stay comparatively constrained, an asymmetry consistent with an account of multimodal performance resting on at least two separable capacities, extracting task-relevant evidence from a given modality, and reasoning over that evidence once it has been mapped into a semantically accessible representation. The preliminary analysis in Section 4.1 speaks primarily to the first capacity, the experimental results discussed in this section speak primarily to the second, and it is the interaction between the two, rather than either taken alone, that determines how well a given model ultimately integrates audio, video, and language in Visual Statement Verification.

ATTENTION ANALYSIS - ANCORA NON PERVENUTA

5. Conclusions

Conclusioni...

6. Future Perspectives

The directions provided should be viewed as a continuous trajectory rather than four separate extensions. The aim is to progressively refine the diagnosis of multimodal integration. This begins with observing behavioral differences across reasoning categories, advances to analyzing the structure of what each model extracts from the video, then delves into the internal dynamics of attention, and ultimately culminates in causal verification.

Expansion to the Twelve Reasoning Categories The experiments in this thesis treat spatial, temporal, and causal reasoning as the primary axes of analysis, without asking whether modality contribution itself varies across the full set of reasoning phenomena covered by *MAIA*. A natural extension is to apply the same behavioural and modality-contribution metrics separately within each of the twelve categories, rather than only comparing aggregate accuracy across them:

$$\text{Model} \times \text{Modality} \times \text{Reasoning Category}. \tag{6.1}$$

This decomposition would make it possible to establish whether different reasoning phenomena induce different forms of modality reliance, for instance whether causal or counterfactual questions depend more heavily on audio than purely spatial ones, rather than assuming a single, uniform pattern of integration across the benchmark.

Richer Preliminary Analysis The existing preliminary analysis divides visual evidence into entities, events, and causal or inferred relationships. To achieve a more detailed representation, it would be beneficial to further segregate the outputs of each model into distinct components: entity attributes, actions, event participants, spatial relations, temporal relations, state changes, and event dependencies. These components should be monitored separately rather than aggregated into a singular event-level description. The main objective of this refinement is diagnostic. It aims to differentiate between a true perceptual failure, in which vital information was not captured at all, and a failure in forming the connections between accurately perceived elements, or even a failure in the ensuing reasoning process. For instance, if a model accurately identifies an entity but erroneously associates it with the wrong event, this should be distinguishable from a scenario where the model entirely fails to perceive the entity.

Joint Analysis of Semantic Availability, Attention, and Behaviour This direction develops directly out of the attention analysis introduced in this thesis. Rather than treating semantic availability, attention allocation, and behavioural contribution as three separate questions, a natural next step is to study their joint relationship:

$$\text{Semantic Availability} \longleftrightarrow \text{Attention Allocation} \longleftrightarrow \text{Behavioural Contribution.} \tag{6.2}$$

The question is no longer only *whether the model found a given piece of information, how much it attends to a given modality, or how much accuracy improves when that modality is added*, but **how these three quantities relate to one another at the level of individual items**. This connects directly back to the question raised by *Lost in Transcription* [4]: whether a modality that is available is also genuinely exploited, and whether its internal use corresponds to an observable contribution to the final decision, rather than being available, attended, and yet behaviourally inert.

From Attention to Causal Attribution This last direction is deliberately the most methodologically demanding, and is placed last for that reason. The attention analysis presented in this thesis remains descriptive: it shows where the model allocates attention, not whether that allocation determines the prediction. Techniques such as activation patching, representation ablation, attention intervention, or gradient-based attribution would make it possible to test whether specific multimodal representations have a causal effect on the model's answer, moving the question from

$$\text{Where does the model attend?} \tag{6.3}$$

to

$$\text{Which information actually affects the prediction?} \tag{6.4}$$

Taken together, these four directions trace a coherent path rather than four independent extensions of the current work. Behavioural differences across reasoning categories identify *where* modality reliance varies, a richer preliminary analysis identifies *what* each model can perceive and relate on its own, the joint analysis of semantic availability, attention, and behaviour identifies *whether* available information is actually used, and causal attribution identifies *whether that use actually matters* to the final decision. Pursued together, they would move the evaluation of multimodal integration from behavioural comparison toward a genuinely mechanistic account of how, and whether, audio-visual models integrate the evidence available to them.

List of Abbreviations

VLM	Visual Language Model
VQA	Visual Question Answering
AGQA	Action Genome Question Answering
MVBench	Multi-modal Video Understanding Benchmark
VILMA	Video Language Model Assessment
MAIA	Multimodal AI Assessment
VSV	Visual Statement Verification
OEVQA	Open-Ended Visual Question Answering
MLM	Multimodal Language Model
AVQA	Audio-Visual Question Answering
Dave	Diagnostic benchmark for Audio Visual Evaluation
MMAU-Pro	Massive Multi-Task Audio Understanding and Reasoning benchmark, Pro version
SONIC-01	SOcial Natural Interaction Corpus (Omnimodal, v1)
GPT	Generative Pre-trained Transformer
MIRAGE	Assessing Hallucination in Multimodal Reasoning Chains of Multimodal Large Language Models

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